

Seminar on Algebraic Geometry 3

Group Schemes

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Reconstruction status

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Editorial Notice

We present here a slightly revised edition of the original Seminar, whose purpose and contents are described in the Introduction. The revision consisted mainly of correcting typographical errors, adding a few supplementary remarks or references in footnotes, dividing the work in its present form into three volumes, each with a detailed table of contents and an index of notation, and adding a terminological index at the end of Volume 3. In addition, Exposé VI_B, by J.-E. Bertin, was partially rewritten by the author, especially Sections 5 and 10, so some references to that Exposé differ from the references to the original version. A list of the Exposés in the Seminar appears at the beginning of this volume.

Since the first edition of the present Seminar appeared, all of Chapter IV of the *Éléments de Géométrie Algébrique* has been published, making certain passages of the Seminar unnecessary. We have occasionally indicated in footnotes the relevant references to EGA IV that allow those passages to be bypassed.

For another treatment of algebraic groups that systematically uses the language of schemes, we refer to M. Demazure and P. Gabriel, *Groupes Algébriques* (North-Holland & Masson et Cie). Unlike the present Seminar, that book assumes no knowledge of algebraic geometry, but includes all the required preliminaries from scheme theory; it can therefore serve as an introduction to the study of this Seminar. (It also treats subjects not covered in the Seminar, such as the Dieudonné-style structure theory of commutative affine algebraic groups, in *loc. cit.*, Chapter V.)

Bures-sur-Yvette, March 1970

Introduction

1. The purpose of this Seminar is twofold.

On the one hand, we aim to provide convenient foundations for the theory of group schemes in general. Exposés I–IV will supply the indispensable preliminary exercises in schematic and categorical syntax. To obtain a language that fits geometric intuition without effort, and to avoid circumlocutions that become intolerable in the long run, we always identify a prescheme X over another prescheme S with the functor

$$(\mathbf{Sch})_{/S}^{\circ} \longrightarrow \mathbf{Set}$$

that it represents*. It is therefore necessary to formulate many definitions so that they apply to arbitrary functors, whether representable or not. Moreover, nearly all the functors we shall use will be “sheaves” (for the “faithfully flat quasi-compact topology”). Exposé IV, which treats groups only incidentally, sketches the language of “localization” and sheaves; this language also proves very convenient in questions concerning the representability of functors. Above all, that Exposé will provide the most convenient framework for the questions of passage to a quotient that arise later.

Exposé V gives some general results on the existence of quotients. They are taken up again in Exposé VI_A for the quotient of an algebraic group over a field (or, more generally, over an Artinian ring) by a subgroup**. That Exposé and the following Exposé VI_B also contain various elementary results specific to algebraic groups over a field that are used routinely later.

Exposé VII studies certain phenomena related to the characteristic of the base field and, in particular, develops with the appropriate generality the correspondence between radicial group schemes of height 1 and restricted p -Lie algebras.

Finally, Exposé XVIII contains the scheme-theoretic generalization of Weil’s theorem on the “birational” definition of algebraic groups.

On the other hand, we intend to generalize the Borel–Chevalley structure theory of affine algebraic groups to groups over an arbitrary base prescheme. In fact, while the Seminar notes were being written, it became apparent that the affine hypothesis was unnecessary for many results of the theory. The most complete results are of course obtained for “semisimple

*Such a viewpoint seems to have been considered for the first time eight or nine years ago, in connection with the theory of formal groups, by P. Cartier, who unfortunately did not take the trouble to state it precisely and systematize it as it deserved.

**For a more thorough study of passage to the quotient, especially by reductive groups, see D. Mumford’s important work *Geometric Invariant Theory*, *Ergebnisse der Mathematik*, vol. 34, Springer, 1965. We note that on one important point the terminology of that book does not agree with ours: over a field of characteristic $p > 0$, the groups Mumford calls “reductive”⁽¹⁾ are the smooth groups of multiplicative type in the sense of the Seminar (cf. Exposé IX). One may reasonably suppose that Mumford adopted this meaning of “reductive,” which loses its significance over a base that is not a field, provisionally and as a kind of stopgap (and this is approximately what Mumford explains, for other reasons, as early as the second paragraph of his preface). ⁽¹⁾ *Editor’s note:* See the remarks added at the end of this Introduction.

group schemes,” or more generally “reductive group schemes,” with which we shall be exclusively concerned beginning with Exposé XIX. Chevalley himself had already constructed the “Tôhoku groups” over the ring of integers, and that construction will be taken up in the present Seminar. The principal uniqueness theorem⁽²⁾ gives a simple characterization of the “twisted” variants of these Tôhoku groups over a base prescheme S : they are the affine, smooth group schemes over S whose geometric fibers are connected semisimple groups in the usual sense^{***}(2).

2.

As in the familiar theory over an algebraically closed field, the subtori of the group schemes under consideration play a crucial role. Thus the preliminary study of tori and, more generally, of “group schemes of multiplicative type”—both intrinsically and as multiplicative-type subgroups of a given group—occupies a fairly large part of this Seminar (Exposés VIII–XII). Their remarkable rigidity (in some respects even greater than that of abelian schemes or semisimple group schemes) makes them highly effective tools for studying certain more general groups.

3.

Beginning with Exposé XII (apart from Exposé XVIII, already mentioned), we shall routinely use the theory of affine algebraic groups over an algebraically closed field, which the reader will find in the 1956 Chevalley Seminar, especially in Exposés IV–IX of that Seminar. We shall also use, though to a lesser extent, the later Exposés of the Chevalley Seminar devoted to the structure of semisimple algebraic groups. Indeed, we shall redevelop Chevalley’s theory directly within the framework of schemes; it will be seen that this approach makes the exposition simpler and more precise, even over a base field.

4.

The principal purpose of this Seminar is, of course, to develop techniques that apply to the study of group schemes over an arbitrary base, that is, essentially, to the study of families of algebraic groups. The infinitesimal properties of such families, and in particular the case of an Artinian base scheme, therefore play an important role. These properties arise even in the study of algebraic groups over a field k when the field is not perfect, especially when applying descent in the non-Galois case. Among the new results obtained in this setting, let us mention the existence of maximal tori and Cartan subgroups in an arbitrary smooth algebraic group, the rationality of the variety of maximal tori, and various related results (Exposé XIV⁽³⁾); another example is the correspondence between the “forms” of a semisimple group and homogeneous principal bundles under a suitable semisimple algebraic group (generally not connected) (Exposé XXIV, 1.16–1.20). In general, one may say that the methods required for work over a nonperfect base field are essentially those used over arbitrary base preschemes and thus lie outside the framework of classical algebraic geometry.

⁽²⁾ *Editor’s note:* This refers to Corollary XXIII.5.6, which follows easily from the uniqueness theorem for split reductive groups (cf. XXIII, Theorem 4.1 and Corollary 5.2), since every semisimple S -group is a “twisted form” (for the étale topology) of a Tôhoku group (cf. XXII, Corollary 2.3).

***This is the essential result of M. Demazure’s thesis, *Schémas en groupes réductifs*, Bull. Soc. Math. France **93** (1965), 369–413.

⁽³⁾ *Editor’s note:* In particular, Theorems 1.1 and 6.1.

5.

It did not seem useful to indicate, at the beginning of the written Exposés, the date or dates of the corresponding oral presentations in the Seminar. We shall merely say that the order of the mimeographed Exposés (I–XXVI) does indeed agree with the order of the oral presentations. Moreover, the final written text was sometimes prepared considerably later than the oral presentation and often differs from it quite substantially: the written text is generally more detailed and more complete (as in Exposés IV and VII_B), or even appreciably more general (as in Exposés XII and VII_B), than the oral presentation. Other written Exposés correspond to no oral presentation (VI_B, VII_A, XV, XVI, XVII, and most of XXVI); they were written and inserted into the mimeographed Seminar either to provide convenient references for various other Exposés (notably in the case of VI_B) or because they form a natural continuation of the concepts and techniques already developed. Consequently, Exposés VII_A, VII_B, XV, XVI, and XVII need not be read in order to study the rest of the Seminar, particularly the part devoted to reductive group schemes.

6.

From scheme theory we shall use primarily the general language of schemes set out in EGA I; the notions of flat, étale, and smooth morphisms set out in SGA 1, I–V; and the theory of faithfully flat descent in SGA 1, VIII. As far as possible, we have avoided imposing unnecessary Noetherian hypotheses. In return, this has required us to replace the usual hypothesis “of finite type” with the hypothesis “of finite presentation.” For the notion of a morphism of finite presentation, the reader should consult EGA IV, 1.4 and 1.6. The results of SGA 1, I–IV, which are most often stated in the Noetherian setting, will be developed in general in EGA IV****; standard methods for reducing certain kinds of statements involving finite-presentation hypotheses to the Noetherian case will also be developed there in detail (EGA IV, Sections 8, 9, and 11). A reader who does not wish to assume these results from EGA IV may simplify certain statements or their proofs by assuming the base prescheme to be locally Noetherian. This does, however, lead to difficulties when a proof proceeds by descent from $\hat{A}$ to A , where $\hat{A}$ is the completion of a Noetherian local ring A , because that method introduces the generally non-Noetherian ring

$$\hat{A} \otimes_A \hat{A}.$$

7.

References follow the usual decimal system: the reference 5.7.11 denotes the proposition (or lemma, definition, and so forth) with that number in the same Exposé; in a reference such as XVII 7.8, the Roman numeral denotes the Exposé. We use the following abbreviations for our standard references:

Bible Chevalley Seminar, *Groupes de Lie algébriques*, 1956/58.

EGA X , $x.y.z$ J. Dieudonné and A. Grothendieck, *Éléments de Géométrie Algébrique*, Chapter X , statement $x.y.z$ (or subsection $x.y$, and so forth).

SGA n , $X y.z$ *Séminaire de Géométrie Algébrique du Bois-Marie*, year n , statement $y.z$ of Exposé X .

****Since this Introduction was written, all four parts (Sections 1–21) of EGA IV have appeared.

TDTE A. Grothendieck, *Techniques de descente et Théorèmes d'existence en Géométrie Algébrique*, Exposés delivered in the Bourbaki Seminar between 1959 and 1962.

⁽¹⁾ *Editor's note.* Concerning quotients by a reductive group, the situation has evolved considerably since A. Grothendieck wrote Note (**). For an affine algebraic group over an arbitrary field k , the appropriate notion, introduced by Mumford, is that of a “geometrically reductive” group. (On the other hand, G is said to be “linearly reductive” if every rational representation of G is completely reducible; as noted in Note (**), however, that condition is too restrictive when $\text{char}(k) > 0$.) By results of M. Nagata and W. J. Haboush ([Na64], [NM64], [Ha75]), the geometrically reductive k -groups are precisely the reductive k -groups in the sense of the present Seminar. For all of this, see the later editions of Mumford’s book ([MF82]). The notion of geometric reductivity over an arbitrary base, together with its consequences for passage to a quotient, was extended to this setting by C. S. Seshadri ([Se77]); additions due to M. Raynaud appear in the article [CTS79] by J.-L. Colliot-Thélène and J.-J. Sansuc. Finally, for more recent developments on passage to the quotient by an algebraic group or, more generally, by a groupoid (cf. Exposé V of the present Seminar), we refer to the articles [Ko97] and [KM97] by J. Kollár and by S. Keel and S. Mori.

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Exposé I

Algebraic Structures. Group Cohomology

by *M. Demazure*

This exposé consists of two parts: the first collects a number of general definitions and establishes notation that will often be used in the sequel; the second treats group cohomology and culminates in Theorem 5.3.3, on the vanishing of the cohomology of diagonalizable groups.¹

We choose once and for all a Universe.² All definitions and constructions will be relative to this Universe. We shall systematically allow ourselves the following abuse of language. To define a functor $f : \mathcal{C} \rightarrow \mathcal{C}'$, we shall be content to define the object $f(S)$ of $\mathcal{C}'$ for every object S of $\mathcal{C}$, whenever there is no ambiguity about how to define $f(h)$ for an arrow h of $\mathcal{C}$. In practice, we shall say: let $f : \mathcal{C} \rightarrow \mathcal{C}'$ be the functor defined by $f(S) = \dots$.

1. Generalities

1.1.

Let $\mathcal{C}$ be a category. We denote by $\widehat{\mathcal{C}}$ the category

$$\mathrm{Hom}(\mathcal{C}^\circ, \mathbf{Set})$$

of contravariant functors from $\mathcal{C}$ to the category $\mathbf{Set}$ of sets.³ There is a canonical functor

$$\mathbf{h} : \mathcal{C} \longrightarrow \widehat{\mathcal{C}}$$

which associates with every $X \in \mathrm{Ob}(\mathcal{C})$ the functor $\mathbf{h}_X$ such that

$$\mathbf{h}_X(S) = \mathrm{Hom}(S, X).$$

For every functor $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathcal{C}})$, one has (see, for example, EGA 0_{III}, 8.1.4) a bijection

$$\mathrm{Hom}(\mathbf{h}_X, \mathbf{F}) \xrightarrow{\sim} \mathbf{F}(X).$$

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¹Editor's note: version of October 13, 2024.

²Editor's note: see SGA 4, Exposé I, §0 and the Appendix; see also the discussion in [DG70], p. xxvi.

³Editor's note: this is called the category of presheaves on $\mathcal{C}$; see IV, 4.3.1.

⁴Editor's note: this result is often called the Yoneda lemma; we use that terminology in other editor's notes.

In particular, for every pair X, Y of objects of $\mathcal{C}$, the canonical map

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{\mathcal{C}}}(\mathbf{h}_X, \mathbf{h}_Y)$$

is bijective; that is, the functor $\mathbf{h}$ is fully faithful. It therefore defines an isomorphism of $\mathcal{C}$ onto a full subcategory of $\widehat{\mathcal{C}}$, and an equivalence of $\mathcal{C}$ with the full subcategory of $\widehat{\mathcal{C}}$ formed by the representable functors (that is, the functors isomorphic to one of the form $\mathbf{h}_X$). In the sequel we shall often identify X with $\mathbf{h}_X$. The following paragraphs are intended to show that this identification can be made safely.

Remark 1.1.1.⁵ We sometimes need the following variant. Let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$, and let $X, Y \in \mathrm{Ob}(\mathcal{D})$. Denote by $\mathbf{h}'_X$ and $\mathbf{h}'_Y$ the restrictions of $\mathbf{h}_X$ and $\mathbf{h}_Y$ to $\mathcal{D}$. Then

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) = \mathrm{Hom}_{\mathcal{D}}(X, Y) = \mathrm{Hom}_{\widehat{\mathcal{D}}}(\mathbf{h}'_X, \mathbf{h}'_Y).$$

Thus, to give a morphism $X \rightarrow Y$ “is the same thing” as to give, for every $T \in \mathrm{Ob}(\mathcal{D})$, a map

$$\phi(T) : \mathrm{Hom}(T, X) \longrightarrow \mathrm{Hom}(T, Y)$$

functorially in T .

1.2.

We shall say that $\mathbf{F}$ is a *subobject* (or a subfunctor) of $\mathbf{G}$ if $\mathbf{F}(S)$ is a subset of $\mathbf{G}(S)$ for every S .⁶

In $\widehat{\mathcal{C}}$, arbitrary inverse limits exist and are computed pointwise:

$$\left(\varprojlim_i \mathbf{F}_i \right) (S) = \varprojlim_i \mathbf{F}_i(S).$$

⁷ In particular, fiber products are defined by

$$(\mathbf{F} \times_{\mathbf{G}} \mathbf{F}')(S) = \mathbf{F}(S) \times_{\mathbf{G}(S)} \mathbf{F}'(S).$$

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As a final object of $\widehat{\mathcal{C}}$, we choose the functor $\mathbf{e}$ such that $\mathbf{e}(S) = \{\emptyset\}$.⁹ Every $\mathbf{F} \in \mathrm{Ob}(\widehat{\mathcal{C}})$ has a unique morphism to $\mathbf{e}$, and we set

$$\mathbf{F} \times \mathbf{F}' = \mathbf{F} \times_{\mathbf{e}} \mathbf{F}'.$$

The functor $\mathbf{h}$ commutes with inverse limits. In particular, for $X \times X'$ to exist, where $X, X' \in \mathrm{Ob}(\mathcal{C})$, respectively for $\mathcal{C}$ to admit a final object e , it is necessary and sufficient that $\mathbf{h}_X \times \mathbf{h}_{X'}$, respectively $\mathbf{e}$, be representable; in that case

$$\mathbf{h}_X \times \mathbf{h}_{X'} \simeq \mathbf{h}_{X \times X'}, \quad \mathbf{h}_e \simeq \mathbf{e}.$$

⁵Editor's note: this remark has been added.

⁶Editor's note: the order has been changed so that fiber products are introduced before monomorphisms; see editor's note (9).

⁷Editor's note: similarly, arbitrary direct limits exist and are computed pointwise: $(\varinjlim_i \mathbf{F}_i)(S) = \varinjlim_i \mathbf{F}_i(S)$. In general, however, the functor $\mathbf{h}$ does not commute with direct limits.

⁸Editor's note: in particular, the equalizer of a pair of morphisms $u, v : \mathbf{F} \rightrightarrows \mathbf{G}$ is the subfunctor $\mathrm{Ker}(u, v)$ of $\mathbf{F}$ defined by $\mathrm{Ker}(u, v)(S) = \{x \in \mathbf{F}(S) \mid u(x) = v(x)\}$.

⁹Editor's note: $\{\emptyset\}$, the power set of the empty set, is a one-element set.

A monomorphism in $\widehat{\mathcal{C}}$ is precisely a morphism $\mathbf{F} \rightarrow \mathbf{G}$ such that, for every $S \in \text{Ob}(\mathcal{C})$, the corresponding map of sets $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is injective.¹⁰

The functor Γ . For every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$, set

$$\Gamma(\mathbf{F}) = \text{Hom}(\mathbf{e}, \mathbf{F}).$$

An element of $\Gamma(\mathbf{F})$ is therefore a family $(\gamma_S)_{S \in \text{Ob}(\mathcal{C})}$, with $\gamma_S \in \mathbf{F}(S)$, such that for every arrow $f : S' \rightarrow S''$ of $\mathcal{C}$,

$$\mathbf{F}(f)(\gamma_{S''}) = \gamma_{S'}.$$

For $X \in \text{Ob}(\mathcal{C})$, set $\Gamma(X) = \Gamma(\mathbf{h}_X)$. If $\mathcal{C}$ has a final object e , then

$$\Gamma(X) \simeq \text{Hom}(e, X).$$

1.3.

Let $S \in \text{Ob}(\mathcal{C})$. We denote by $\mathcal{C}_{/S}$ the category of objects of $\mathcal{C}$ over S : its objects are the arrows $f : T \rightarrow S$ of $\mathcal{C}$, and $\text{Hom}(f, f')$ is the subset of $\text{Hom}(T, T')$ formed by the arrows u such that $f = f' \circ u$. If $\mathcal{C}$ has a final object e , then $\mathcal{C}_{/e}$ is isomorphic to $\mathcal{C}$. The category $\mathcal{C}_{/S}$ has a final object, namely the identity arrow $S \rightarrow S$.

If $f : T \rightarrow S$ is an object of $\mathcal{C}_{/S}$, one may form the category $(\mathcal{C}_{/S})_{/f}$, which by abuse of language we denote by $(\mathcal{C}_{/S})_{/T}$, and there is a canonical isomorphism

$$\mathcal{C}_{/T} \simeq (\mathcal{C}_{/S})_{/T}.$$

This construction also applies to $\widehat{\mathcal{C}}$; in particular, one obtains the category $\widehat{\mathcal{C}}_{/\mathbf{h}_S}$. On the other hand, one may form the category $\widehat{\mathcal{C}}_{/S}$.

If $f : T \rightarrow S$ is an object of $\mathcal{C}_{/S}$, then $\Gamma(f)$ is identified with the set $\Gamma(T/S)$ of sections of T over S , that is, arrows $S \rightarrow T$ that are right inverses to f . Moreover, $\mathbf{h}_f : \mathbf{h}_T \rightarrow \mathbf{h}_S$ is then an object of $\widehat{\mathcal{C}}_{/\mathbf{h}_S}$, and

$$\Gamma(\mathbf{h}_f) \simeq \Gamma(\mathbf{h}_T/\mathbf{h}_S) \simeq \Gamma(T/S) \simeq \Gamma(f).$$

1.4.

We now define an equivalence between the categories $\widehat{\mathcal{C}}_{/S}$ and $\widehat{\mathcal{C}}_{/\mathbf{h}_S}$. In other words, we shall prove that to give a functor on the category of objects of $\mathcal{C}$ over S is “the same thing” as to give a functor on $\mathcal{C}$ equipped with a morphism to $\mathbf{h}_S$.

(i) **Construction of $\alpha_S : \widehat{\mathcal{C}}_{/\mathbf{h}_S} \rightarrow \widehat{\mathcal{C}}_{/S}$.**

First let $H : \mathbf{F} \rightarrow \mathbf{h}_S$ be an object of $\widehat{\mathcal{C}}_{/\mathbf{h}_S}$. We must define a functor $\alpha_S(H)$ on $\mathcal{C}_{/S}$. For an object $f : T \rightarrow S$ of $\mathcal{C}_{/S}$, define $\alpha_S(H)(f)$ to be the inverse image of $f \in \mathbf{h}_S(T)$ under the map

$$H(T) : \mathbf{F}(T) \longrightarrow \mathbf{h}_S(T).$$

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¹⁰Editor's note: if $\mathbf{F}(S) \rightarrow \mathbf{G}(S)$ is injective for every S , it is clear that $\mathbf{F} \rightarrow \mathbf{G}$ is a monomorphism. The converse follows by considering $\mathbf{F} \times_{\mathbf{G}} \mathbf{F} \rightrightarrows \mathbf{F} \rightarrow \mathbf{G}$. Thus $\mathbf{F} \rightarrow \mathbf{G}$ is a monomorphism if and only if the diagonal morphism $\mathbf{F} \rightarrow \mathbf{F} \times_{\mathbf{G}} \mathbf{F}$ is an isomorphism (see EGA I, 5.3.8). Similarly, a pointwise surjective morphism is an epimorphism, and the converse follows by considering the pushout $\mathbf{G} \amalg_{\mathbf{F}} \mathbf{G}$; see the proof of Lemma 4.4.4 in Exposé IV.

¹¹Editor's note: for example, if $\mathbf{F} = \mathbf{h}_X$, then H corresponds to a morphism $h : X \rightarrow S$, and $H(T) : \text{Hom}_{\mathcal{C}}(T, X) \rightarrow \text{Hom}_{\mathcal{C}}(T, S)$ is the map $g \mapsto h \circ g$. Thus $\alpha_S(\mathbf{h}_X) = \text{Hom}_{\mathcal{C}_{/S}}(-, X)$.

Next, let $u : f \rightarrow f'$ be an arrow of $\mathcal{C}_{/S}$. Then $\mathbf{F}(u) : \mathbf{F}(T') \rightarrow \mathbf{F}(T)$ induces a map from $\alpha_S(H)(f')$ to $\alpha_S(H)(f)$, denoted $\alpha_S(H)(u)$. It is immediate that the assignments

$$f \mapsto \alpha_S(H)(f), \quad u \mapsto \alpha_S(H)(u)$$

define a functor on $\mathcal{C}_{/S}$, hence an object $\alpha_S(H)$ of $\widehat{\mathcal{C}_{/S}}$.

Finally, let $H : \mathbf{F} \rightarrow \mathbf{h}_S$ and $H' : \mathbf{F}' \rightarrow \mathbf{h}_S$ be two objects of $\widehat{\mathcal{C}_{/\mathbf{h}_S}}$, and let $U : H \rightarrow H'$ be a morphism in that category:

$$\begin{array}{ccc} \mathbf{F} & \xrightarrow{U} & \mathbf{F}' \\ & \searrow H & \swarrow H' \\ & \mathbf{h}_S & \end{array} .$$

For every $f : T \rightarrow S$, the map $U(T) : \mathbf{F}(T) \rightarrow \mathbf{F}'(T)$ induces a map

$$\alpha_S(U)(f) : \alpha_S(H)(f) \rightarrow \alpha_S(H')(f),$$

which defines a morphism of functors

$$\alpha_S(U) : \alpha_S(H) \rightarrow \alpha_S(H').$$

It is easy to verify that the assignments

$$H \mapsto \alpha_S(H), \quad U \mapsto \alpha_S(U)$$

define a functor

$$\alpha_S : \widehat{\mathcal{C}_{/\mathbf{h}_S}} \rightarrow \widehat{\mathcal{C}_{/S}}.$$

(ii) **Proposition 1.4.1.** *The functor $\alpha_S : \widehat{\mathcal{C}_{/\mathbf{h}_S}} \rightarrow \widehat{\mathcal{C}_{/S}}$ is an equivalence of categories.*

We indicate only the principle of the construction of a quasi-inverse functor

$$\beta_S : \widehat{\mathcal{C}_{/S}} \rightarrow \widehat{\mathcal{C}_{/\mathbf{h}_S}}.$$

Let $\mathbf{G}$ be a functor on $\mathcal{C}_{/S}$. For every object T of $\mathcal{C}$, set

$$\beta_S(\mathbf{G})(T) = \coprod_{f \in \text{Hom}(T, S)} \mathbf{G}(f), \quad \text{Hom}(T, S) = \mathbf{h}_S(T).$$

This defines a functor $\beta_S(\mathbf{G})$ on $\mathcal{C}$, equipped with the evident projection to $\mathbf{h}_S$.

1.5.

The equivalence α_S commutes with the functors Γ . In other words, if $H : \mathbf{F} \rightarrow \mathbf{h}_S$ is an object of $\widehat{\mathcal{C}_{/\mathbf{h}_S}}$ and $\alpha_S(H)$ is the corresponding object of $\widehat{\mathcal{C}_{/S}}$, then

$$\Gamma(\alpha_S(H)) \simeq \Gamma(H) \simeq \Gamma(\mathbf{F}/\mathbf{h}_S).$$

The equivalence α_S also commutes with the functors $\mathbf{h}$. That is, if $f : T \rightarrow S$ is an object of $\mathcal{C}_{/S}$, then $\mathbf{h}_f : \mathbf{h}_T \rightarrow \mathbf{h}_S$ is an object of $\widehat{\mathcal{C}_{/\mathbf{h}_S}}$, and its image under α_S is precisely $\mathbf{h}_{\mathcal{C}_{/S}}(f)$, where

$$\mathbf{h}_{\mathcal{C}_{/S}} : \mathcal{C}_{/S} \rightarrow \widehat{\mathcal{C}_{/S}}$$

is the canonical functor.¹² Consequently:

Proposition 1.5.1. *Let $H : \mathbf{F} \rightarrow \mathbf{h}_S$ be an object of $\widehat{\mathcal{C}}_{\mathbf{h}_S}$. The functor*

$$\alpha_S(H) : (\mathcal{C}/S)^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if $\mathbf{F} : \mathcal{C}^\circ \rightarrow \mathbf{Set}$ is representable. If $\mathbf{F} \simeq \mathbf{h}_T$, then $\alpha_S(H)$ is represented by the object $T \rightarrow S$ of $\mathcal{C}/S$.

The equivalence α_S is transitive in S . If $f : T \rightarrow S$ is an object of $\mathcal{C}/S$, there is a commutative diagram of equivalences:

$$\begin{array}{ccccc} (\widehat{\mathcal{C}}_{\mathbf{h}_S})_{/\mathbf{h}_T} & \xrightarrow{\alpha_S/\mathbf{h}_T} & (\widehat{\mathcal{C}/S})_{/\mathbf{h}_T} & \xrightarrow{\alpha_f} & (\widehat{\mathcal{C}/S})_{/T} \\ & \searrow \cong & & & \swarrow \cong \\ & & \widehat{\mathcal{C}}_{/\mathbf{h}_T} & \xrightarrow{\alpha_T} & \widehat{\mathcal{C}}_{/T} \end{array},$$

Here $\alpha_S/\mathbf{h}_T$ denotes, provisionally, the restriction (see 1.6) of the functor α_S to the objects over $\mathbf{h}_T$.

1.6. Base change in a functor

For every $S \in \text{Ob}(\mathcal{C})$, there is a canonical functor

$$i_S : \mathcal{C}/S \longrightarrow \mathcal{C}$$

defined by $i_S(f) = T$ when f is the arrow $T \rightarrow S$. If $f : T \rightarrow S$ is an object of $\mathcal{C}/S$, we denote by $i_{T/S} = i_f$ the functor

$$i_{T/S} : (\mathcal{C}/S)_{/T} \longrightarrow \mathcal{C}/S,$$

and there is a commutative diagram:

$$\begin{array}{ccccc} (\mathcal{C}/S)_{/T} & \xrightarrow{i_{T/S}} & \mathcal{C}/S & \xrightarrow{i_S} & \mathcal{C} \\ & \searrow \cong & & & \swarrow i_T \\ & & \mathcal{C}_{/T} & & \end{array},$$

Thus, identifying $(\mathcal{C}/S)_{/T}$ with $\mathcal{C}_{/T}$, as we shall henceforth do,

$$i_S \circ i_{T/S} = i_T.$$

Likewise, if $\mathcal{C}$ is identified with $\mathcal{C}_{/e}$ when $\mathcal{C}$ has a final object e , then $i_{S/e} : \mathcal{C}/S \rightarrow \mathcal{C}_{/e}$ is identified with i_S .

For $X \in \text{Ob}(\mathcal{C})$ (respectively, $Y \in \text{Ob}(\mathcal{C}/S)$), let $p_S(X)$ (respectively, $p_{T/S}(Y)$) be the object of $\mathcal{C}/S$ (respectively, of $\mathcal{C}_{/T}$), when it exists, defined by $X \times_S$ (respectively, $Y \times_S T$) equipped with its second projection:

¹²Editor's note: see editor's note (10).

$$\begin{array}{ccc}
X \times S & & Y \times_S T \\
p_S(X) \downarrow & \text{resp.} & \downarrow p_{T/S}(Y) \\
S & & T
\end{array}
.$$

The partially defined functor p_S (respectively, $p_{T/S}$) is called the *base-change functor*. By the definition of a product (respectively, a fiber product), it is right adjoint to i_S (respectively, to $i_{T/S}$).¹³ We also write

$$p_S(X) = X_S, \quad p_{T/S}(Y) = Y_T.$$

The functor i_S defines a restriction functor

$$i_S^* : \widehat{\mathcal{C}} \longrightarrow \widehat{\mathcal{C}}_S.$$

We write

$$\mathbf{F}_S = i_S^*(\mathbf{F}) = \mathbf{F} \circ i_S.$$

Plainly,

$$i_{T/S}^* \circ i_S^* = i_T^*,$$

that is, for every functor $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$,

$$(\mathbf{F}_S)_T = \mathbf{F}_T.$$

The notation is justified by the following result.

Proposition 1.6.1. *The functor*

$$(\mathbf{h}_X)_S : (\mathcal{C}_S)^\circ \longrightarrow \mathbf{Set}$$

is representable if and only if the product $X \times S$ exists. In that case

$$(\mathbf{h}_X)_S \simeq \mathbf{h}_{X_S}.$$

This shows that $\mathbf{F}_S$ has two interpretations: it is the restriction of $\mathbf{F}$ to $\mathcal{C}_S$, and it is the functor obtained by the base change $e \leftarrow S$. This leads to the following notation:

$$\begin{array}{ccccc}
\mathbf{F} & \xrightarrow{\quad} & \mathbf{F}_S & \xrightarrow{\quad} & \mathbf{F}_T \\
\downarrow & & \downarrow & & \downarrow \\
e & \xleftarrow{\quad} & S & \xleftarrow{\quad} & T
\end{array}$$

¹³Editor's note: that is, if $g : U \rightarrow S$ (respectively, $h : V \rightarrow T$) is an object of $\mathcal{C}_S$ (respectively, of $\mathcal{C}_T$), then $i_S(g) = U$, while $i_{T/S}(h)$ is the object $f \circ h : V \rightarrow S$ of $\mathcal{C}_S$, and

$$\text{Hom}_{\mathcal{C}_S}(U, X \times S) \simeq \text{Hom}_{\mathcal{C}}(U, X),$$

respectively,

$$\text{Hom}_{\mathcal{C}_T}(V, X \times_S T) \simeq \text{Hom}_{\mathcal{C}_S}(V, X).$$

This diagram reflects both of the preceding interpretations. Notice that

$$\Gamma(\mathbf{F}_S) \simeq \text{Hom}(\mathbf{h}_S, \mathbf{F}) \simeq \mathbf{F}(S),$$

and, in particular,

$$\Gamma(X_S) \simeq \text{Hom}(S, X).$$

1.7.0.

Let $\mathbf{E}$ be an object of $\widehat{\mathcal{C}}$.¹⁴ Consider the category $\mathcal{C}/\mathbf{E}$ of *objects of $\mathcal{C}$ over $\mathbf{E}$* . Its objects are pairs (V, ρ) formed by an object V of $\mathcal{C}$ and a morphism $\rho : \mathbf{h}_V \rightarrow \mathbf{E}$ in $\widehat{\mathcal{C}}$, that is, $\rho \in \mathbf{E}(V)$. A morphism from (V, ρ) to (V', ρ') is a morphism $f : V \rightarrow V'$ in $\mathcal{C}$ such that

$$\rho' \circ f = \rho, \quad \text{that is,} \quad \mathbf{E}(f)(\rho') = \rho.$$

Let $\mathbf{L}$ be the functor

$$\mathbf{L} = \varinjlim_{(V, \rho) \in \mathcal{C}/\mathbf{E}} \mathbf{h}_V.$$

Thus, for every $S \in \text{Ob}(\mathcal{C})$,

$$\mathbf{L}(S) = \varinjlim_{(V, \rho) \in \mathcal{C}/\mathbf{E}} \mathbf{h}_V(S)$$

is the set of equivalence classes of triples (V, ρ, v) , where $v : S \rightarrow V$ is a morphism in $\mathcal{C}$, and where (V, ρ, v) is identified with $(V', \rho', f \circ v)$ for every morphism $f : V \rightarrow V'$ in $\mathcal{C}$ such that $\rho' \circ f = \rho$.

The map $\phi_{\mathbf{E}}(S)$ which associates with the class of (V, ρ, v) the element $\rho \circ v$ of $\mathbf{E}(S)$ is well defined and gives a morphism of functors

$$\phi_{\mathbf{E}} : \varinjlim_{(V, \rho) \in \mathcal{C}/\mathbf{E}} \mathbf{h}_V \longrightarrow \mathbf{E}.$$

Lemma. *The morphism $\phi_{\mathbf{E}}$ is an isomorphism.*

Indeed, let $S \in \text{Ob}(\mathcal{C})$. Every $x \in \mathbf{E}(S)$ is the image under $\phi_{\mathbf{E}}(S)$ of the triple (S, x, id_S) ; hence $\phi_{\mathbf{E}}(S)$ is surjective. Conversely, let $\ell_1 = (V_1, \rho_1, v_1)$ and $\ell_2 = (V_2, \rho_2, v_2)$ be two elements of $\mathbf{L}(S)$ having the same image in $\mathbf{E}(S)$, and set

$$\gamma = \rho_1 \circ v_1 = \rho_2 \circ v_2.$$

Then ℓ_1 and ℓ_2 are both equal in $\mathbf{L}(S)$ to the class of (S, γ, id_S) . Thus $\phi_{\mathbf{E}}(S)$ is injective.

Corollary. *For every object $\mathbf{F}$ of $\widehat{\mathcal{C}}$,*

$$\text{Hom}(\mathbf{E}, \mathbf{F}) = \varprojlim_{(V, \rho) \in \mathcal{C}/\mathbf{E}} \mathbf{F}(V).$$

¹⁴Editor's note: the lemma below has been added (see SGA 4, I.3.4); it is used in the proof of 1.7.1 and will be useful repeatedly in the sequel.

1.7. The objects $\underline{\text{Hom}}$, $\underline{\text{Isom}}$, etc.

Let $\mathbf{F}$ and $\mathbf{G}$ be two objects of $\widehat{\mathcal{C}}$. We define another object of $\widehat{\mathcal{C}}$ by

$$\begin{aligned}\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})(S) &= \text{Hom}_{\widehat{\mathcal{C}}/S}(\mathbf{F}_S, \mathbf{G}_S) \\ &\simeq \text{Hom}_{\widehat{\mathcal{C}}/\mathbf{h}_S}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G} \times \mathbf{h}_S) \\ &\simeq \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F} \times \mathbf{h}_S, \mathbf{G}).\end{aligned}$$

The object $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ has the following properties:

- (i) $\underline{\text{Hom}}(\mathbf{e}, \mathbf{G}) \simeq \mathbf{G}$.¹⁵
- (ii) The formation of $\underline{\text{Hom}}$ commutes with base extension:

$$\underline{\text{Hom}}(\mathbf{F}_S, \mathbf{G}_S) \simeq \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})_S.$$

- (iii) $(\mathbf{F}, \mathbf{G}) \mapsto \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ is a bifunctor, contravariant in $\mathbf{F}$ and covariant in $\mathbf{G}$.

These three properties follow immediately from the definitions.

We shall prove that, for every object $\mathbf{E}$ of $\widehat{\mathcal{C}}$,

$$\text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

Let $\phi : \mathbf{E} \times \mathbf{F} \rightarrow \mathbf{G}$. We must associate with it a morphism from $\mathbf{E}$ to $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$. For an arrow $S' \rightarrow S$ of $\mathcal{C}$, there are maps

$$\mathbf{E}(S) \times \mathbf{F}(S') \longrightarrow \mathbf{E}(S') \times \mathbf{F}(S') \xrightarrow{\phi(S')} \mathbf{G}(S').$$

Every element e of $\mathbf{E}(S)$ therefore defines, for each $S' \rightarrow S$, a map $\mathbf{F}(S') \rightarrow \mathbf{G}(S')$ functorial in S' ; this is an element $\theta_\phi(e)$ of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})(S)$. We have thus obtained a map

$$\phi \longmapsto \theta_\phi, \quad \text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \longrightarrow \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})),$$

which is functorial in $\mathbf{E}$.

Proposition 1.7.1.¹⁶ *Let $\mathbf{E}, \mathbf{F}, \mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$.*

- (a) *The map $\phi \mapsto \theta_\phi$ is a bijection:*

$$\text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G}) \xrightarrow{\sim} \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})).$$

- (b) *Moreover, there is an isomorphism of functors*

$$\underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G}).$$

¹⁵Editor's note: if $\mathbf{E}$ is a third object of $\widehat{\mathcal{C}}$, then

$$\underline{\text{Hom}}(\mathbf{E}, \mathbf{F} \times \mathbf{G}) \simeq \underline{\text{Hom}}(\mathbf{E}, \mathbf{F}) \times \underline{\text{Hom}}(\mathbf{E}, \mathbf{G}).$$

¹⁶Editor's note: part (b), which will be useful in II.1 and II.3.11, has been added. A second, more direct proof of part (a) has also been sketched.

For (a), regard both sides as functors of $\mathbf{E}$. The asserted result is true when $\mathbf{E} = \mathbf{h}_X$, since it is then precisely the definition of the functor $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$. Moreover, both sides, as functors of $\mathbf{E}$, take direct limits to inverse limits. Finally, by Lemma 1.7.0, every object $\mathbf{E}$ of $\mathcal{C}$ is isomorphic to the direct limit of the $\mathbf{h}_X$, as X ranges over $\mathcal{C}_{/\mathbf{E}}$. This proves (a).

Let us also sketch a direct proof of (a). To every

$$\theta \in \text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G}))$$

associate $\phi_\theta \in \text{Hom}(\mathbf{E} \times \mathbf{F}, \mathbf{G})$ as follows. For every $S \in \text{Ob}(\mathcal{C})$, there is a map

$$\theta(S) : \mathbf{E}(S) \longrightarrow \text{Hom}(\mathbf{F} \times S, \mathbf{G}),$$

functorial in S . If $(e, f) \in \mathbf{E}(S) \times \mathbf{F}(S)$, then f is a morphism $S \rightarrow \mathbf{F}$, so $f \times \text{id}_S$ is a morphism $S \rightarrow \mathbf{F} \times S$. On the other hand, $\theta(S)(e)$ is a morphism $\mathbf{F} \times S \rightarrow \mathbf{G}$. Composition therefore gives a morphism

$$\theta(S)(e) \circ (f \times \text{id}_S) : S \longrightarrow \mathbf{G},$$

that is, an element $\phi_\theta(S)(e, f)$ of $\mathbf{G}(S)$. It is easy to check that $S \mapsto \phi_\theta(S)$ is functorial in S , and therefore defines a morphism ϕ_θ from $\mathbf{E} \times \mathbf{F}$ to $\mathbf{G}$. We leave to the reader the verification that $\theta \mapsto \phi_\theta$ and $\phi \mapsto \theta_\phi$ are mutually inverse bijections.

For (b), if $S \in \text{Ob}(\mathcal{C})$, then by 1.7(ii) and part (a), applied to $\mathcal{C}_{/S}$,

$$\begin{aligned} & \underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G}))(S) \\ & \simeq \text{Hom}_S(\mathbf{E}_S, \underline{\text{Hom}}_S(\mathbf{F}_S, \mathbf{G}_S)) \\ & \simeq \text{Hom}_S(\mathbf{E}_S \times_S \mathbf{F}_S, \mathbf{G}_S) \\ & \simeq \text{Hom}(\mathbf{E} \times \mathbf{F} \times S, \mathbf{G}) \\ & \simeq \underline{\text{Hom}}(\mathbf{E} \times \mathbf{F}, \mathbf{G})(S), \end{aligned}$$

and these isomorphisms are functorial in S .

Corollary 1.7.2. *There are isomorphisms*

$$\text{Hom}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \text{Hom}(\mathbf{F}, \underline{\text{Hom}}(\mathbf{E}, \mathbf{G})),$$

and

$$\underline{\text{Hom}}(\mathbf{E}, \underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \underline{\text{Hom}}(\mathbf{F}, \underline{\text{Hom}}(\mathbf{E}, \mathbf{G})).$$

In particular, taking $\mathbf{E} = \underline{\mathbf{e}}$ and using $\underline{\text{Hom}}(\underline{\mathbf{e}}, \mathbf{G}) \simeq \mathbf{G}$, we obtain

$$\Gamma(\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})) \simeq \text{Hom}(\mathbf{F}, \mathbf{G}).$$

Composition supplies functorial morphisms

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{H}) \longrightarrow \underline{\text{Hom}}(\mathbf{F}, \mathbf{H}).$$

If $\mathbf{F}$ and $\mathbf{G}$ are two objects of $\widehat{\mathcal{C}}$, let $\text{Isom}(\mathbf{F}, \mathbf{G})$ denote the subset of $\text{Hom}(\mathbf{F}, \mathbf{G})$ formed by the isomorphisms from $\mathbf{F}$ onto $\mathbf{G}$. Define a subobject $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$ of $\underline{\text{Hom}}(\mathbf{F}, \mathbf{G})$ by

$$\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})(S) = \text{Isom}(\mathbf{F}_S, \mathbf{G}_S).$$

Then

$$\Gamma(\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})) \simeq \text{Isom}(\mathbf{F}, \mathbf{G}), \quad \underline{\text{Isom}}(\mathbf{F}, \mathbf{G}) \simeq \underline{\text{Isom}}(\mathbf{G}, \mathbf{F}).$$

In the special case $\mathbf{F} = \mathbf{G}$, set

$$\begin{aligned}\underline{\text{End}}(\mathbf{F}) &= \underline{\text{Hom}}(\mathbf{F}, \mathbf{F}), & \text{End}(\mathbf{F}) &= \text{Hom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\text{End}}(\mathbf{F})), \\ \underline{\text{Aut}}(\mathbf{F}) &= \underline{\text{Isom}}(\mathbf{F}, \mathbf{F}), & \text{Aut}(\mathbf{F}) &= \text{Isom}(\mathbf{F}, \mathbf{F}) \simeq \Gamma(\underline{\text{Aut}}(\mathbf{F})).\end{aligned}$$

Formation of the objects $\underline{\text{Hom}}$, $\underline{\text{Isom}}$, $\underline{\text{Aut}}$, and $\underline{\text{End}}$ commutes with base change.

Remark 1.7.3. ¹⁷ An object isomorphic to $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$ may also be constructed as follows. There is a morphism

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\text{End}}(\mathbf{F}).$$

Interchanging $\mathbf{F}$ and $\mathbf{G}$ gives a morphism

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F}) \longrightarrow \underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G}).$$

On the other hand, the identity morphism of $\mathbf{F}$ is an element of $\text{End}(\mathbf{F})$, and hence defines a morphism $\underline{\mathbf{e}} \rightarrow \underline{\text{End}}(\mathbf{F})$. Doing the same for $\mathbf{G}$ and taking the product gives a morphism

$$\underline{\mathbf{e}} \longrightarrow \underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G}).$$

It is then immediate that the fiber product of $\underline{\mathbf{e}}$ and

$$\underline{\text{Hom}}(\mathbf{F}, \mathbf{G}) \times \underline{\text{Hom}}(\mathbf{G}, \mathbf{F})$$

over $\underline{\text{End}}(\mathbf{F}) \times \underline{\text{End}}(\mathbf{G})$ is isomorphic to $\underline{\text{Isom}}(\mathbf{F}, \mathbf{G})$.

All these definitions apply, in particular, when $\mathbf{F} = \mathbf{h}_X$ and $\mathbf{G} = \mathbf{h}_Y$. If $\underline{\text{Hom}}(\mathbf{h}_X, \mathbf{h}_Y)$ is representable by an object of $\mathcal{C}$, we denote that object by $\underline{\text{Hom}}(X, Y)$. It has the following property: if $Z \times X$ exists, then

$$\text{Hom}(Z, \underline{\text{Hom}}(X, Y)) \simeq \text{Hom}(Z \times X, Y).$$

This property characterizes it when products exist in $\mathcal{C}$.

Similarly, one defines, when they exist, the objects

$$\underline{\text{Isom}}(X, Y), \quad \underline{\text{End}}(X), \quad \underline{\text{Aut}}(X).$$

We merely note that the construction above shows that $\underline{\text{Isom}}(X, Y)$ exists whenever fiber products exist in $\mathcal{C}$ and the objects

$$\underline{\text{Hom}}(X, Y), \quad \underline{\text{Hom}}(Y, X), \quad \underline{\text{End}}(X), \quad \text{and} \quad \underline{\text{End}}(Y)$$

exist.

Everything above also applies to categories of the form $\mathcal{C}_{/S}$. The corresponding objects will be denoted as explicitly as possible by appropriate symbols. For example, if T and T' are two objects of $\mathcal{C}$ over S , we denote by $\underline{\text{Hom}}_S(T, T')$ the object

$$\underline{\text{Hom}}_{\mathcal{C}_{/S}}(T/S, T'/S).$$

¹⁷Editor's note: the numbering 1.7.3 has been added for later references.

1.8. Constant objects

Let $\mathcal{C}$ be a category in which direct sums and fiber products exist and in which direct sums commute with base change (for example, the category of schemes).¹⁸ For every set E and every object S of $\mathcal{C}$, set

$$E_S = \{\text{the direct sum of a family } (S_i)_{i \in E} \text{ of objects of } \mathcal{C}, \text{ all isomorphic to } S. \quad (1.8.1)$$

This object is characterized by the formula

$$\mathrm{Hom}_{\mathcal{C}}(E_S, T) = \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}_{\mathcal{C}}(S, T)), \quad (1.8.2)$$

for every $T \in \mathrm{Ob}(\mathcal{C})$.

The object E_S has a canonical projection to S , so that $E \mapsto E_S$ is in fact a functor from $\mathbf{Set}$ to $\mathcal{C}/_S$.¹⁹ If $S' \rightarrow S$ is an arrow of $\mathcal{C}$, then, because direct sums commute with base change,

$$E_{S'} = (E_S)_{S'}.$$

In particular, if $\mathcal{C}$ has a final object e , then

$$E_S = (E_e)_S.$$

The functor $E \mapsto E_S/S$, from $\mathbf{Set}$ to $\mathcal{C}/_S$, commutes with finite products. It is enough to prove that

$$E_S \times_S F_S = (E \times F)_S. \quad (\times)$$

Indeed, by the results of 1.7 applied to $\mathcal{C}/_S$, for every $T \in \mathrm{Ob}(\mathcal{C}/_S)$ there are natural isomorphisms (all unspecified Hom 's being taken in $\mathcal{C}/_S$)

$$\begin{aligned} \mathrm{Hom}((E \times F)_S, T) &\simeq \mathrm{Hom}_{\mathbf{Set}}(E \times F, \mathrm{Hom}(S, T)) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}_{\mathbf{Set}}(F, \mathrm{Hom}(S, T))) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}(F_S, T)), \end{aligned}$$

and

$$\begin{aligned} \mathrm{Hom}(E_S \times_S F_S, T) &\simeq \mathrm{Hom}(E_S, \underline{\mathrm{Hom}}(F_S, T)) \\ &\simeq \mathrm{Hom}_{\mathbf{Set}}(E, \mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T))). \end{aligned}$$

But

$$\mathrm{Hom}(S, \underline{\mathrm{Hom}}(F_S, T)) \simeq \mathrm{Hom}(F_S, T),$$

which proves $(\times)$.

Suppose that, with $\emptyset$ denoting an initial object of $\mathcal{C}$, the following diagram is cartesian:

$$\begin{array}{ccc} \emptyset & \xrightarrow{\quad} & S \\ \downarrow & & \downarrow \\ S & \xrightarrow{\quad} & S \amalg S \end{array} \quad (*)$$

¹⁸Editor's note: the former terminology "preschemes/schemes" has been replaced by the current terminology "schemes/separated schemes."

¹⁹Editor's note: in the next three paragraphs, the order of the sentences has been changed and a few details have been added concerning the role of condition $(*)$ below.

This is the case for the category of schemes.²⁰ Then the functor $E \mapsto E_S$ commutes with finite inverse limits.

Indeed, in view of $(\times)$, it is enough to show that $E \mapsto E_S$ commutes with fiber products. Let $u : E \rightarrow G$ and $v : F \rightarrow G$ be two maps of sets. Since direct sums in $\mathcal{C}$ commute with base change, one has

$$F_S \times_{G_S} E_S \simeq \coprod_{f \in F} (S_f \times_{G_S} E_S) \simeq \coprod_{\substack{f \in F \\ x \in E}} (S_f \times_{G_S} S_x). \quad (1)$$

If $v(f) \neq u(x)$, there is a morphism in $\mathcal{C}/_S$

$$S_f \times_{G_S} S_x \longrightarrow S_f \times_{S_{v(f)} \amalg S_{u(x)}} S_x.$$

By condition $(*)$, the term on the right is $\emptyset$. Hence

$$S_f \times_{G_S} S_x \simeq \emptyset \quad \text{if } v(f) \neq u(x). \quad (2)$$

On the other hand, if $v(f) = u(x)$, there is a morphism in $\mathcal{C}/_S$

$$S \longrightarrow S_f \times_{G_S} S_x.$$

Since $S \xrightarrow{\text{id}} S$ is the final object of $\mathcal{C}/_S$, it follows that

$$S_f \times_{G_S} S_x \simeq S \quad \text{if } v(f) = u(x). \quad (3)$$

Combining (1), (2), and (3), we obtain an isomorphism functorial in S :

$$F_S \times_{G_S} E_S \simeq \coprod_{F \times_G E} S = (F \times_G E)_S.$$

An object of the form E_S is called a *constant object*. There is a morphism, functorial in E ,

$$E \longrightarrow \Gamma(E_S/S),$$

which associates with each $i \in E$ the section of E_S over S defined by the isomorphism from S onto S_i . Suppose condition $(*)$ is satisfied for every object S of $\mathcal{C}$. Then $E \rightarrow \Gamma(E_S/S)$ is a monomorphism whenever $S \not\simeq \emptyset$.

If $\mathcal{C}$ is the category of schemes, then $\Gamma(E_S/S)$ is identified with the locally constant maps from the topological space S to the set E , the preceding map associating with each element of E the corresponding constant map. It follows that E_S may equivalently be defined as representing the functor that assigns to every S' over S the set of locally constant functions from the topological space S' to E .²¹

2. Algebraic structures

Given a species of algebraic structure in the category of sets, we shall extend it to the category $\mathcal{C}$. We first treat the example of groups.

²⁰Editor's note: Condition $(*)$ is not satisfied if $\mathcal{C}$ is the category whose arrows are $A \rightarrow B$, id_A , and id_B . In that case $B \amalg B = B$ and $B \times_B B = B \not\simeq A$.

²¹Editor's note: the diagonal morphism $E_S \rightarrow E_S \times_S E_S = (E \times E)_S$ is also a closed immersion; that is, E_S is separated over S .

2.1. Group structures

We retain the notation of the preceding paragraph.

Definition 2.1.1. Let $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$. A $\widehat{\mathcal{C}}$ -group structure on $\mathbf{G}$ is the datum, for every $S \in \text{Ob}(\mathcal{C})$, of a group structure on the set $\mathbf{G}(S)$, such that for every arrow $f : S' \rightarrow S''$ of $\mathcal{C}$, the map

$$\mathbf{G}(f) : \mathbf{G}(S'') \longrightarrow \mathbf{G}(S')$$

is a group homomorphism. If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, a *morphism of $\widehat{\mathcal{C}}$ -groups* from $\mathbf{G}$ to $\mathbf{H}$ is a morphism $u \in \text{Hom}(\mathbf{G}, \mathbf{H})$ such that $u(S) : \mathbf{G}(S) \rightarrow \mathbf{H}(S)$ is a group homomorphism for every $S \in \text{Ob}(\mathcal{C})$.

We denote by $\text{Hom}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ the set of these morphisms and by $(\widehat{\mathcal{C}}\text{-Gr})$ their category.

Examples. For $\mathbf{E} \in \text{Ob}(\widehat{\mathcal{C}})$, the object $\text{Aut}(\mathbf{E})$ has an evident $\widehat{\mathcal{C}}$ -group structure. The final object $\mathbf{e}$ has a unique $\widehat{\mathcal{C}}$ -group structure, making it a final object of $(\widehat{\mathcal{C}}\text{-Gr})$.

For every $S \in \text{Ob}(\mathcal{C})$, let $e_{\mathbf{G}}(S)$ be the identity element of $\mathbf{G}(S)$. The family of elements $e_{\mathbf{G}}(S)$ defines

$$e_{\mathbf{G}} \in \Gamma(\mathbf{G}) = \text{Hom}(\mathbf{e}, \mathbf{G}).$$

It is a morphism of $\widehat{\mathcal{C}}$ -groups $\mathbf{e} \rightarrow \mathbf{G}$, called the *unit section* of $\mathbf{G}$.

To give a $\widehat{\mathcal{C}}$ -group structure on $\mathbf{G}$ is equivalently to give a composition law on $\mathbf{G}$, that is, a $\widehat{\mathcal{C}}$ -morphism

$$\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}$$

such that $\pi_{\mathbf{G}}(S)$ equips $\mathbf{G}(S)$ with a group structure for every $S \in \text{Ob}(\mathcal{C})$.

Similarly, $f : \mathbf{G} \rightarrow \mathbf{H}$ is a morphism of $\widehat{\mathcal{C}}$ -groups if and only if the following diagram is commutative:

$$\begin{array}{ccc} \mathbf{G} \times \mathbf{G} & \xrightarrow{\pi_{\mathbf{G}}} & \mathbf{G} \\ (f, f) \downarrow & & \downarrow f \\ \mathbf{H} \times \mathbf{H} & \xrightarrow{\pi_{\mathbf{H}}} & \mathbf{H} \end{array}$$

A subobject $\mathbf{H}$ of $\mathbf{G}$ such that $\mathbf{H}(S)$ is a subgroup of $\mathbf{G}(S)$ for every S has the evident $\widehat{\mathcal{C}}$ -group structure induced by that of $\mathbf{G}$. It is the unique structure for which the monomorphism $\mathbf{H} \rightarrow \mathbf{G}$ is a morphism of $\widehat{\mathcal{C}}$ -groups. With this structure, $\mathbf{H}$ is called a *sub- $\widehat{\mathcal{C}}$ -group* of $\mathbf{G}$.

If $\mathbf{G}$ and $\mathbf{H}$ are two $\widehat{\mathcal{C}}$ -groups, then $\mathbf{G} \times \mathbf{H}$ has the evident product structure: for every S , the set $\mathbf{G}(S) \times \mathbf{H}(S)$ is given the product of the prescribed group structures. The resulting object is called the *product $\widehat{\mathcal{C}}$ -group* of $\mathbf{G}$ and $\mathbf{H}$; it is their product in the category of $\widehat{\mathcal{C}}$ -groups.

If $\mathbf{G}$ is a $\widehat{\mathcal{C}}$ -group, then $\mathbf{G}_S$ is a $\widehat{\mathcal{C}}_S$ -group for every $S \in \text{Ob}(\mathcal{C})$. For two $\widehat{\mathcal{C}}$ -groups $\mathbf{G}$ and $\mathbf{H}$, define the object $\underline{\text{Hom}}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})$ of $\widehat{\mathcal{C}}$ by

$$\underline{\text{Hom}}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H})(S) = \text{Hom}_{\widehat{\mathcal{C}}_S\text{-Gr}}(\mathbf{G}_S, \mathbf{H}_S).$$

Notice that $\underline{\text{Hom}}_{\widehat{\mathcal{C}}\text{-Gr}}$ is not in general a $\widehat{\mathcal{C}}$ -group, still less the internal $\underline{\text{Hom}}$ object in the category $(\widehat{\mathcal{C}}\text{-Gr})$.

One similarly defines

$$\underline{\text{Isom}}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}, \mathbf{H}), \quad \underline{\text{End}}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}), \quad \underline{\text{Aut}}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}).$$

Definition 2.1.2. Let $G \in \text{Ob}(\mathcal{C})$. A $\mathcal{C}$ -group structure on G is a $\widehat{\mathcal{C}}$ -group structure on $\mathbf{h}_G \in \text{Ob}(\widehat{\mathcal{C}})$. A morphism from the $\mathcal{C}$ -group G to the $\mathcal{C}$ -group H is an element

$$u \in \text{Hom}(G, H) \simeq \text{Hom}(\mathbf{h}_G, \mathbf{h}_H)$$

that defines a morphism of $\widehat{\mathcal{C}}$ -groups from $\mathbf{h}_G$ to $\mathbf{h}_H$.

We denote the category of $\mathcal{C}$ -groups by $(\mathcal{C}\text{-Gr})$. There is a cartesian square in **Cat**:

$$\begin{array}{ccc} (\mathcal{C}\text{-Gr}) & \longrightarrow & (\widehat{\mathcal{C}}\text{-Gr}) \\ \downarrow & & \downarrow \\ \mathcal{C} & \xrightarrow{\mathbf{h}} & \widehat{\mathcal{C}} \end{array} .$$

All the preceding definitions and constructions therefore carry over at once to $(\mathcal{C}\text{-Gr})$ whenever the functors they use (products, internal Hom objects, and so on) are representable. They also apply to the categories $\mathcal{C}_{/S}$. In that case we write $\underline{\text{Hom}}_{S\text{-Gr}}$ for $\underline{\text{Hom}}_{\mathcal{C}_{/S}\text{-Gr}}$, and similarly for the other objects.

2.2.

More generally, suppose (T) is a species of structure on n base sets defined by finite inverse limits—for example, by commutative diagrams constructed from cartesian products, as in the structures of a monoid, a group, a set with operators, a module over a ring, or a Lie algebra over a ring. The preceding construction defines the notion of a “structure of species (T) ” on n objects $\mathbf{F}_1, \dots, \mathbf{F}_n$ of $\widehat{\mathcal{C}}$. Such a structure consists, for every $S \in \text{Ob}(\mathcal{C})$, of a structure of species (T) on the sets $\mathbf{F}_1(S), \dots, \mathbf{F}_n(S)$, in such a way that for every arrow $S' \rightarrow S''$ of $\mathcal{C}$, the family of maps

$$(\mathbf{F}_i(S''))_i \longrightarrow (\mathbf{F}_i(S'))_i$$

is a polyhomomorphism for the species (T) .

Morphisms of structures of species (T) are defined similarly, giving a category

$$(\widehat{\mathcal{C}} \times \dots \times \widehat{\mathcal{C}})^{(T)}.$$

The fully faithful functor $\mathbf{h} \times \dots \times \mathbf{h}$ then defines, by inverse image, the category

$$(\mathcal{C} \times \dots \times \mathcal{C})^{(T)}.$$

Since it commutes with inverse limits, all the functorial properties, notions, and notation introduced in $\widehat{\mathcal{C}}$ carry over to this category.

Now suppose fiber products exist in $\mathcal{C}$, and let (T) be a species of algebraic structure specified by certain morphisms between cartesian products, subject to axioms consisting of commutative diagrams formed with those morphisms. A structure of species (T) on a family of objects of $\mathcal{C}$ is then defined by corresponding morphisms between cartesian products satisfying the corresponding commutativity conditions. Consequently, if $\mathcal{C}$ and $\mathcal{C}'$ are categories with products and $f : \mathcal{C} \rightarrow \mathcal{C}'$ is a functor commuting with products, then a structure of species (T) on a family (F_i) of objects of $\mathcal{C}$ induces one on the family $(f(F_i))$ in $\mathcal{C}'$. Thus every $\mathcal{C}$ -group becomes a $\mathcal{C}'$ -group, every pair consisting of a $\mathcal{C}$ -ring and a module over it becomes the analogous pair in $\mathcal{C}'$, and so on.

In particular, let $\mathcal{C}$ satisfy the conditions of 1.8.²² The functor $E \mapsto E_S$ defined there commutes with finite inverse limits. It therefore carries groups to S -groups (that is, $\mathcal{C}/S$ -groups), rings to S -rings, and so forth.

Remark. It is worth observing that the preceding construction, when applied to $\widehat{\mathcal{C}}$, gives back precisely the notions already defined there. In other words, to give an object of $\widehat{\mathcal{C}}$ a structure of species (T) when it is viewed as a functor on $\mathcal{C}$ is the same as to give the representable functor on $\widehat{\mathcal{C}}$ defined by that object a structure of species (T) .²³

We shall treat two further special cases of this construction: structures with operator groups and modules.

2.3. Structures with operator groups

Definition 2.3.1. Let $\mathbf{E} \in \text{Ob}(\widehat{\mathcal{C}})$ and $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}}\text{-Gr})$. A structure on $\mathbf{E}$ of an object with $\widehat{\mathcal{C}}$ -operator group $\mathbf{G}$ (or of a $\mathbf{G}$ -object) consists in giving $\mathbf{E}(S)$, for every $S \in \text{Ob}(\mathcal{C})$, the structure of a set with operator group $\mathbf{G}(S)$, in such a way that, for every arrow $S' \rightarrow S''$ in $\mathcal{C}$, the map of sets $\mathbf{E}(S'') \rightarrow \mathbf{E}(S')$ is compatible with the homomorphism of operator groups $\mathbf{G}(S'') \rightarrow \mathbf{G}(S')$.

As usual, this is equivalent to giving a morphism

$$\mu : \mathbf{G} \times \mathbf{E} \longrightarrow \mathbf{E}$$

which, for every S , equips $\mathbf{E}(S)$ with the structure of a set on which $\mathbf{G}(S)$ acts. But

$$\text{Hom}(\mathbf{G} \times \mathbf{E}, \mathbf{E}) \simeq \text{Hom}(\mathbf{G}, \underline{\text{End}}(\mathbf{E})),$$

so μ defines a morphism $\mathbf{G} \rightarrow \underline{\text{End}}(\mathbf{E})$. It is immediate that this morphism maps $\mathbf{G}$ into $\underline{\text{Aut}}(\mathbf{E})$ and is a morphism of $\widehat{\mathcal{C}}$ -groups. Consequently, giving $\mathbf{E}$ the structure of an object with $\widehat{\mathcal{C}}$ -operator group $\mathbf{G}$ is equivalent to giving a morphism of $\widehat{\mathcal{C}}$ -groups

$$\rho : \mathbf{G} \longrightarrow \underline{\text{Aut}}(\mathbf{E}).$$

In particular, every element $g \in \mathbf{G}(S)$ defines an automorphism $\rho(g)$ of the functor $\mathbf{E}_S$, that is, an automorphism of $\mathbf{E} \times \mathbf{h}_S$ commuting with the projection $\mathbf{E} \times \mathbf{h}_S \rightarrow \mathbf{h}_S$, and hence, in particular, an automorphism of the set $\mathbf{E}(S')$ for every $S' \rightarrow S$.

Definition 2.3.2. We denote by $\mathbf{E}^{\mathbf{G}}$ the subobject of $\mathbf{E}$ defined as follows:

$$\mathbf{E}^{\mathbf{G}}(S) = \{x \in \mathbf{E}(S) \mid x_{S'} \text{ is invariant under } \mathbf{G}(S') \text{ for every } S' \rightarrow S\},$$

where $x_{S'}$ denotes the image of x under $\mathbf{E}(S) \rightarrow \mathbf{E}(S')$.

Thus $\mathbf{E}^{\mathbf{G}}$, the “subobject of invariants of $\mathbf{G}$,” is the largest subobject of $\mathbf{E}$ on which $\mathbf{G}$ acts trivially.

²²Editor’s note: including condition (*).

²³Editor’s note: for group structures, for example, let $\mathbf{G} \in \text{Ob}(\widehat{\mathcal{C}})$. If the functor

$$\widehat{\mathcal{C}} \longrightarrow \mathbf{Set}, \quad \mathbf{F} \longmapsto \text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F}, \mathbf{G})$$

has a group structure, then so does its restriction to $\mathcal{C}$, namely $X \mapsto \mathbf{G}(X)$. Conversely, if $\mathbf{G}$ is a $\widehat{\mathcal{C}}$ -group, its multiplication morphism $\pi_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$ induces, for every $\mathbf{F} \in \text{Ob}(\widehat{\mathcal{C}})$, a group structure on $\text{Hom}_{\widehat{\mathcal{C}}}(\mathbf{F}, \mathbf{G})$, functorially in $\mathbf{F}$.

Definition 2.3.3. Let $\mathbf{F}$ be a subobject of $\mathbf{E}$. We denote by $\underline{\text{Norm}}_{\mathbf{G}}\mathbf{F}$ and $\underline{\text{Centr}}_{\mathbf{G}}\mathbf{F}$ the sub- $\widehat{\mathcal{C}}$ -groups of $\mathbf{G}$ defined by

$$\begin{aligned} (\underline{\text{Norm}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}_S = \mathbf{F}_S\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)\mathbf{F}(S') = \mathbf{F}(S') \text{ for every } S' \rightarrow S\}, \\ (\underline{\text{Centr}}_{\mathbf{G}}\mathbf{F})(S) &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}_S} = \text{id}\} \\ &= \{g \in \mathbf{G}(S) \mid \rho(g)|_{\mathbf{F}(S')} = \text{id} \text{ for every } S' \rightarrow S\}. \end{aligned}$$

Here the vertical bar after $\rho(g)$ denotes restriction.²⁴

Scholium 2.3.3.1.²⁵ In particular, let $x \in \Gamma(\mathbf{E})$, that is (cf. 1.2), a collection of elements $x_S \in \mathbf{E}(S)$, $S \in \text{Ob}(\mathcal{C})$, such that for every arrow $f : S' \rightarrow S$ one has $\mathbf{E}(f)(x_S) = x_{S'}$. If $\mathcal{C}$ has a final object S_0 , then $\Gamma(\mathbf{E}) = \mathbf{E}(S_0)$. The element x then defines a subfunctor of $\mathbf{E}$, denoted by $\mathbf{x}$, and

$$\underline{\text{Norm}}_{\mathbf{G}}\mathbf{x} = \underline{\text{Centr}}_{\mathbf{G}}\mathbf{x}.$$

We denote this functor by $\underline{\text{Stab}}_{\mathbf{G}}(x)$ and call it the *stabilizer* of x ; thus, for every $S \in \text{Ob}(\mathcal{C})$,

$$\underline{\text{Stab}}_{\mathbf{G}}(x)(S) = \{g \in \mathbf{G}(S) \mid \rho(g)x_S = x_S\}.$$

Suppose that fiber products exist in $\mathcal{C}$. If $\mathbf{G} = \mathbf{h}_G$ (respectively, $\mathbf{E} = \mathbf{h}_E$), where G is a $\mathcal{C}$ -group (respectively, $E \in \text{Ob}(\mathcal{C})$), and if $\mathcal{C}$ has a final object S_0 , so that x is a morphism $S_0 \rightarrow E$, then $\underline{\text{Stab}}_{\mathbf{G}}(x)$ is represented by the fiber product $G \times_E S_0$, where the morphism $G \rightarrow E$ is the composite of

$$\text{id}_G \times x : G = G \times S_0 \longrightarrow G \times E$$

and

$$\mu : G \times E \longrightarrow E.$$

Remark 2.3.3.2.²⁵ The formation of $\mathbf{E}^{\mathbf{G}}$, $\underline{\text{Norm}}_{\mathbf{G}}\mathbf{F}$, and $\underline{\text{Centr}}_{\mathbf{G}}\mathbf{F}$ commutes with base change; that is, for every $S \in \text{Ob}(\mathcal{C})$,

$$(\mathbf{E}^{\mathbf{G}})_S = (\mathbf{E}_S)^{\mathbf{G}_S}, \quad (\underline{\text{Norm}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\text{Norm}}_{\mathbf{G}_S}\mathbf{F}_S, \quad (\underline{\text{Centr}}_{\mathbf{G}}\mathbf{F})_S \simeq \underline{\text{Centr}}_{\mathbf{G}_S}\mathbf{F}_S.$$

Definition 2.3.4. If G is a $\mathcal{C}$ -group and $\mathbf{E}$ is an object of $\widehat{\mathcal{C}}$ (respectively, E is an object of $\mathcal{C}$), a G -object structure on $\mathbf{E}$ (respectively, on E) is an $\mathbf{h}_G$ -object structure on $\mathbf{E}$ (respectively, on $\mathbf{h}_E$).

In view of this definition, all the notions and notation defined above carry over to $\mathcal{C}$ whenever they involve only representable functors. For example, if $\underline{\text{Norm}}_{\mathbf{h}_G}(\mathbf{h}_F)$ is representable, then there is a unique subobject of G representing it; it is then a sub- $\mathcal{C}$ -group of G , and is denoted by $\text{Norm}_G(F)$, and so forth.

Definition 2.3.5.

- (a) We say that the $\widehat{\mathcal{C}}$ -group $\mathbf{G}$ acts on the $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ if $\mathbf{H}$ is equipped with a $\mathbf{G}$ -object structure such that, for every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{H}(S)$ defined by g is a group automorphism.

Equivalently, for every $g \in \mathbf{G}(S)$, the automorphism $\rho(g)$ of $\mathbf{H}_S$ is an automorphism of $\widehat{\mathcal{C}}_S$ -groups; or equivalently, the morphism of $\widehat{\mathcal{C}}$ -groups $\mathbf{G} \rightarrow \underline{\text{Aut}}(\mathbf{H})$ maps $\mathbf{G}$ into

$$\underline{\text{Aut}}_{\widehat{\mathcal{C}}_{\mathbf{Gr}}}(\mathbf{H}).$$

²⁴Editor's note: the original has been corrected by replacing the inclusion $\rho(g)\mathbf{F}_S \subset \mathbf{F}_S$ with an equality, so as to ensure that $\underline{\text{Norm}}_{\mathbf{G}}(\mathbf{F})$ is indeed a group. See VI_B, 6.4 for conditions under which the “transporter” coincides with the “strict transporter.”

²⁵Editor's note: Scholium 2.3.3.1 and Remark 2.3.3.2 have been added.

- (b) In the situation above, there is a unique $\widehat{\mathcal{C}}$ -group structure on $\mathbf{H} \times \mathbf{G}$ such that, for every S , $(\mathbf{H} \times \mathbf{G})(S)$ is the semidirect product of the groups $\mathbf{H}(S)$ and $\mathbf{G}(S)$ for the given action of $\mathbf{G}(S)$ on $\mathbf{H}(S)$. We denote this $\widehat{\mathcal{C}}$ -group by $\mathbf{H} \cdot \mathbf{G}$ and call it the *semidirect product of $\mathbf{H}$ by $\mathbf{G}$* . Thus, by definition,

$$(\mathbf{H} \cdot \mathbf{G})(S) = \mathbf{H}(S) \cdot \mathbf{G}(S).$$

Let $\mathbf{G}$ be a $\widehat{\mathcal{C}}$ -group. For every arrow $S' \rightarrow S$ in $\mathcal{C}$ and every $g \in \mathbf{G}(S)$, let $\text{Int}(g)$ be the automorphism of $\mathbf{G}(S')$ defined by

$$\text{Int}(g)h = ghg^{-1}.$$

This definition extends to a morphism of $\widehat{\mathcal{C}}$ -groups

$$\text{Int} : \mathbf{G} \longrightarrow \underline{\text{Aut}}_{\widehat{\mathcal{C}}\text{-Gr}}(\mathbf{G}) \subset \underline{\text{Aut}}(\mathbf{G}).$$

Definition 2.3.3 therefore applies, and for every subobject $\mathbf{E}$ of $\mathbf{G}$ one has the sub- $\widehat{\mathcal{C}}$ -groups of $\mathbf{G}$

$$\underline{\text{Norm}}_{\mathbf{G}}(\mathbf{E}) \quad \text{and} \quad \underline{\text{Centr}}_{\mathbf{G}}(\mathbf{E}).$$

Definition 2.3.6. The *center* of $\mathbf{G}$, denoted $\underline{\text{Centr}}(\mathbf{G})$, is the sub- $\widehat{\mathcal{C}}$ -group $\underline{\text{Centr}}_{\mathbf{G}}(\mathbf{G})$ of $\mathbf{G}$. We say that $\mathbf{G}$ is commutative if $\underline{\text{Centr}}(\mathbf{G}) = \mathbf{G}$, or, equivalently, if $\mathbf{G}(S)$ is commutative for every S .

We say that the sub- $\widehat{\mathcal{C}}$ -group $\mathbf{H}$ of $\mathbf{G}$ is invariant in $\mathbf{G}$ if $\underline{\text{Norm}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is invariant in $\mathbf{G}(S)$ for every S .²⁶

Definition 2.3.6.1.²⁷ Let $f : \mathbf{G} \rightarrow \mathbf{G}'$ be a morphism of $\widehat{\mathcal{C}}$ -groups. The *kernel* of f , denoted $\text{Ker } f$, is the sub- $\widehat{\mathcal{C}}$ -group of $\mathbf{G}$ defined by

$$(\text{Ker } f)(S) = \{x \in \mathbf{G}(S) \mid f(S)(x) = 1\} = \text{Ker } f(S)$$

for every $S \in \text{Ob}(\mathcal{C})$; it is an invariant sub- $\widehat{\mathcal{C}}$ -group.

Notice that if $\mathbf{G} = \mathbf{h}_G$ and $\mathbf{G}' = \mathbf{h}_{G'}$, if $\mathcal{C}$ has a final object S_0 , and if fiber products exist in $\mathcal{C}$, then $\text{Ker}(f)$ is represented by $S_0 \times_{G'} G$.

Definition 2.3.6.2.²⁷ Let $\mathbf{E} \in \widehat{\mathcal{C}}$, and let $\mathbf{G}$ be a $\widehat{\mathcal{C}}$ -group acting on $\mathbf{E}$. We say that the action of $\mathbf{G}$ on $\mathbf{E}$ is *faithful* if the kernel of the morphism $\mathbf{G} \rightarrow \underline{\text{Aut}}(\mathbf{E})$ is trivial; that is, if, for every $S \in \text{Ob}(\mathcal{C})$ and $g \in \mathbf{G}(S)$, the condition

$$g_{S'} \cdot x = x \quad \text{for every } S' \rightarrow S \text{ and every } x \in \mathbf{E}(S')$$

implies $g = 1$.

Many definitions and propositions from elementary group theory carry over readily. We record only the following one, which will be useful to us.

Proposition 2.3.7. Let $f : \mathbf{W} \rightarrow \mathbf{G}$ be a morphism of $\widehat{\mathcal{C}}$ -groups. Put $\mathbf{H}(S) = \text{Ker } f(S)$. Let $u : \mathbf{G} \rightarrow \mathbf{W}$ be a morphism of $\widehat{\mathcal{C}}$ -groups that is a section of f (and is therefore necessarily a monomorphism). Then $\mathbf{W}$ is identified with the semidirect product of $\mathbf{H}$ by $\mathbf{G}$ for the action of $\mathbf{G}$ on $\mathbf{H}$ defined by

$$(g, h) \longmapsto \text{Int}(u(g))h$$

²⁶Editor's note: in addition, $\mathbf{H}$ is said to be *central* in $\mathbf{G}$ if $\underline{\text{Centr}}_{\mathbf{G}}(\mathbf{H}) = \mathbf{G}$, or, equivalently, if $\mathbf{H}(S)$ is central in $\mathbf{G}(S)$ for every S .

²⁷Editor's note: Definitions 2.3.6.1 and 2.3.6.2 have been added.

for $g \in \mathbf{G}(S)$, $h \in \mathbf{H}(S)$, and $S \in \text{Ob}(\mathcal{C})$.

All these definitions and propositions carry over, as usual, to $\mathcal{C}$. In particular, the semidirect product of two $\mathcal{C}$ -groups H and G , with G acting on H , is defined whenever the Cartesian product $H \times G$ exists, and the analog of Proposition 2.3.7 takes the following form.

Proposition 2.3.8. *Let*

$$H \xrightarrow{i} W \xrightarrow{f} G$$

be a sequence of morphisms of $\mathcal{C}$ -groups such that, for every $S \in \text{Ob}(\mathcal{C})$, the pair $(H(S), i(S))$ is a kernel of $f(S) : W(S) \rightarrow G(S)$. Let $u : G \rightarrow W$ be a morphism of $\mathcal{C}$ -groups that is a section of f . Then W is identified with the semidirect product of H by G for the action of G on H such that, if $S \in \text{Ob}(\mathcal{C})$, $g \in G(S)$, and $h \in H(S)$, then

$$\text{Int}(u(g))i(h) = i(g^h).$$

3. The category of $\mathbf{O}$ -modules, the category of $\mathbf{G}$ - $\mathbf{O}$ -modules

Definition 3.1. Let $\mathbf{O}$ and $\mathbf{F}$ be two objects of $\widehat{\mathcal{C}}$. We say that $\mathbf{F}$ is a $\widehat{\mathcal{C}}$ -module over the $\widehat{\mathcal{C}}$ -ring $\mathbf{O}$, or, for short, an $\mathbf{O}$ -module, if, for every $S \in \text{Ob}(\mathcal{C})$, the set $\mathbf{O}(S)$ has been endowed with a ring structure and $\mathbf{F}(S)$ with a module structure over that ring, in such a way that for every arrow $S' \rightarrow S''$ in $\mathcal{C}$, the map $\mathbf{O}(S'') \rightarrow \mathbf{O}(S')$ is a ring homomorphism and $\mathbf{F}(S'') \rightarrow \mathbf{F}(S')$ is a homomorphism of abelian groups compatible with that ring homomorphism.

Once $\mathbf{O}$ is fixed, one defines in the usual way a morphism between two $\mathbf{O}$ -modules $\mathbf{F}$ and $\mathbf{F}'$, hence the abelian group $\text{Hom}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and the category of $\mathbf{O}$ -modules, denoted $(\mathbf{O}\text{-Mod.})$.

Lemma 3.1.1.²⁸ *The category $(\mathbf{O}\text{-Mod.})$ has the structure of an abelian category, defined objectwise. Moreover, $(\mathbf{O}\text{-Mod.})$ satisfies axiom (AB 5) (see [Gr57, 1.5]); that is, arbitrary direct sums exist, and if $\mathbf{M}$ is an $\mathbf{O}$ -module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i \in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then*

$$\bigcup_{i \in I} (\mathbf{F}_i \cap \mathbf{N}) = \left(\bigcup_{i \in I} \mathbf{F}_i \right) \cap \mathbf{N}.$$

Indeed, let $f : \mathbf{F} \rightarrow \mathbf{F}'$ be a morphism of $\mathbf{O}$ -modules. Define the $\mathbf{O}$ -modules $\text{Ker } f$ (respectively, $\text{Im } f$ and $\text{Coker } f$) by setting, for every $S \in \text{Ob}(\mathcal{C})$,

$$(\text{Ker } f)(S) = \text{Ker } f(S)$$

(respectively, the analogous formulas). Then $\text{Ker } f$ (respectively, $\text{Coker } f$) is a kernel (respectively, a cokernel) of f , and there is an isomorphism of $\mathbf{O}$ -modules

$$\mathbf{F} / \text{Ker } f \xrightarrow{\sim} \text{Im } f.$$

This proves that $(\mathbf{O}\text{-Mod.})$ is an abelian category.

Arbitrary direct sums exist and are defined objectwise. Finally, if $\mathbf{M}$ is an $\mathbf{O}$ -module, $\mathbf{N}$ a submodule, and $(\mathbf{F}_i)_{i \in I}$ an increasing filtered family of submodules of $\mathbf{M}$, then the inclusion

$$\bigcup_{i \in I} (\mathbf{F}_i \cap \mathbf{N}) \subset \left(\bigcup_{i \in I} \mathbf{F}_i \right) \cap \mathbf{N}$$

is an equality. Indeed, if $S \in \text{Ob}(\mathcal{C})$ and

$$x \in \mathbf{N}(S) \cap \bigcup_i \mathbf{F}_i(S),$$

there is an $i \in I$ such that $x \in \mathbf{N}(S) \cap \mathbf{F}_i(S)$.

Proposition 3.1.2.²⁸ *Suppose that the category $\mathcal{C}$ is small, that is, that $\text{Ob}(\mathcal{C})$ is a set. Then $(\mathbf{O}\text{-Mod.})$ has a generator, and consequently has enough injective objects.*

Proof. For every object S of $\mathcal{C}$, define the $\mathbf{O}$ -module $\mathbf{O}_S$ as follows. For every object S' , $\mathbf{O}_S(S')$ is a direct sum of copies of $\mathbf{O}(S')$ indexed by $\text{Hom}(S', S)$, that is,

$$\mathbf{O}_S(S') = \bigoplus_{g:S' \rightarrow S} \mathbf{O}(S')g.$$

For every morphism $f : S'' \rightarrow S'$, writing f^* for the ring homomorphism $\mathbf{O}(S') \rightarrow \mathbf{O}(S'')$, the morphism

$$\mathbf{O}_S(f) : \mathbf{O}_S(S') \rightarrow \mathbf{O}_S(S'')$$

sends each element ag of $\mathbf{O}_S(S')$, where $a \in \mathbf{O}(S')$ and $g : S' \rightarrow S$, to the element $f^*(a)(g \circ f)$ of $\mathbf{O}_S(S'')$. It is readily checked that this defines a functor in $\mathbf{O}$ -modules.

Since $\text{Ob}(\mathcal{C})$ is a set by hypothesis, we may form the direct sum

$$\mathbf{G} = \bigoplus_{S \in \text{Ob}(\mathcal{C})} \mathbf{O}_S.$$

Lemma 3.1.2.1. *Let $\mathbf{F}$ be an $\mathbf{O}$ -module. For every $S \in \text{Ob}(\mathcal{C})$, every element $x \in \mathbf{F}(S)$ defines a unique morphism of $\mathbf{O}$ -modules*

$$\mathbf{x} : \mathbf{O}_S \rightarrow \mathbf{F}$$

such that $\mathbf{x}(1 \text{id}_S) = x$.

Proof. Let $\phi : \mathbf{O}_S \rightarrow \mathbf{F}$ be a morphism of $\mathbf{O}$ -modules such that $\phi(1 \text{id}_S) = x$. Let $g : S' \rightarrow S$ and $a \in \mathbf{O}(S')$. The $\mathbf{O}(S')$ -linearity and the commutativity of the diagram

$$\begin{array}{ccc} \mathbf{O}_S(S) & \xrightarrow{\psi} & \mathbf{F}(S) \\ \mathbf{O}_S(g) \downarrow & & \downarrow \mathbf{F}(g) \\ \mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \end{array}$$

imply that

$$\phi(ag) = a\phi(1g) = a\mathbf{F}(g)(\phi(1 \text{id}_S)) = a\mathbf{F}(g)(x).$$

Thus ϕ , if it exists, is determined by the preceding equality. Conversely, if ϕ is defined in this way, then for every $f : S'' \rightarrow S'$ one has

$$\begin{aligned} \phi \circ \mathbf{O}_S(f)(ag) &= \phi(f^*(a)(g \circ f)) \\ &= f^*(a)\mathbf{F}(g \circ f)(x) \\ &= \mathbf{F}(f)(a\mathbf{F}(g)(x)) \\ &= \mathbf{F}(f) \circ \phi(ag), \end{aligned}$$

that is, the diagram

²⁸Editor's note: Statements 3.1.1 and 3.1.2 have been added to show that the category $(\mathbf{O}\text{-Mod.})$ is abelian and satisfies axiom (AB 5), and moreover has enough injective objects when the category $\mathcal{C}$ is small. An error in 3.1.2, pointed out in 2016 by Linyuan Liu, is corrected here.

$$\begin{array}{ccc}
\mathbf{O}_S(S') & \xrightarrow{\phi} & \mathbf{F}(S') \\
\mathbf{O}_S(f) \downarrow & & \downarrow \mathbf{F}(f) \\
\mathbf{O}_S(S'') & \xrightarrow{\phi} & \mathbf{F}(S'')
\end{array}$$

is commutative. This proves the lemma.

Now let $\mathbf{F}$ be an $\mathbf{O}$ -module. For each $S \in \text{Ob}(\mathcal{C})$, let $F_0(S)$ be a set of generators of the $\mathbf{O}(S)$ -module $\mathbf{F}(S)$, and let $\mathbf{L}_S$ be the direct sum, indexed by $F_0(S)$, of copies of $\mathbf{O}_S$. By the lemma, we obtain a morphism of $\mathbf{O}$ -modules

$$\mathbf{L}_S \longrightarrow \mathbf{F}$$

such that the morphism $\mathbf{L}_S(S) \rightarrow \mathbf{F}(S)$ is surjective, and hence is an epimorphism in the category of $\mathbf{O}(S)$ -modules.

Set

$$I = \bigsqcup_{S \in \text{Ob}(\mathcal{C})} F_0(S).$$

We then obtain a morphism of $\mathbf{O}$ -modules

$$\mathbf{G}^{\oplus I} \longrightarrow \bigoplus_{S \in \text{Ob}(\mathcal{C})} \mathbf{L}_S \longrightarrow \mathbf{F}$$

which is an epimorphism objectwise, and hence an epimorphism of $(\mathbf{O}\text{-Mod.})$. This shows that $\mathbf{G}$ is a generator of $(\mathbf{O}\text{-Mod.})$ (see [Gr57, 1.9.1]). Since $(\mathbf{O}\text{-Mod.})$ satisfies (AB 5), it follows from [Gr57, 1.10.1] that $(\mathbf{O}\text{-Mod.})$ has enough injective objects.

Remark 3.1.3. If $\mathbf{O}_0$ is the $\widehat{\mathcal{C}}$ -ring defined by

$$\mathbf{O}_0(S) = \mathbb{Z}$$

(which must not be confused with the functor associated with the constant object $\mathbb{Z}$), then the category of $\mathbf{O}_0$ -modules is isomorphic to the category of commutative $\widehat{\mathcal{C}}$ -groups.

Definition 3.1.4. Notice that if $\mathbf{F}$ is an $\mathbf{O}$ -module, then for every $S \in \text{Ob}(\mathcal{C})$, $\mathbf{F}_S$ is an $\mathbf{O}_S$ -module. We may therefore define an abelian $\widehat{\mathcal{C}}$ -group $\underline{\text{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')$ by

$$\underline{\text{Hom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}')(S) = \text{Hom}_{\mathbf{O}_S}(\mathbf{F}_S, \mathbf{F}'_S).$$

In the same way one defines the objects

$$\underline{\text{Isom}}_{\mathbf{O}}(\mathbf{F}, \mathbf{F}'), \quad \underline{\text{End}}_{\mathbf{O}}(\mathbf{F}), \quad \text{and} \quad \underline{\text{Aut}}_{\mathbf{O}}(\mathbf{F}),$$

the last of which is a $\widehat{\mathcal{C}}$ -group under the structure induced by composition of automorphisms.

Definition 3.2. Let $\mathbf{O}$ be a $\widehat{\mathcal{C}}$ -ring, $\mathbf{F}$ an $\mathbf{O}$ -module, and $\mathbf{G}$ a $\widehat{\mathcal{C}}$ -group. A $\mathbf{G}$ - $\mathbf{O}$ -module structure on $\mathbf{F}$ is a $\mathbf{G}$ -object structure such that, for every $S \in \text{Ob}(\mathcal{C})$ and every $g \in \mathbf{G}(S)$, the automorphism of $\mathbf{F}(S)$ defined by g is an automorphism of its $\mathbf{O}(S)$ -module structure.

Equivalently, the morphism of $\widehat{\mathcal{C}}$ -groups

$$\rho : \mathbf{G} \longrightarrow \underline{\text{Aut}}(\mathbf{F})$$

defined in 2.3 maps $\mathbf{G}$ into the $\widehat{\mathcal{C}}$ -subgroup $\underline{\text{Aut}}_{\mathbf{O}}(\mathbf{F})$ of $\underline{\text{Aut}}(\mathbf{F})$. Thus, giving a $\mathbf{G}$ - $\mathbf{O}$ -module structure on the $\mathbf{O}$ -module $\mathbf{F}$ is the same as giving a morphism of $\widehat{\mathcal{C}}$ -groups

$$\rho : \mathbf{G} \longrightarrow \underline{\text{Aut}}_{\mathbf{O}}(\mathbf{F}).$$

One defines naturally the abelian group $\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}}(\mathbf{F}, \mathbf{F}')$, and hence the additive category of $\mathbf{G}$ - $\mathbf{O}$ -modules, denoted $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$.

Remark 3.2.1. The reader will notice that $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ may also be defined as the category of $\mathbf{O}[\mathbf{G}]$ -modules, where $\mathbf{O}[\mathbf{G}]$ is the algebra of the $\widehat{\mathcal{C}}$ -group $\mathbf{G}$ over the $\widehat{\mathcal{C}}$ -ring $\mathbf{O}$, whose definition is clear.²⁹ Consequently, by 3.1.1 and 3.1.2, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ is an abelian category satisfying axiom (AB 5); moreover, if $\mathcal{C}$ is small, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects.

All the constructions of this section specialize immediately to the case where $\mathbf{G}$ (or $\mathbf{O}$, or both) is representable by an object of $\mathcal{C}$, which is thereby endowed with the corresponding algebraic structure.

We have treated briefly the principal algebraic structures that occur later in this seminar. For the others (the structure of an $\mathbf{O}$ -Lie algebra, for example), we believe that the reader has now seen enough examples in this section to make the general mechanism sketched in 2.2 work in each particular case.

We shall now apply what we have just done to the category of schemes, denoted $(\mathbf{Sch})$, and more generally to the categories $(\mathbf{Sch})_S$, also denoted $(\mathbf{Sch})_S$.

4. Algebraic structures in the category of schemes

Whenever there is no ambiguity, we shall permit ourselves the following abuses of language. We shall denote by T the object $T \xrightarrow{f} S$ of $\mathcal{C}_S$, with the datum of f (the structural morphism of T) understood, and we shall identify $\mathcal{C}$ with a subcategory of $\widehat{\mathcal{C}}$. In view of the compatibilities stated in the preceding sections, these identifications may be made without danger.

We shall also simplify terminology according to the following pattern: a $(\mathbf{Sch})$ -group will also be called a *group scheme*; a $(\mathbf{Sch})_S$ -group will be called a *group scheme over S* , an *S -group scheme*, or an *S -group*; and it will be called an *A -group* when S is the spectrum of the ring A .

4.1. Constant schemes

The category of schemes satisfies the conditions required in 1.8. Constant objects may therefore be defined in it. For every set E , there is a scheme $E_{\mathbb{Z}}$, and for every scheme S , an S -scheme

$$E_S = (E_{\mathbb{Z}})_S.$$

Recall (cf. *loc. cit.*) that for every S -scheme T ,

$$\text{Hom}_S(T, E_S)$$

is the set of locally constant maps from the topological space underlying T to E .

The functor $E \mapsto E_S$ commutes with finite inverse limits. In particular, if G is a group, then G_S is an S -group scheme; if O is a ring, then O_S is an S -ring scheme; and similarly for other algebraic structures.

²⁹Editor's note: the sentence which follows has been added; it will be used in Section 5.

4.2. Affine S -groups

Let us recall some facts about affine S -schemes (EGA II, Section 1). An S -scheme T is said to be *affine over S* if the inverse image of every affine open subset of S is affine. The $\mathcal{O}_S$ -algebra $f_*\mathcal{O}_T$, denoted by $\mathcal{A}(T)$, is then quasi-coherent, where f denotes the structural morphism of T . Conversely, to every quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{A}$, one can associate an S -scheme affine over S , denoted $\mathrm{Spec}(\mathcal{A})$. The functors

$$T \longmapsto \mathcal{A}(T), \quad \mathcal{A} \longmapsto \mathrm{Spec}(\mathcal{A})$$

are quasi-inverse equivalences between the category of S -schemes affine over S and the category opposite to that of quasi-coherent $\mathcal{O}_S$ -algebras.

It follows that giving an algebraic structure on an S -scheme T affine over S is equivalent to giving the corresponding structure on $\mathcal{A}(T)$ in the category opposite to that of quasi-coherent $\mathcal{O}_S$ -algebras. In particular, if G is an S -group affine over S , then $\mathcal{A}(G)$ is endowed with the structure of an augmented $\mathcal{O}_S$ -bialgebra: there are two morphisms of $\mathcal{O}_S$ -algebras

$$\Delta : \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G), \quad \varepsilon : \mathcal{A}(G) \longrightarrow \mathcal{O}_S,$$

corresponding to the morphisms of S -schemes

$$\pi : G \times G \longrightarrow G, \quad e_G : S \longrightarrow G.$$

The maps Δ and ε satisfy the following conditions, which express that G is an S -monoid.³⁰

(HA 1) Δ is coassociative: the following diagram is commutative.

$$\begin{array}{ccccc}
 & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) & & \\
 & \nearrow \Delta & & \searrow \mathrm{Id} \otimes \Delta & \\
 \mathcal{A}(G) & & & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\
 & \searrow \Delta & & \nearrow \Delta \otimes \mathrm{Id} & \\
 & & \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) & &
 \end{array}$$

Figure 1: Coassociativity of the diagonal map Δ .

Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 22; combined-reader p. 32.

(HA 2) Δ is compatible with ε : the following two composites are the identity:

$$\begin{aligned}
 \mathcal{A}(G) &\xrightarrow{\Delta} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\mathrm{Id} \otimes \varepsilon} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{O}_S \xrightarrow{\sim} \mathcal{A}(G), \\
 \mathcal{A}(G) &\xrightarrow{\Delta} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\varepsilon \otimes \mathrm{Id}} \mathcal{O}_S \otimes_{\mathcal{O}_S} \mathcal{A}(G) \xrightarrow{\sim} \mathcal{A}(G).
 \end{aligned}$$

Let us take this opportunity to note once again that it follows from the definition of an S -group structure that, in order to give such a structure on an S -scheme G affine over S , it is not necessary to verify anything on $\mathcal{A}(G)$; one need only endow every $G(S')$, for S' over S , with a group structure functorial in S' . This observation applies *mutatis mutandis* to morphisms.

³⁰Editor's note: of course, the inversion morphism $G \rightarrow G$ induces a morphism of $\mathcal{O}_S$ -algebras $\tau : \mathcal{A}(G) \rightarrow \mathcal{A}(G)$ which, together with Δ and ε , makes $\mathcal{A}(G)$ into a Hopf $\mathcal{O}_S$ -algebra.

4.3. The groups $\mathbb{G}_a$ and $\mathbb{G}_m$. The ring $\mathbf{O}$

4.3.1.

Let $\mathbb{G}_a$ be the functor from $(\mathbf{Sch})^\circ$ to $\mathbf{Set}$ defined by

$$\mathbb{G}_a(S) = \Gamma(S, \mathcal{O}_S),$$

endowed with the $\widehat{(\mathbf{Sch})}$ -group structure defined by the additive-group structure of the ring $\Gamma(S, \mathcal{O}_S)$. It is representable by an affine scheme, which we denote by $\mathbb{G}_a$, and which is therefore a group scheme:

$$\mathbb{G}_a = \mathrm{Spec} \mathbb{Z}[T].$$

Indeed,

$$\mathbb{G}_a(S) = \mathrm{Hom}(S, \mathbb{G}_a) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[T], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S).$$

For every scheme S , one therefore has an S -group affine over S , denoted $\mathbb{G}_{a,S}$, corresponding to the $\mathcal{O}_S$ -bialgebra $\mathcal{O}_S[T]$, whose diagonal map and augmentation are defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \quad \varepsilon(T) = 0.$$

4.3.2.

Let $\mathbb{G}_m$ be the functor from $(\mathbf{Sch})^\circ$ to $\mathbf{Set}$ defined by

$$\mathbb{G}_m(S) = \Gamma(S, \mathcal{O}_S)^\times,$$

where $\Gamma(S, \mathcal{O}_S)^\times$ denotes the multiplicative group of invertible elements of the ring $\Gamma(S, \mathcal{O}_S)$, endowed with its natural $\widehat{(\mathbf{Sch})}$ -group structure. It is representable by an affine scheme denoted by $\mathbb{G}_m$:

$$\mathbb{G}_m = \mathrm{Spec} \mathbb{Z}[T, T^{-1}] = \mathrm{Spec} \mathbb{Z}[\mathbb{Z}],$$

where $\mathbb{Z}[\mathbb{Z}]$ denotes the group algebra of $\mathbb{Z}$ over the ring $\mathbb{Z}$. Indeed,

$$\mathbb{G}_m(S) = \mathrm{Hom}_{\mathrm{Alg.}}(\mathbb{Z}[T, T^{-1}], \Gamma(S, \mathcal{O}_S)) \simeq \Gamma(S, \mathcal{O}_S)^\times.$$

For every scheme S , one therefore has an S -group affine over S , denoted $\mathbb{G}_{m,S}$, corresponding to the $\mathcal{O}_S$ -algebra $\mathcal{O}_S[\mathbb{Z}]$, whose diagonal map and augmentation are defined by

$$\Delta(x) = x \otimes x, \quad \varepsilon(x) = 1, \quad x \in \mathbb{Z}.$$

4.3.3.

Let $\mathbf{O}$ be the functor $\mathbb{G}_a$ endowed with its $\widehat{(\mathbf{Sch})}$ -ring structure. It is represented by the scheme $\mathrm{Spec} \mathbb{Z}[T]$, which we denote by $\mathbf{O}$ when it is regarded as endowed with its ring-scheme structure.

For every scheme S ,

$$\mathbf{O}_S = S \times_{\mathrm{Spec} \mathbb{Z}} \mathrm{Spec} \mathbb{Z}[T] = \mathrm{Spec}(\mathcal{O}_S[T])$$

is therefore an S -ring scheme affine over S . (Note: in EGA II, 1.7.13, $\mathbf{O}_S$ is denoted by $S[T]$.)

4.3.3.1. ³¹ For every object X of $\widehat{(\mathbf{Sch})}$,

$$\mathbf{O}(X) = \mathrm{Hom}(X, \mathbf{O})$$

³¹Editor's note: this paragraph has been added; it will be useful later (cf. II, 1.3).

has a ring structure, functorial in X . In particular, if X' is a scheme and morphisms $x : X' \rightarrow X$ and $f : X \rightarrow \mathbf{O}$ are given (that is, $f \in \mathbf{O}(X)$), then

$$f(x) = f \circ x \in \mathbf{O}(X') = \Gamma(X', \mathcal{O}_{X'}).$$

Definition. Let $\pi : M \rightarrow X$ be a morphism in $\widehat{(\mathbf{Sch})}$, and set

$$\mathbf{O}_X = \mathbf{O} \times X.$$

We say that M is an $\mathbf{O}_X$ -module if, for every X -scheme X' , the set

$$\mathrm{Hom}_X(X', M)$$

has been endowed with an $\mathbf{O}(X')$ -module structure, functorial in X' .

This is equivalent to giving an X -abelian-group law

$$\mu : M \times_X M \longrightarrow M$$

and an *external law*

$$\mathbf{O} \times M = \mathbf{O}_X \times_X M \longrightarrow M, \quad (f, m) \longmapsto f \cdot m,$$

which is an X -morphism, that is, $\pi(f \cdot m) = \pi(m)$, and which, for every $x \in X(S)$, endows

$$M(x) = \{m \in M(S) \mid \pi(m) = x\}$$

with an $\mathbf{O}(S)$ -module structure.

In that case, for every $Y \in \mathrm{Ob}(\widehat{(\mathbf{Sch})}_X)$, not necessarily representable,

$$\mathrm{Hom}_X(Y, M) = \Gamma(M_Y/Y)$$

is an $\mathbf{O}(Y)$ -module, functorially in Y .

4.4. Diagonalizable groups

4.4.1. The construction of $\mathbb{G}_m$ generalizes as follows. Let M be a commutative group, and let $M_{\mathbb{Z}}$ be the group scheme associated with it by the procedure of 4.1. Consider the functor defined by

$$\mathbf{D}(M)(S) = \mathrm{Hom}_{\mathrm{groups}}(M, \mathbb{G}_m(S)) \simeq \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This is a commutative $\widehat{(\mathbf{Sch})}$ -group represented by a group scheme denoted $D(M)$; thus, by definition,

$$D(M) \simeq \underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m).$$

Indeed, set

$$D(M) = \mathrm{Spec} \mathbb{Z}[M],$$

where $\mathbb{Z}[M]$ is the group algebra of M over $\mathbb{Z}$. Then

$$D(M)(S) = \mathrm{Hom}_{\mathrm{Alg}}(\mathbb{Z}[M], \Gamma(S, \mathcal{O}_S)) \simeq \mathrm{Hom}_{\mathrm{groups}}(M, \Gamma(S, \mathcal{O}_S)^\times)$$

by the very definition of the algebra $\mathbb{Z}[M]$.

4.4.2. For every scheme S , one therefore has an S -group scheme affine over S ,

$$D_S(M) = D(M)_S = (\underline{\mathrm{Hom}}_{(\mathbf{Sch})\text{-}\mathbf{Gr}}(M_{\mathbb{Z}}, \mathbb{G}_m))_S = \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

It is associated with the $\mathcal{O}_S$ -bialgebra $\mathcal{O}_S[M]$, endowed with the diagonal map and augmentation defined by

$$\Delta(x) = x \otimes x, \quad \varepsilon(x) = 1, \quad x \in M.$$

4.4.3. If $f : M \rightarrow N$ is a homomorphism of commutative groups, one defines in the evident way a morphism of S -groups

$$D_S(f) : D_S(N) \longrightarrow D_S(M),$$

and hence a functor

$$D_S : M \longmapsto D_S(M)$$

from the category of abelian groups to the category of S -groups affine over S . It may also be defined as the composite of the functor $M \mapsto M_S$ and the functor

$$M_S \longmapsto \underline{\mathrm{Hom}}_{S\text{-}\mathbf{Gr}}(M_S, \mathbb{G}_{m,S}).$$

This functor commutes with base extension.

An S -group isomorphic to a group of the form $D_S(M)$ is said to be *diagonalizable*. Notice that the elements of M are interpreted as certain characters of $D_S(M)$, that is, as certain elements of

$$\mathrm{Hom}_{S\text{-}\mathbf{Gr}}(D_S(M), \mathbb{G}_{m,S}).$$

(Compare VIII, 1.)

4.4.4. Let us give some examples of diagonalizable groups. First,

$$D(\mathbb{Z}) = \mathbb{G}_m, \quad D(\mathbb{Z}^r) = (\mathbb{G}_m)^r.$$

Set

$$\mu_n = D(\mathbb{Z}/n\mathbb{Z}),$$

and call it the *group of n -th roots of unity*. Indeed,

$$\mu_n(S) = \mathrm{Hom}_{\mathrm{groups}}(\mathbb{Z}/n\mathbb{Z}, \Gamma(S, \mathcal{O}_S)^\times) = \{f \in \Gamma(S, \mathcal{O}_S) \mid f^n = 1\}.$$

The S -group $\mu_{n,S}$ corresponds to the $\mathcal{O}_S$ -algebra $\mathcal{O}_S[T]/(T^n - 1)$. Suppose, in particular, that S is the spectrum of a field k of characteristic $p = n$. On setting $T - 1 = s$, one obtains

$$k[T]/(T^p - 1) = k[s]/(s^p),$$

which shows that the underlying topological space of $\mu_{p,k}$ consists of one point, whose local ring is the Artinian k -algebra $k[s]/(s^p)$. In the same vein, note that the S -schemes $\mathbb{G}_{a,S}$, $\mathbb{G}_{m,S}$, and $\mathbf{O}_S$ are smooth over S , and that $D_S(M)$ is

flat over S . Moreover, it is formally smooth (respectively, smooth) over S if and only if no residue characteristic of S divides the torsion of M (respectively, and if in addition M is of finite type); see Exposé VIII, 2.1.

4.5. Other examples of groups

The preceding procedure applies to the “classical groups” (linear groups GL_n , symplectic groups Sp_n , orthogonal groups O_n , and so forth). For example, one defines GL_n as representing the functor $\mathbf{GL}_n$ such that

$$\mathbf{GL}_n(S) = \mathrm{GL}(n, \Gamma(S, \mathcal{O}_S)) = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{O}_S^n).$$

It may be constructed, for example, as the open subscheme of $\mathrm{Spec} \mathbb{Z}[T_{ij}]$, $1 \leq i, j \leq n$, defined by the function

$$f = \det((T_{ij})_{i,j=1}^n),$$

that is, as $\mathrm{Spec} \mathbb{Z}[T_{ij}, f^{-1}]$.

4.6. Functor-modules in the category of schemes

We shall associate with every $\mathcal{O}_S$ -module on a scheme S a $\mathbf{O}_S$ -module, where $\mathbf{O}_S$ denotes the functor-ring introduced in 4.3.3. This can be done in two different ways, as follows.

Definition 4.6.1. Let S be a scheme. For every $\mathcal{O}_S$ -module $\mathcal{F}$, denote by $\mathbf{V}(\mathcal{F})$ and $\mathbf{W}(\mathcal{F})$ the contravariant functors on $(\mathbf{Sch})/S$ defined by

$$\begin{aligned}\mathbf{V}(\mathcal{F})(S') &= \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}, \mathcal{O}_{S'}), \\ \mathbf{W}(\mathcal{F})(S') &= \Gamma(S', \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),\end{aligned}$$

where $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ denotes the inverse image on S' of the $\mathcal{O}_S$ -module $\mathcal{F}$.

Then $\mathbf{V}(\mathcal{F})$ and $\mathbf{W}(\mathcal{F})$ are endowed in the evident way with $\mathbf{O}_S$ -module structures (recall that $\mathbf{O}_S(S') = \Gamma(S', \mathcal{O}_{S'}) = \mathbf{W}(\mathcal{O}_S)(S')$). Thus one has in fact defined two functors $\mathbf{V}$ and $\mathbf{W}$ from the category of $\mathcal{O}_S$ -modules to the category of $\mathbf{O}_S$ -modules, with $\mathbf{V}$ contravariant and $\mathbf{W}$ covariant.

For the remainder of this paragraph, we restrict to the case in which the $\mathcal{O}_S$ -modules in question are quasi-coherent. Thus we regard $\mathbf{V}$ and $\mathbf{W}$ as functors

$$\mathbf{V} : (\mathcal{O}_S\text{-Mod.q.c.})^\circ \longrightarrow (\mathbf{O}_S\text{-Mod.}), \quad \mathbf{W} : (\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathbf{O}_S\text{-Mod.}).$$

Remark 4.6.1.1. ³² The reader will notice that, in what follows, every proposition involving only the functor $\mathbf{W}$ remains valid for arbitrary $\mathcal{O}_S$ -modules, not necessarily quasi-coherent.

Proposition 4.6.2.

- (i) The functors $\mathbf{V}$ and $\mathbf{W}$ commute with base extension: if S' lies over S and $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$ -module, then

$$\mathbf{V}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \simeq \mathbf{V}(\mathcal{F})_{S'}, \quad \mathbf{W}(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \simeq \mathbf{W}(\mathcal{F})_{S'}.$$

- (ii) The functors $\mathbf{V}$ and $\mathbf{W}$ are fully faithful: the canonical maps

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F})),$$

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') \longrightarrow \mathrm{Hom}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}'))$$

are bijective.

- (iii) The functors $\mathbf{V}$ and $\mathbf{W}$ are additive:

$$\mathbf{V}(\mathcal{F} \oplus \mathcal{F}') \simeq \mathbf{V}(\mathcal{F}) \times_S \mathbf{V}(\mathcal{F}'), \quad \mathbf{W}(\mathcal{F} \oplus \mathcal{F}') \simeq \mathbf{W}(\mathcal{F}) \times_S \mathbf{W}(\mathcal{F}').$$

Parts (i) and (iii) are immediate from the definitions. For (ii), take the S' to be open subschemes of S . We leave the proof to the reader (for $\mathbf{V}$, use EGA II, 1.7.14).

Recall (cf. 3.1.4) that if $\mathbf{F}$ and $\mathbf{F}'$ are $\mathbf{O}_S$ -modules, then $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{F}, \mathbf{F}')$ denotes the S -functor in abelian groups that assigns to every $S' \rightarrow S$ the group

$$\mathrm{Hom}_{\mathbf{O}_{S'}}(\mathbf{F}_{S'}, \mathbf{F}'_{S'}).$$

Proposition 4.6.3. *There are canonical morphisms in $(\mathbf{O}_S\text{-Mod.})$:* ³³

³²Editor's note: the numbering 4.6.1.1 has been added for later references.

³³Editor's note: the isomorphism $\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) \simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F}))$ has been added.

$$\begin{array}{ccc}
\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}')) & \xlongequal{\quad} & \underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F})) \\
& \nwarrow \quad \nearrow & \\
& \mathbf{W}(\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}')) &
\end{array}$$

Figure 2: Canonical morphisms of Proposition 4.6.3.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 26;
combined-reader p. 36.*

This follows immediately from 4.6.2(i) and (ii).

Notation 4.6.3.1. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$ -module. By EGA II, 1.7.8, the S -functor $\mathbf{V}(\mathcal{F})$ is represented by an S -scheme affine over S , denoted $\mathbf{V}(\mathcal{F})$, and called the *vector fibration*³⁴ defined by $\mathcal{F}$:

$$\mathbf{V}(\mathcal{F}) = \mathrm{Spec}(\mathcal{S}(\mathcal{F})),$$

where $\mathcal{S}(\mathcal{F})$ denotes the symmetric algebra of the $\mathcal{O}_S$ -module $\mathcal{F}$.³⁵

Proposition 4.6.4. *Let $\mathcal{F}$ and $\mathcal{F}'$ be two quasi-coherent $\mathcal{O}_S$ -modules, and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_S$ -algebra. There is a functorial isomorphism*

$$\mathrm{Hom}_S(\mathrm{Spec}(\mathcal{A}), \underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{F}'), \mathbf{W}(\mathcal{F}))) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}', \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}).$$

Indeed, put $X = \mathrm{Spec}(\mathcal{A})$. The left-hand side is canonically isomorphic to

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{F}'), \mathbf{W}(\mathcal{F}))(X),$$

that is, by definition, to

$$\begin{aligned}
& \mathrm{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{F}')_X, \mathbf{W}(\mathcal{F})_X) \\
& \simeq \mathrm{Hom}_{\mathcal{O}_X}(\mathbf{W}(\mathcal{F}' \otimes \mathcal{O}_X), \mathbf{W}(\mathcal{F} \otimes \mathcal{O}_X))
\end{aligned}$$

(cf. 4.6.2(i)). By 4.6.2(ii), this may also be written as

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}' \otimes \mathcal{O}_X, \mathcal{F} \otimes \mathcal{O}_X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{F}', \pi_* \pi^*(\mathcal{F})),$$

where $\pi : X \rightarrow S$ is the structural morphism. But EGA II, 1.4.7 gives

$$\pi_* \pi^*(\mathcal{F}) \simeq \mathcal{F} \otimes \mathcal{A},$$

which completes the proof.

Corollary 4.6.4.1. *There is a canonical isomorphism*

$$\mathbf{W}(\mathcal{F} \otimes \mathcal{A}) \simeq \underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathcal{A}), \mathbf{W}(\mathcal{F})).$$

Indeed,³⁶ let $f : S' \rightarrow S$ be an S -scheme and put $X' = X \times_S S'$. There is a Cartesian square

³⁴Editor's note: “vector bundle” has been replaced by “vector fibration.” Current usage calls a vector fibration that is locally trivial of rank r a “vector bundle of rank r ,” that is, one whose sheaf of sections is locally isomorphic to $\mathcal{O}_S^{\oplus r}$.

³⁵Editor's note: the articles [Ni04] and [Ni02], respectively, show that if S is Noetherian and $\mathcal{F}$ is a coherent $\mathcal{O}_S$ -module, then $\mathbf{W}(\mathcal{F})$, respectively the S -group assigning $\mathrm{Aut}_{\mathcal{O}_T}(\mathcal{F} \otimes \mathcal{O}_T)$ to every $T \rightarrow S$, is representable if and only if $\mathcal{F}$ is locally free.

³⁶Editor's note: the original has been expanded in what follows.

$$\begin{array}{ccc} X' & \xrightarrow{f'} & X \\ \pi' \downarrow & & \downarrow \pi \\ S' & \xrightarrow{f} & S \end{array}$$

Figure 3: The Cartesian base-change square used in Corollary 4.6.4.1.

Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 27; combined-reader p. 37.

By EGA II, 1.5.2, X' is affine over S' and $\pi'_* \mathcal{O}_{X'} = f^*(\mathcal{A})$. Hence

$$\begin{aligned} \underline{\mathrm{Hom}}_S(\mathrm{Spec}(\mathcal{A}), \mathbf{W}(\mathcal{F}))(S') \\ = \mathrm{Hom}_{S'}(\mathrm{Spec}(f^*\mathcal{A}), \mathbf{W}(f^*\mathcal{F})). \end{aligned}$$

By 4.6.4, applied to $f^*\mathcal{F}$, $\mathcal{F}' = \mathcal{O}_{S'}$, and $f^*\mathcal{A}$, this is equal to

$$\begin{aligned} \Gamma(S', f^*\mathcal{F} \otimes f^*\mathcal{A}) &= \Gamma(S', f^*(\mathcal{F} \otimes \mathcal{A})) \\ &= \mathbf{W}(\mathcal{F} \otimes \mathcal{A})(S'). \end{aligned}$$

Proposition 4.6.5. *If $\mathcal{F}$ and $\mathcal{F}'$ are two $\mathcal{O}_S$ -modules locally free of finite type, the morphisms of 4.6.3 are isomorphisms.*

Indeed, for every $S' \rightarrow S$,

$$\begin{aligned} \mathbf{W}(\mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}'))(S') \\ = \Gamma(S', \mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{F}') \otimes \mathcal{O}_{S'}) \\ = \mathrm{Hom}_{\mathcal{O}_{S'}}(\mathcal{F} \otimes \mathcal{O}_{S'}, \mathcal{F}' \otimes \mathcal{O}_{S'}). \end{aligned}$$

The last member is isomorphic both to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{W}(\mathcal{F}'))(S')$$

and to

$$\underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}'), \mathbf{V}(\mathcal{F}))(S')$$

by 4.6.2(i) and (ii).

Corollary 4.6.5.1. *Let $\mathcal{F}$ be an $\mathcal{O}_S$ -module locally free of finite type. Put $\mathcal{F}^\vee = \mathcal{H}om_{\mathcal{O}_S}(\mathcal{F}, \mathcal{O}_S)$. There are canonical isomorphisms*

$$\begin{aligned} \mathbf{W}(\mathcal{F}^\vee) &\simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}), \mathbf{O}_S) \simeq \mathbf{V}(\mathcal{F}), \\ \mathbf{V}(\mathcal{F}^\vee) &\simeq \underline{\mathrm{Hom}}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F}), \mathbf{O}_S) \simeq \mathbf{W}(\mathcal{F}). \end{aligned}$$

Finally, we have the following proposition.

Proposition 4.6.6. *Let $f : \mathcal{F} \rightarrow \mathcal{F}'$ be a morphism of $\mathcal{O}_S$ -modules locally free of finite rank. The morphism*

$$\mathbf{W}(f) : \mathbf{W}(\mathcal{F}) \longrightarrow \mathbf{W}(\mathcal{F}')$$

is a monomorphism if and only if f identifies $\mathcal{F}$ locally with a direct summand of $\mathcal{F}'$.

The direct assertion is essentially contained in EGA 0_I, 5.5.5. ³⁷ Conversely, if $\mathcal{F}$ is locally a direct summand of $\mathcal{F}'$, then for every $\pi : S' \rightarrow S$, the module $\pi^*\mathcal{F}$ is a submodule of $\pi^*\mathcal{F}'$, since this holds locally as a direct summand. Thus

$$\mathbf{W}(\mathcal{F})(S') = \Gamma(S', \pi^*\mathcal{F})$$

is a submodule of

$$\mathbf{W}(\mathcal{F}')(S') = \Gamma(S', \pi^*\mathcal{F}').$$

³⁷Editor's note: the following sentence has been expanded.

4.7. The category of $G\text{-}\mathcal{O}_S$ -modules

Let G be an S -group and $\mathcal{F}$ an $\mathcal{O}_S$ -module; $\mathbf{W}(\mathcal{F})$ has a $\mathbf{O}_S$ -module structure.

Definition 4.7.1. A $G\text{-}\mathcal{O}_S$ -module structure on $\mathcal{F}$ is an $\mathbf{h}_G\text{-}\mathbf{O}_S$ -module structure on $\mathbf{W}(\mathcal{F})$ (cf. 3.2). A morphism of $G\text{-}\mathcal{O}_S$ -modules is, by definition, a morphism of the associated $\mathbf{h}_G\text{-}\mathbf{O}_S$ -modules. This gives the category $(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.})$; we denote by $(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.})$ its full subcategory consisting of those $G\text{-}\mathcal{O}_S$ -modules that are quasi-coherent as $\mathcal{O}_S$ -modules.

By 3.2, giving a $G\text{-}\mathcal{O}_S$ -module structure on $\mathcal{F}$ is therefore equivalent to giving a morphism of $(\mathbf{Sch})_{/S}$ -groups

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathbf{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F})).$$

Remark 4.7.1.0. ³⁸ Since 4.6.2 gives an anti-isomorphism of S -functors in groups

$$(\dagger) \quad \underline{\mathbf{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F})) \simeq \underline{\mathbf{Aut}}_{\mathbf{O}_S}(\mathbf{V}(\mathcal{F})),$$

it is equivalent to give an $\mathbf{h}_G\text{-}\mathbf{O}_S$ -module structure on $\mathbf{W}(\mathcal{F})$ or on $\mathbf{V}(\mathcal{F})$. Indeed, let

$$\rho : \mathbf{h}_G \longrightarrow \underline{\mathbf{Aut}}_{\mathbf{O}_S}(\mathbf{W}(\mathcal{F}))$$

be as above. For every $T \rightarrow S$ and $g \in G(T)$, denote by $\rho^*(g)$ the image of $\rho(g)$ under the anti-isomorphism $(\dagger)$. Then

$$\rho^*(gh) = \rho^*(h) \circ \rho^*(g),$$

so ρ^* defines a *right* $\mathbf{h}_G\text{-}\mathbf{O}_S$ -module structure on $\mathbf{V}(\mathcal{F})$. Setting

$$\rho^\vee(g) = \rho^*(g^{-1}),$$

one obtains an $\mathbf{h}_G\text{-}\mathbf{O}_S$ -module structure on $\mathbf{V}(\mathcal{F})$, whose datum is equivalent to that of ρ .

Remark 4.7.1.1. One may say that the categories just defined have been constructed by the following Cartesian squares.

$$\begin{array}{ccccc} (G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) & \hookrightarrow & (G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.}) & \longrightarrow & (\mathbf{h}_G\text{-}\mathbf{O}_S\text{-}\mathbf{Mod.}) \\ \downarrow & & \downarrow & & \downarrow \text{oubli} \\ (\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) & \hookrightarrow & (\mathcal{O}_S\text{-}\mathbf{Mod.}) & \xrightarrow{\mathbf{W}} & (\mathbf{O}_S\text{-}\mathbf{Mod.}). \end{array}$$

Figure 4: Cartesian squares defining the categories of $G\text{-}\mathcal{O}_S$ -modules.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 28;
combined-reader p. 38.*

The categories $(\mathcal{O}_S\text{-}\mathbf{Mod.})$ and $(\mathbf{O}_S\text{-}\mathbf{Mod.})$ are abelian, but one should remember that in general the functor $\mathbf{W}$ is neither left nor right exact. ³⁹

Remark 4.7.1.2. ⁴⁰ Let $\mathcal{F}$ be a $G\text{-}\mathcal{O}_S$ -module. The subsheaf of invariants $\mathcal{F}^G$ is defined as follows: for every open subset U of S ,

$$\mathcal{F}^G(U) = \mathbf{W}(\mathcal{F})^G(U) = \{x \in \mathcal{F}(U) \mid g \cdot x_{S'} = x_{S'} \text{ for every } S' \xrightarrow{f} U, g \in G(S')\},$$

³⁸Editor's note: this remark has been added from [DG], II, Section 2, 1.1.

³⁹Editor's note: the original has been corrected by deleting the assertion that the category $(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.})$ is abelian; see 4.7.2.1 below.

⁴⁰Editor's note: this remark has been added.

where $x_{S'}$ denotes the image of x in

$$\Gamma(S', f^* \mathcal{F}) = \Gamma(U, f_* f^* \mathcal{F}).$$

One should note that the natural morphism

$$\mathbf{W}(\mathcal{F}^G) \longrightarrow \mathbf{W}(\mathcal{F})^G$$

is not an isomorphism in general. For example, let $S = \operatorname{Spec}(\mathbb{Z})$, and let G be the constant group $\mathbb{Z}/2\mathbb{Z} = \{1, \tau\}$ acting on $\mathcal{F} = \mathcal{O}_S$ by $\tau \cdot 1 = -1$. Then $\mathcal{F}^G = 0$, whereas, if R is an $\mathbb{F}_2$ -algebra,

$$\mathbf{W}(\mathcal{F})^G(\operatorname{Spec}(R)) = R.$$

4.7.2. From now until the end of 4.7, suppose that G is affine over S .⁴¹ By 4.6.4, giving a morphism of S -functors

$$\rho : \mathbf{h}_G \longrightarrow \underline{\operatorname{End}}_{\mathcal{O}_S}(\mathbf{W}(\mathcal{F}))$$

is equivalent to giving a morphism of $\mathcal{O}_S$ -modules

$$\mu : \mathcal{F} \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

Saying that ρ is a morphism of $(\widehat{\mathbf{Sch}})_{/S}$ -groups is then equivalent to saying that μ satisfies the following axioms.

(CM 1) The following diagram is commutative.

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\mu} & \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ \mu \downarrow & & \downarrow \operatorname{id} \otimes \Delta \\ \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) & \xrightarrow{\mu \otimes \operatorname{id}} & \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \end{array}.$$

Figure 5: The coassociativity axiom for the coaction μ .

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 29;
combined-reader p. 39.*

(CM 2) The following composite is the identity.

$$\mathcal{F} \xrightarrow{\mu} \mathcal{F} \otimes \mathcal{A}(G) \xrightarrow{\operatorname{id} \otimes \varepsilon} \mathcal{F} \otimes \mathcal{O}_S \xrightarrow{\sim} \mathcal{F}.$$

Figure 6: The counit axiom for the coaction μ .

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 29;
combined-reader p. 39.*

These axioms (CM 1) and (CM 2) are those of a right *comodule* structure over the bialgebra $\mathcal{A}(G)$.⁴²

Put $\mathcal{A} = \mathcal{A}(G)$. If $\mathcal{F}$ and $\mathcal{F}'$ are $\mathcal{A}$ -comodules, a morphism of comodules $f : \mathcal{F} \rightarrow \mathcal{F}'$ is a morphism of $\mathcal{O}_S$ -modules such that the following diagram is

⁴¹Editor's note: see VI_B, Sections 11.1–11.6 for the extension of the results of 4.7.2 to the case where G is not necessarily affine but G and $\mathcal{F}$ are assumed flat over S .

⁴²Editor's note: left G - $\mathcal{O}_S$ -modules correspond naturally to right $\mathcal{A}(G)$ -comodules.

commutative.

$$\begin{array}{ccc}
 \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \\
 \mu_{\mathcal{F}} \downarrow & & \downarrow \mu_{\mathcal{F}'} \\
 \mathcal{F} \otimes \mathcal{A} & \xrightarrow{f \otimes \text{id}} & \mathcal{F}' \otimes \mathcal{A}.
 \end{array}$$

Figure 7: Compatibility required of a morphism of comodules.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 30;
combined-reader p. 40.*

This gives the category $(\mathcal{A}\text{-}\mathbf{Comod})$; we denote by $(\mathcal{A}\text{-}\mathbf{Comod.q.c.})$ its full subcategory consisting of those $\mathcal{A}$ -comodules that are quasi-coherent as $\mathcal{O}_S$ -modules. We have therefore proved the following statement.

Proposition 4.7.2. *Let G be an S -group affine over S . There are equivalences of categories*

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.}) \simeq (\mathcal{A}(G)\text{-}\mathbf{Comod.}),$$

$$(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.}) \simeq (\mathcal{A}(G)\text{-}\mathbf{Comod.q.c.}).$$

If, in addition, $S = \text{Spec}(\Lambda)$ is affine and one sets $\Lambda[G] = \Gamma(S, \mathcal{A}(G))$, there is an equivalence of categories

$$(\mathcal{A}(G)\text{-}\mathbf{Comod.q.c.}) \simeq (\Lambda[G]\text{-}\mathbf{Comod.}).$$

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Suppose, moreover, that $\mathcal{A} = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module.⁴⁴ Let $\mathcal{E}$ be an $\mathcal{A}$ -comodule and $\mathcal{F}$ an $\mathcal{O}_S$ -submodule of $\mathcal{E}$. Since $\mathcal{A}$ is flat over $\mathcal{O}_S$, one may identify $\mathcal{F} \otimes \mathcal{A}$ (respectively, $\mathcal{F} \otimes \mathcal{A} \otimes \mathcal{A}$) with an $\mathcal{O}_S$ -submodule of $\mathcal{E} \otimes \mathcal{A}$ (respectively, of $\mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A}$). Suppose that $\mu_{\mathcal{E}}$ maps $\mathcal{F}$ into $\mathcal{F} \otimes \mathcal{A}$. Then its restriction

$$\mu_{\mathcal{F}} : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}$$

endows $\mathcal{F}$ with an $\mathcal{A}$ -comodule structure; one says that $\mathcal{F}$ is a *subcomodule* of $\mathcal{E}$. Passing to the quotient, $\mu_{\mathcal{E}}$ defines a morphism of $\mathcal{O}_S$ -modules

$$\mathcal{E}/\mathcal{F} \longrightarrow (\mathcal{E}/\mathcal{F}) \otimes \mathcal{A},$$

which endows $\mathcal{E}/\mathcal{F}$ with an $\mathcal{A}$ -comodule structure. If $f : \mathcal{E} \rightarrow \mathcal{E}'$ is a morphism of $\mathcal{A}$ -comodules, then $\text{Ker } f$ (respectively, $\text{Im } f$) is an $\mathcal{A}$ -subcomodule of $\mathcal{E}$ (respectively, of $\mathcal{E}'$), and f induces an isomorphism of $\mathcal{A}$ -comodules

$$\mathcal{E}/\text{Ker } f \xrightarrow{\sim} \text{Im } f.$$

Moreover, if $\mathcal{E}$ and $\mathcal{E}'$ are quasi-coherent $\mathcal{O}_S$ -modules, then so are $\text{Ker } f$ and $\text{Im } f$. Consequently, $(\mathcal{A}\text{-}\mathbf{Comod.})$ and $(\mathcal{A}\text{-}\mathbf{Comod.q.c.})$ are abelian categories.

Corollary 4.7.2.1. *Suppose that G is affine and flat over S . Then the category $(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.q.c.})$ (respectively, $(G\text{-}\mathcal{O}_S\text{-}\mathbf{Mod.})$), equivalent to the category of $\mathcal{A}(G)$ -comodules that are quasi-coherent over $\mathcal{O}_S$ (respectively, of all $\mathcal{A}(G)$ -comodules), is abelian.*

⁴³Editor's note: the preceding sentence has been added.

⁴⁴Editor's note: the following paragraph has been added from [Se68, Section 1.3].

4.7.3. Suppose now that G is a *diagonalizable group*, that is, that $\mathcal{A}(G)$ is the algebra of a commutative group M over the sheaf of rings $\mathcal{O}_S$. If $\mathcal{F}$ is an $\mathcal{O}_S$ -module, then

$$\mathcal{F} \otimes \mathcal{A}(G) = \coprod_{m \in M} \mathcal{F} \otimes m\mathcal{O}_S.$$

Giving a morphism of $\mathcal{O}_S$ -modules

$$\mu : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}(G)$$

is therefore equivalent to giving $\mathcal{O}_S$ -endomorphisms $(\mu_m)_{m \in M}$ of $\mathcal{F}$ such that, for every section x of $\mathcal{F}$ on an open subset of S , the family $(\mu_m(x))$ is a section of the direct sum $\coprod_{m \in M} \mathcal{F}$. This means that on every sufficiently small open subset, only finitely many restrictions of the $\mu_m(x)$ are nonzero.

For μ defined by

$$\mu(x) = \sum_{m \in M} \mu_m(x) \otimes m,$$

condition (CM 1), respectively (CM 2), holds if and only if

$$\mu_m \circ \mu_{m'} = \delta_{mm'} \mu_m, \quad \left(\text{respectively, } \sum_{m \in M} \mu_m = \text{Id}_{\mathcal{F}} \right).$$

Thus the μ_m are pairwise orthogonal projectors whose sum is the identity. We have proved the following result.

Proposition 4.7.3. *If $G = D_S(M)$ is a diagonalizable S -group, the category of quasi-coherent G - $\mathcal{O}_S$ -modules (respectively, of all G - $\mathcal{O}_S$ -modules) is equivalent to the category of quasi-coherent $\mathcal{O}_S$ -modules (respectively, of all $\mathcal{O}_S$ -modules) graded by M .*

Remark. If $\mathcal{F}$ has an $\mathcal{O}_S$ -algebra structure preserved by the operations of G , then the grading of $\mathcal{F}$ is an algebra grading. More precisely:

Corollary 4.7.3.1. *The functor $\mathcal{A} \mapsto \text{Spec}(\mathcal{A})$ induces an equivalence between the category of quasi-coherent $\mathcal{O}_S$ -algebras graded by M and the category opposite to that of S -schemes affine over S with S -operator group $G = D_S(M)$.*

Proposition 4.7.4. *Let G be a diagonalizable S -group. If*

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

is an exact sequence of quasi-coherent G - $\mathcal{O}_S$ -modules that splits as a sequence of $\mathcal{O}_S$ -modules, then it also splits as a sequence of G - $\mathcal{O}_S$ -modules.

Indeed, if $G = D_S(M)$, each $\mathcal{F}_i$ is graded by the $(\mathcal{F}_i)_m$, and for every $m \in M$ the sequence

$$0 \longrightarrow (\mathcal{F}_1)_m \longrightarrow (\mathcal{F}_2)_m \longrightarrow (\mathcal{F}_3)_m \longrightarrow 0$$

of $\mathcal{O}_S$ -modules is split. The preceding proposition then gives the result.

5. Group cohomology

5.1. The standard complex

Let $\mathcal{C}$ be a category, $\mathbf{G}$ a $\widehat{\mathcal{C}}$ -group, $\mathbf{O}$ a $\widehat{\mathcal{C}}$ -ring, and $\mathbf{F}$ a $\mathbf{G}$ - $\mathbf{O}$ -module.⁴⁵ For $n \geq 0$, set

$$C^n(\mathbf{G}, \mathbf{F}) = \text{Hom}(\mathbf{G}^n, \mathbf{F}), \quad \underline{C}^n(\mathbf{G}, \mathbf{F}) = \underline{\text{Hom}}(\mathbf{G}^n, \mathbf{F}),$$

⁴⁵Editor's note: This complex is often called the Hochschild complex; see, for example, Section II.3 of [DG70].

where $\mathbf{G}^0$ is the final object $\mathbf{e}$. Then $\underline{C}^n(\mathbf{G}, \mathbf{F})$ (respectively, $C^n(\mathbf{G}, \mathbf{F})$) is endowed in the evident way with an $\mathbf{O}$ -module structure (respectively, a $\Gamma(\mathbf{O})$ -module structure), and

$$C^n(\mathbf{G}, \mathbf{F}) \simeq \Gamma(\underline{C}^n(\mathbf{G}, \mathbf{F})), \quad \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S).$$

To give an element of $C^n(\mathbf{G}, \mathbf{F})$ is to give, for every $S \in \text{Ob}(\mathcal{C})$, an n -cochain of $\mathbf{G}(S)$ with values in $\mathbf{F}(S)$, functorially in S . Recall that the boundary operator

$$\partial : C^n(\mathbf{G}(S), \mathbf{F}(S)) \longrightarrow C^{n+1}(\mathbf{G}(S), \mathbf{F}(S))$$

is given by

$$\begin{aligned} \partial f(g_1, \dots, g_{n+1}) &= g_1 f(g_2, \dots, g_{n+1}) \\ &\quad + \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+1} f(g_1, \dots, g_n). \end{aligned}$$

It is functorial in S , and hence defines a homomorphism

$$\partial : C^n(\mathbf{G}, \mathbf{F}) \longrightarrow C^{n+1}(\mathbf{G}, \mathbf{F})$$

such that $\partial \circ \partial = 0$. We have thus defined a complex of abelian groups (indeed, of $\Gamma(\mathbf{O})$ -modules), denoted $C^*(\mathbf{G}, \mathbf{F})$. In the same way one defines the complex of $\mathbf{O}$ -modules $\underline{C}^*(\mathbf{G}, \mathbf{F})$, and

$$C^*(\mathbf{G}, \mathbf{F}) = \Gamma(\underline{C}^*(\mathbf{G}, \mathbf{F})).$$

We denote by $H^n(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{H}^n(\mathbf{G}, \mathbf{F})$) the groups (respectively, the $\widehat{\mathcal{C}}$ -groups) of cohomology of $C^*(\mathbf{G}, \mathbf{F})$ (respectively, $\underline{C}^*(\mathbf{G}, \mathbf{F})$). In particular,

$$\underline{H}^0(\mathbf{G}, \mathbf{F}) = \mathbf{F}^{\mathbf{G}}, \quad H^0(\mathbf{G}, \mathbf{F}) = \Gamma(\mathbf{F}^{\mathbf{G}}).$$

Remark 5.1.1. The “set-theoretic” description of ∂ given above is convenient for checking that $\partial \circ \partial = 0$. One may also define ∂ in terms of the multiplication $m : \mathbf{G} \times \mathbf{G} \rightarrow \mathbf{G}$ and the action $\mu : \mathbf{G} \times \mathbf{F} \rightarrow \mathbf{F}$ as follows: for every $f \in C^n(\mathbf{G}, \mathbf{F})$,

$$\partial f = \mu \circ (\text{id}_{\mathbf{G}} \times f) + \sum_{i=1}^n (-1)^i f \circ (\text{id}_{\mathbf{G}^{i-1}} \times m \times \text{id}_{\mathbf{G}^{n-i}}) + (-1)^{n+1} f \circ \text{pr}_{[1, n]},$$

where $\text{pr}_{[1, n]}$ denotes the projection of $\mathbf{G}^{n+1} = \mathbf{G}^n \times \mathbf{G}$ onto $\mathbf{G}^n$. Similarly, for every $S \in \text{Ob}(\mathcal{C})$ and $f \in \underline{C}^n(\mathbf{G}, \mathbf{F})(S) = C^n(\mathbf{G}_S, \mathbf{F}_S)$,

$$\partial f = \mu_S \circ (\text{id}_{\mathbf{G}_S} \times f) + \sum_{i=1}^n (-1)^i f \circ (\text{id}_{\mathbf{G}_S^{i-1}} \times m_S \times \text{id}_{\mathbf{G}_S^{n-i}}) + (-1)^{n+1} f \circ \text{pr}_{[1, n]},$$

where m_S and μ_S are obtained from m and μ by base change. ⁴⁶

⁴⁶Editor’s note: This remark has been added.

5.2.

Recall (see Section 3) that $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ is an abelian category, with its structure defined objectwise. Thus

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

is an exact sequence of $\mathbf{G}\text{-}\mathbf{O}$ -modules if and only if the sequence of abelian groups

$$0 \longrightarrow \mathbf{F}'(S) \longrightarrow \mathbf{F}(S) \longrightarrow \mathbf{F}''(S) \longrightarrow 0$$

is exact for every $S \in \text{Ob}(\mathcal{C})$.⁴⁷

Suppose that $\mathcal{C}$ is small. Then, by 3.2.1, $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects, so the derived functors of the left exact functors $\underline{H}^0$ and H^0 are defined. We shall now show that the functors $\underline{H}^n$ (respectively, H^n) are indeed the derived functors of $\underline{H}^0$ (respectively, H^0).⁴⁸

Definition 5.2.0. For every $\mathbf{O}$ -module $\mathbf{P}$, let $E(\mathbf{P})$ denote the object $\underline{\text{Hom}}(\mathbf{G}, \mathbf{P})$ of $\widehat{\mathcal{C}}$, endowed with the following $\mathbf{G}\text{-}\mathbf{O}$ -module structure. For every $S \in \text{Ob}(\mathcal{C})$,

$$\underline{\text{Hom}}(\mathbf{G}, \mathbf{P})(S) = \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S),$$

and $g \in \mathbf{G}(S)$ and $a \in \mathbf{O}(S)$ act on $\phi \in \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ by

$$(g \cdot \phi)(h) = \phi(hg), \quad (a \cdot \phi)(h) = a\phi(h),$$

for every $h \in \mathbf{G}(S')$, $S' \rightarrow S$. Moreover, for every $\phi \in \text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$, set

$$\varepsilon(\phi) = \phi(1) \in \mathbf{P}(S),$$

where 1 denotes the unit element of $\mathbf{G}(S)$. This defines a functor

$$E : (\mathbf{O}\text{-Mod.}) \longrightarrow (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$$

and a natural transformation $\varepsilon : E \rightarrow \text{Id}$, where Id denotes the identity functor of $(\mathbf{O}\text{-Mod.})$.⁴⁹

Remark 5.2.0.1. In what follows, let $\mathbf{G}_1$ and $\mathbf{G}_2$ denote two copies of $\mathbf{G}$. The morphism

$$\mathbf{G}_1 \times E(\mathbf{P}) \longrightarrow E(\mathbf{P}), \quad (g_1, \phi) \longmapsto (g_2 \longmapsto \phi(g_2 g_1))$$

corresponds, via the isomorphisms

$$\begin{aligned} \text{Hom}(\mathbf{G}_1 \times E(\mathbf{P}), E(\mathbf{P})) &\simeq \text{Hom}\left(E(\mathbf{P}), \underline{\text{Hom}}(\mathbf{G}_1, \underline{\text{Hom}}(\mathbf{G}_2, \mathbf{P}))\right) \\ &\simeq \text{Hom}\left(E(\mathbf{P}), \underline{\text{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})\right), \end{aligned}$$

to the morphism

$$\phi \longmapsto ((g_2, g_1) \longmapsto \phi(g_2 g_1)),$$

that is, to the morphism

$$\underline{\text{Hom}}(\mathbf{G}, \mathbf{P}) \longrightarrow \underline{\text{Hom}}(\mathbf{G}_2 \times \mathbf{G}_1, \mathbf{P})$$

⁴⁷Editor's note: The preceding reminder has been added.

⁴⁸Editor's note: The preceding sentence has been added.

⁴⁹Editor's note: The original has been modified so as to introduce 5.2.0.1 and 5.2.0.2, which will be useful in the proof of Theorem 5.3.1.

induced by the multiplication

$$\mu_{\mathbf{G}} : \mathbf{G} \times \mathbf{G} \longrightarrow \mathbf{G}, \quad (g_2, g_1) \longmapsto g_2 g_1.$$

Lemma 5.2.0.2. (i) *The functor E is right adjoint to the forgetful functor*

$$(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}) \longrightarrow (\mathbf{O}\text{-Mod.}).$$

More precisely, $\varepsilon : E \rightarrow \text{Id}$ induces, for every $\mathbf{M} \in (\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ and $\mathbf{P} \in \text{Ob}(\mathbf{O}\text{-Mod.})$, a bijection

$$\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}\text{-Mod.}}(\mathbf{M}, \mathbf{P})$$

functorial in $\mathbf{M}$ and $\mathbf{P}$.

(ii) *Consequently, if $\mathbf{I}$ is an injective object of $(\mathbf{O}\text{-Mod.})$, then $E(\mathbf{I})$ is an injective object of $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$.*

Proof. To every $\mathbf{O}$ -morphism $f : \mathbf{M} \rightarrow \mathbf{P}$, associate the element ϕ_f of $\text{Hom}_{\mathbf{O}}(\mathbf{M}, E(\mathbf{P}))$ defined as follows. For every $S \in \text{Ob}(\mathcal{C})$ and $m \in \mathbf{M}(S)$, the element $\phi_f(m)$ of $\text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $g \in \mathbf{G}(S')$, $S' \rightarrow S$, by

$$\phi_f(m)(g) = f(gm) \in \mathbf{P}(S').$$

Then, for every $h \in \mathbf{G}(S)$, one has $\phi_f(hm) = h \cdot f(m)$; that is, ϕ_f is an element of

$$\text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P})).$$

If $\phi \in \text{Hom}_{\mathbf{G}\text{-}\mathbf{O}\text{-Mod.}}(\mathbf{M}, E(\mathbf{P}))$, and if for every $m \in \mathbf{M}(S)$ we write $f(m) = \phi(m)(1)$, then

$$\phi_f(m)(g) = f(gm) = \phi(gm)(1) = (g \cdot \phi(m))(1) = \phi(m)(g);$$

thus $\phi_f = \phi$. Conversely, it is clear that $\phi_f(m)(1) = f(m)$. This proves (i), and (ii) follows at once.

Definition 5.2.0.3. Let $\mathbf{M}$ be a $\mathbf{G}\text{-}\mathbf{O}$ -module. The identity map of $\mathbf{M}$, considered as an $\mathbf{O}$ -module, corresponds under the adjunction to the morphism of $\mathbf{G}\text{-}\mathbf{O}$ -modules

$$\mu_{\mathbf{M}} : \mathbf{M} \longrightarrow E(\mathbf{M})$$

such that, for every $S \in \text{Ob}(\mathcal{C})$ and $m \in \mathbf{M}(S)$, $\mu_{\mathbf{M}}(m)$ is the morphism $\mathbf{G}_S \rightarrow \mathbf{M}_S$ defined as follows: for every $S' \rightarrow S$ and $g \in \mathbf{G}(S')$,

$$\mu_{\mathbf{M}}(m)(g) = g \cdot m_{S'} \in \mathbf{M}(S').$$

Notice that $\mu_{\mathbf{M}}$ is a monomorphism. Indeed, $\varepsilon_{\mathbf{M}} : E(\mathbf{M}) \rightarrow \mathbf{M}$ is a morphism of $\mathbf{O}$ -modules such that $\varepsilon_{\mathbf{M}} \circ \mu_{\mathbf{M}} = \text{id}_{\mathbf{M}}$. Consequently, as an $\mathbf{O}$ -module, $\mathbf{M}$ is a direct summand of $E(\mathbf{M})$.

Proposition 5.2.1. *Suppose that $\mathcal{C}$ is small, that finite products exist in $\mathcal{C}$, and that $\mathbf{G}$ is representable. Then the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) are the derived functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$) on the category of $\mathbf{G}\text{-}\mathbf{O}$ -modules.*

By the well-known general results,⁵⁰ it suffices to verify that the $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0 . Let

$$0 \longrightarrow \mathbf{F}' \longrightarrow \mathbf{F} \longrightarrow \mathbf{F}'' \longrightarrow 0$$

⁵⁰Editor's note: See [Gr57, 2.2.1] and [Gr57, 2.3]. The original has also been expanded in what follows.

be an exact sequence of $\mathbf{G}$ - $\mathbf{O}$ -modules, and let $S \in \text{Ob}(\mathcal{C})$. By hypothesis, $\mathbf{G}$ is represented by an object $G \in \text{Ob}(\mathcal{C})$, and finite products exist in $\mathcal{C}$; in particular, $\mathcal{C}$ has a final object e . Hence every $\mathbf{G}^n \times \mathbf{h}_S$ is represented by $G^n \times S$ (with $G^0 = e$), and the sequence

$$0 \longrightarrow \mathbf{F}'(G^n \times S) \longrightarrow \mathbf{F}(G^n \times S) \longrightarrow \mathbf{F}''(G^n \times S) \longrightarrow 0$$

is exact. This shows that the sequence of $\mathbf{O}$ -modules

$$0 \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}') \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}) \longrightarrow \underline{C}^n(\mathbf{h}_G, \mathbf{F}'') \longrightarrow 0$$

is exact. It follows that $\underline{C}^*(\mathbf{G}, -)$, regarded as a functor on $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ with values in the category of complexes of $(\mathbf{O}\text{-Mod.})$, is exact. Thus the functors $\underline{H}^n(\mathbf{G}, -)$ are cohomological.

Since Γ is exact, the same is true of the functors $H^n(\mathbf{G}, -)$. It remains only to prove the following.

Lemma 5.2.2. *For every $\mathbf{P} \in \text{Ob}(\mathbf{O}\text{-Mod.})$, one has*

$$\underline{H}^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \quad H^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) = 0, \quad n > 0.$$

It is enough to show that $\underline{C}^*(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$ and $C^*(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$ are homotopically trivial in degrees > 0 . It is enough, in fact, to do this for the second complex, since the corresponding result for the first follows by base change.⁵¹ For every $n \geq 0$, define a morphism

$$\sigma : C^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P})) \longrightarrow C^n(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$$

as follows. Let $f \in C^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$. For every $S \in \text{Ob}(\mathcal{C})$ and $g_1, \dots, g_n \in \mathbf{G}(S)$, the element $\sigma(f)(g_1, \dots, g_n)$ of $\text{Hom}_S(\mathbf{G}_S, \mathbf{P}_S)$ is defined, for every $S' \rightarrow S$ and $x \in \mathbf{G}(S')$, by

$$\sigma(f)(g_1, \dots, g_n)(x) = f(x, g_1, \dots, g_n)(e) \in \mathbf{P}(S'),$$

where e denotes the unit element of $\mathbf{G}(S')$. Then σ is a homotopy operator in degrees > 0 . Indeed, for every $g_1, \dots, g_{n+1} \in \mathbf{G}(S)$ and $x \in \mathbf{G}(S')$, on the one hand,

$$\begin{aligned} \partial\sigma(f)(g_1, \dots, g_{n+1})(x) &= f(xg_1, g_2, \dots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^n (-1)^i f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1})(e) \\ &\quad + (-1)^{n+1} f(x, g_1, \dots, g_n)(e), \end{aligned}$$

and, on the other hand,

$$\begin{aligned} \sigma(\partial f)(g_1, \dots, g_{n+1})(x) &= (xf(g_1, g_2, \dots, g_{n+1}))(e) - f(xg_1, g_2, \dots, g_{n+1})(e) \\ &\quad + \sum_{i=1}^n (-1)^{i+1} f(x, g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+2} f(x, g_1, \dots, g_n)(e). \end{aligned}$$

Consequently,

$$(\partial\sigma(f) + \sigma(\partial f))(g_1, \dots, g_{n+1})(x) = f(g_1, \dots, g_{n+1})(x);$$

that is, $\partial\sigma + \sigma\partial$ is the identity map of $C^{n+1}(\mathbf{G}, \underline{\text{Hom}}(\mathbf{G}, \mathbf{P}))$ for every $n \geq 0$.

Remark 5.2.3. The hypothesis that $\mathcal{C}$ be small is used only to ensure the existence of the derived functors $R^n \underline{H}^0$ and $R^n H^0$. In every case, what precedes shows that the functors $\underline{H}^n(\mathbf{G}, -)$ (respectively, $H^n(\mathbf{G}, -)$) form an effaceable cohomological functor in degrees > 0 , and hence are the (right) satellite functors of the left exact functor $\underline{H}^0(\mathbf{G}, -)$ (respectively, $H^0(\mathbf{G}, -)$), in the sense of [Gr57, 2.2]. If $(\mathbf{G}\text{-}\mathbf{O}\text{-Mod.})$ has enough injective objects (as it does when $\mathcal{C}$ is small), they coincide with the derived functors (*loc. cit.*, 2.3).⁵²

⁵¹Editor's note: The original has been expanded in what follows.

⁵²Editor's note: This remark has been added.

5.3. Cohomology of $G\text{-}\mathcal{O}_S$ -modules

Let S be a scheme, G an S -group, and $\mathcal{F}$ a quasi-coherent $G\text{-}\mathcal{O}_S$ -module. The cohomology groups of G with values in $\mathcal{F}$ are defined by

$$H^n(G, \mathcal{F}) = H^n(\mathbf{h}_G, \mathbf{W}(\mathcal{F}))$$

(for the notation, see 4.6).

Suppose that G is affine over S . In view of Proposition 4.6.4, this cohomology is computed as follows: $H^n(G, \mathcal{F})$ is the n -th cohomology group of the complex $C^*(G, \mathcal{F})$, whose n -th term is

$$C^n(G, \mathcal{F}) = \Gamma\left(S, \mathcal{F} \otimes \underbrace{\mathcal{A}(G) \otimes \cdots \otimes \mathcal{A}(G)}_{n \text{ times}}\right).$$

If f (respectively, a_i) is a section of $\mathcal{F}$ (respectively, of $\mathcal{A}(G)$) over an open subset of S , then

$$\begin{aligned} \partial(f \otimes a_1 \otimes \cdots \otimes a_n) &= \mu_{\mathcal{F}}(f) \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \\ &\quad + \sum_{i=1}^n (-1)^i f \otimes a_1 \otimes \cdots \otimes \Delta a_i \otimes \cdots \otimes a_n \\ &\quad + (-1)^{n+1} f \otimes a_1 \otimes a_2 \otimes \cdots \otimes a_n \otimes 1, \end{aligned}$$

where

$$\Delta : \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes \mathcal{A}(G), \quad \mu_{\mathcal{F}} : \mathcal{F} \longrightarrow \mathcal{F} \otimes \mathcal{A}(G)$$

describe the coalgebra structure of $\mathcal{A}(G)$ and the comodule structure of $\mathcal{F}$. Notice in passing that the cohomology of G with values in $\mathcal{F}$ therefore depends only on the comodule structure of $\mathcal{F}$, and in particular only on the S -monoid structure of G .

In particular,

$$H^0(G, \mathcal{F}) = \Gamma(S, \mathcal{F}^G),$$

where $\mathcal{F}^G$, the *sheaf of invariants of $\mathcal{F}$* , is the sheaf whose sections over an open subset U of S are those sections of $\mathcal{F}$ over U whose inverse image in every S' over U is invariant under $G(S')$ (see 4.7.1.2).

Theorem 5.3.1. *Let S be an affine scheme and G an S -group that is affine and flat over S . The functors $H^n(G, -)$ are the derived functors of $H^0(G, -)$ on the category of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules.*

Proof. Since G is affine and flat over S , 4.7.2.1 shows that the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ is equivalent to the category $(\mathcal{A}(G)\text{-Comod.q.c.})$ of quasi-coherent $\mathcal{A}(G)$ -comodules over $\mathcal{O}_S$, and is therefore abelian. Moreover, since $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, each functor

$$\mathcal{F} \longmapsto \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G)^{\otimes n}$$

is exact; since S is affine, it follows that $C^*(G, -)$ is an exact functor on $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

Let Δ (respectively, η) denote the comultiplication (respectively, the augmentation) of $\mathcal{A}(G)$. For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{P}$, let

$$\text{Ind}(\mathcal{P}) = \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G),$$

⁵³Editor's note: The original has been modified to show that the category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ is abelian and has enough injective objects. Compare [Ja03], Part I, 3.3–3.4, 3.9, 4.2, and 4.14–4.16 (bearing in mind that “ k -group scheme” there means “affine k -group scheme”; see *loc. cit.*, 2.1).

endowed with the $\mathcal{A}(G)$ -comodule structure defined by

$$\mathrm{id}_{\mathcal{P}} \otimes \Delta : \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

This defines a functor

$$\mathrm{Ind} : (\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (G\text{-}\mathcal{O}_S\text{-Mod.q.c.}).$$

It follows from 4.6.4.1, 5.2.0, and 5.2.0.1 that there is an isomorphism of $\mathbf{G}\text{-}\mathbf{O}_S$ -modules

$$\mathbf{W}(\mathrm{Ind}(\mathcal{P})) \simeq E(\mathbf{W}(\mathcal{P})) = \underline{\mathrm{Hom}}(\mathbf{G}, \mathbf{W}(\mathcal{P})). \quad (*)$$

Under this identification, the morphism $\varepsilon : E(\mathbf{W}(\mathcal{P})) \rightarrow \mathbf{W}(\mathcal{P})$ corresponds to the morphism of $\mathcal{O}_S$ -modules

$$\mathrm{id}_{\mathcal{P}} \otimes \eta : \mathrm{Ind}(\mathcal{P}) \longrightarrow \mathcal{P}.$$

We have already used the fact that the functor

$$\mathbf{W} : (\mathcal{O}_S\text{-Mod.}) \longrightarrow (\mathbf{O}_S\text{-Mod.})$$

is fully faithful. By Definition 4.7.1, the same is true of its restriction to $(G\text{-}\mathcal{O}_S\text{-Mod.})$: if $\mathcal{M}$ and $\mathcal{M}'$ are $G\text{-}\mathcal{O}_S$ -modules, there is a functorial isomorphism

$$\mathrm{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathcal{M}, \mathcal{M}') \simeq \mathrm{Hom}_{\mathbf{G}\text{-}\mathbf{O}_S\text{-Mod.}}(\mathbf{W}(\mathcal{M}), \mathbf{W}(\mathcal{M}')).$$

Consequently, Lemma 5.2.0.2 gives the following result.

Corollary 5.3.1.1. *(i) The functor Ind is right adjoint to the forgetful functor*

$$(G\text{-}\mathcal{O}_S\text{-Mod.q.c.}) \longrightarrow (\mathcal{O}_S\text{-Mod.q.c.}).$$

More precisely, the map $\mathrm{id}_{\mathcal{P}} \otimes \eta : \mathrm{Ind}(\mathcal{P}) \rightarrow \mathcal{P}$ induces, for every object $\mathcal{M}$ of $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$, a bijection

$$\mathrm{Hom}_{G\text{-}\mathcal{O}_S\text{-Mod.}}(\mathcal{M}, \mathrm{Ind}(\mathcal{P})) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{M}, \mathcal{P}).$$

(ii) Consequently, if $\mathcal{I}$ is an injective object of $(\mathcal{O}_S\text{-Mod.q.c.})$, then $\mathrm{Ind}(\mathcal{I})$ is an injective object of $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$.

Let $\mathcal{F}$ be a $G\text{-}\mathcal{O}_S$ -module, and let $\mu_{\mathcal{F}} : \mathcal{F} \rightarrow \mathrm{Ind}(\mathcal{F})$ be the map defining its $\mathcal{A}(G)$ -comodule structure. It follows from 5.2.0.3 (or directly from axioms (CM 1) and (CM 2) of 4.7.2) that $\mu_{\mathcal{F}}$ is a morphism of $G\text{-}\mathcal{O}_S$ -modules, and that

$$(\mathrm{id}_{\mathcal{F}} \otimes \eta) \circ \mu_{\mathcal{F}} = \mathrm{id}_{\mathcal{F}}.$$

Thus $\mathcal{F}$ is a direct summand of $\mathrm{Ind}(\mathcal{F})$ as an $\mathcal{O}_S$ -module; in particular, $\mu_{\mathcal{F}}$ is a monomorphism. By (*) and 5.2.2,

$$H^n(\mathbf{G}, \mathbf{W}(\mathrm{Ind}(\mathcal{F}))) \simeq H^n(\mathbf{G}, \underline{\mathrm{Hom}}_S(\mathbf{G}, \mathbf{W}(\mathcal{F}))) = 0 \quad (n > 0).$$

Therefore $H^n(G, -)$ is effaceable for $n > 0$.

Finally, since S is affine, $(\mathcal{O}_S\text{-Mod.q.c.})$ has enough injective objects. Let $\mathcal{F} \hookrightarrow \mathcal{I}$ be a monomorphism of $\mathcal{O}_S$ -modules, where $\mathcal{I}$ is injective in $(\mathcal{O}_S\text{-Mod.q.c.})$. Since $\mathcal{A}(G)$ is flat over $\mathcal{O}_S$, $\mathrm{Ind}(\mathcal{F})$ is a $G\text{-}\mathcal{O}_S$ -submodule of $\mathrm{Ind}(\mathcal{I})$. Hence:

Corollary 5.3.1.2. *The abelian category $(G\text{-}\mathcal{O}_S\text{-Mod.q.c.})$ has enough injective objects.*

Taking account of [Gr57, 2.2.1 and 2.3], already used in the proof of 5.2.1, completes the proof of Theorem 5.3.1.

Remark 5.3.1.3. Corollary 5.3.1.1 may also be proved by the following calculation. To every morphism of $G\text{-}\mathcal{O}_S$ -modules

$$\phi : \mathcal{M} \longrightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

associate the $\mathcal{O}_S$ -morphism

$$(\mathrm{id}_{\mathcal{P}} \otimes \eta) \circ \phi : \mathcal{M} \longrightarrow \mathcal{P}.$$

Conversely, to every $\mathcal{O}_S$ -morphism $f : \mathcal{M} \rightarrow \mathcal{P}$, associate the morphism of $G\text{-}\mathcal{O}_S$ -modules

$$(f \otimes \mathrm{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} : \mathcal{M} \longrightarrow \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

One sees at once that

$$(\mathrm{id}_{\mathcal{P}} \otimes \eta) \circ (f \otimes \mathrm{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} = (f \otimes \mathrm{id}_{\mathcal{O}_S}) \circ (\mathrm{id}_{\mathcal{P}} \otimes \eta) \circ \mu_{\mathcal{M}} = f.$$

On the other hand, for every ϕ , the following diagram is commutative.

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{\phi} & \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \\ \mu_{\mathcal{M}} \downarrow & & \downarrow \mathrm{id}_{\mathcal{P}} \otimes \Delta \\ \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{A}(G) & \xrightarrow{\phi \otimes \mathrm{id}_{\mathcal{A}(G)}} & \mathcal{P} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G). \end{array}$$

Figure 8: The adjunction square for ϕ , the comodule map $\mu_{\mathcal{M}}$, and the comultiplication Δ .

Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local/printed p. 38; combined-reader p. 48.

It follows that

$$\begin{aligned} & \left((\mathrm{id}_{\mathcal{P}} \otimes \eta) \circ \phi \right) \otimes \mathrm{id}_{\mathcal{A}(G)} \circ \mu_{\mathcal{M}} \\ &= (\mathrm{id}_{\mathcal{P}} \otimes \eta \otimes \mathrm{id}_{\mathcal{A}(G)}) \circ (\phi \otimes \mathrm{id}_{\mathcal{A}(G)}) \circ \mu_{\mathcal{M}} \\ &= (\mathrm{id}_{\mathcal{P}} \otimes \eta \otimes \mathrm{id}_{\mathcal{A}(G)}) \circ (\mathrm{id}_{\mathcal{P}} \otimes \Delta) \circ \phi = \phi. \end{aligned}$$

This proves Corollary 5.3.1.1(i), and (ii) follows.

Let $\mathcal{F}$ be a $G\text{-}\mathcal{O}_S$ -module. We saw above that axiom (CM 2) of 4.7.2 shows that, as an $\mathcal{O}_S$ -module, $\mathcal{F}$ is a direct summand of $E(\mathcal{F})$. This gives:

Proposition 5.3.2. *Let S be an affine scheme and G an S -group that is affine and flat. Suppose that every exact sequence*

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \mathcal{F}_3 \longrightarrow 0$$

of quasi-coherent $G\text{-}\mathcal{O}_S$ -modules which splits as a sequence of $\mathcal{O}_S$ -modules also splits as a sequence of $G\text{-}\mathcal{O}_S$ -modules. Then the functors $H^n(G, -)$ vanish for $n > 0$ (or, equivalently, the functor $H^0(G, -)$ is exact).

Indeed, by hypothesis, the sequence of $G\text{-}\mathcal{O}_S$ -modules

$$0 \longrightarrow \mathcal{F} \longrightarrow E(\mathcal{F}) \longrightarrow E(\mathcal{F})/\mathcal{F} \longrightarrow 0$$

is split. Thus $\mathcal{F}$ is a direct summand of $E(\mathcal{F})$ as a $G\text{-}\mathcal{O}_S$ -module, while the cohomology of the latter vanishes.

Combining this with Proposition 4.7.4 immediately gives:

Theorem 5.3.3. *Let S be an affine scheme and G a diagonalizable S -group. For every quasi-coherent $G\text{-}\mathcal{O}_S$ -module $\mathcal{F}$, one has*

$$H^n(G, \mathcal{F}) = 0 \quad (n > 0).$$

Remark. Proposition 5.3.2 remains valid when G is not necessarily flat over S ; the proof then uses relative cohomology.⁵⁴

6. G -equivariant objects and modules

Let $\mathcal{C}$ be a category having a final object and in which fiber products exist.⁵⁵ Let G be a $\mathcal{C}$ -group, $\pi : M \rightarrow X$ a morphism in $\mathcal{C}$, and $\lambda = \lambda_X : G \times X \rightarrow X$ an action of G on X . In what follows, $Y \times_f M$ denotes the fiber product of $\pi : M \rightarrow X$ and an X -functor $f : Y \rightarrow X$.

For every $U \in \text{Ob}(\mathcal{C})$ and $x \in X(U)$, write $M_x = U \times_x M$; thus, for every $\phi : U' \rightarrow U$,

$$M_x(U') = \{m \in M(U') \mid \pi(m) = x_{U'} = \phi^*(x)\}.$$

Finally, if $g \in G(U)$, we also write gx for the element $\lambda(g, x)$ of $X(U)$.

Definition 6.1. (a) *We say that M is a G -equivariant X -object if an action $\Lambda : G \times M \rightarrow M$ of G on M , lifting λ , has been given; that is, if the following square commutes.*

$$\begin{array}{ccc} G \times M & \xrightarrow{\Lambda} & M \\ \text{id}_G \times \pi \downarrow & & \downarrow \pi \\ G \times X & \xrightarrow{\lambda} & X \end{array}$$

Figure 9: The action of G on M lifts its action on X .

Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 39; combined-reader p. 49.

Equivalently, for every morphism $(g, x) : U \rightarrow G \times X$, one is given maps

$$\Lambda_x^U(g) : M_x(U) \longrightarrow M_{gx}(U), \quad m \longmapsto g \cdot m,$$

which satisfy $1 \cdot m = m$ and $g \cdot (h \cdot m) = (gh) \cdot m$, and which are functorial in the $(G \times X)$ -object U . Equivalently again, one is given morphisms of U -objects

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}$$

⁵⁴Editor's note: The editors have not attempted to develop this remark.

⁵⁵Editor's note: This section has been added.

satisfying $\Lambda_x(1) = \text{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$.

(b) Now let $\mathbf{O}$ be a $\widehat{\mathcal{C}}$ -ring and put $\mathbf{O}_X = \mathbf{O} \times X$. Under the hypotheses of (a), we say that M is a G -equivariant $\mathbf{O}_X$ -module if it is an $\mathbf{O}_X$ -module (cf. Definition 4.3.3.1, which is valid for every ring functor on a category $\mathcal{C}$) and if the action Λ is compatible with the $\mathbf{O}_X$ -module structure on M . Thus, for every $(g, x) \in G(U) \times X(U)$, $U \in \text{Ob}(\mathcal{C})$, the map

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \quad m \longmapsto g \cdot m,$$

is a morphism of $\mathbf{O}_U$ -modules.

Remark 6.2. (i) In (a), the conditions $\Lambda_x(1) = \text{id}$ and $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ plainly imply that each $\Lambda_x(g)$ is an isomorphism, with inverse $\Lambda_{gx}(g^{-1})$. Conversely, if each $\Lambda_x(g)$ is an isomorphism, applying the second condition with $h = 1$ gives $\Lambda_x(1) = \text{id}$.

(ii) Let M be an $\mathbf{O}_X$ -module. Giving a morphism $\Lambda : G \times M \rightarrow M$ which makes the diagram of Definition 6.1 commute and for which every morphism

$$\Lambda_x(g) : M_x \longrightarrow M_{gx}, \quad m \longmapsto g \cdot m,$$

is an isomorphism of $\mathbf{O}_U$ -modules is equivalent to giving an isomorphism of $\mathbf{O}_{G \times X}$ -modules

$$\theta : G \times M = (G \times X) \times_{\text{pr}_X} M \xrightarrow{\sim} (G \times X) \times_{\lambda} M, \quad (g, x, m) \longmapsto (g, x, g \cdot m).$$

Since each $\Lambda_x(g)$ is assumed to be an isomorphism, the equality $\Lambda_x(1) = \text{id}$ follows from $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$ with $h = 1$. Consequently, Λ is an action of G on M , that is, $g \cdot (h \cdot m) = (gh) \cdot m$, if and only if the following diagram of $(G \times G \times X)$ -isomorphisms commutes. Here μ denotes multiplication in G , and $f^*(\theta)$ is obtained from θ by base change along $f : G \times G \times X \rightarrow G \times X$.

$$\begin{array}{ccc}
 (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ \text{pr}_{2,3}} \mathbf{M} & \xrightarrow[\sim]{\text{pr}_{2,3}^*(\theta)} & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ \text{pr}_{2,3}} \mathbf{M} \\
 \parallel & & \parallel \\
 (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ (\mu \times \text{id}_{\mathbf{X}})} \mathbf{M} & & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\text{pr}_{\mathbf{X}} \circ (\text{id}_{\mathbf{G}} \times \lambda)} \mathbf{M} \\
 \downarrow (\mu \times \text{id}_{\mathbf{X}})^*(\theta) \wr & & \downarrow (\text{id}_{\mathbf{G}} \times \lambda)^*(\theta) \wr \\
 (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ (\mu \times \text{id}_{\mathbf{X}})} \mathbf{M} & \xlongequal{\quad} & (\mathbf{G} \times \mathbf{G} \times \mathbf{X}) \times_{\lambda \circ (\text{id}_{\mathbf{G}} \times \lambda)} \mathbf{M} .
 \end{array}$$

Figure 10: The cocycle condition for the isomorphism θ .

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 40;
combined-reader p. 50.*

Thus, giving a G -equivariant $\mathbf{O}_X$ -module structure on M is equivalent to giving an isomorphism θ of $\mathbf{O}_{G \times X}$ -modules as above for which this diagram commutes.

(iii) Everything above extends to the case in which G is merely a $\widehat{\mathcal{C}}$ -monoid. In that case, giving an action $\Lambda : G \times M \rightarrow M$ lifting λ , such that every $\Lambda_x(g) : M_x \rightarrow M_{gx}$ is a

morphism of $\mathbf{O}_U$ -modules, is equivalent to giving a morphism θ of $\mathbf{O}_{G \times X}$ -modules as in (ii), for which the diagram above commutes without the isomorphism signs under the arrows and which satisfies

$$p_M \circ \theta \circ (\varepsilon_G \times \text{id}_M) = \text{id}_M,$$

where ε_G is the unit section of G and p_M is the projection onto M .

6.3. G -equivariant morphisms

Let Y be a second object of $\widehat{\mathcal{C}}$, endowed with an action $\lambda_Y : G \times Y \rightarrow Y$, and let N be a second G -equivariant $\mathbf{O}_X$ -module. A morphism $f : Y \rightarrow X$ in $\widehat{\mathcal{C}}$ (respectively, a morphism of $\mathbf{O}_X$ -modules $\phi : M \rightarrow N$) is called *G -equivariant* if it commutes with the action of G . In set-theoretic notation, this says

$$f(g \cdot y) = g \cdot f(y) \quad (\text{respectively, } \phi(g \cdot m) = g \cdot \phi(m)),$$

or, equivalently,

$$f \circ \lambda_Y = \lambda_X \circ (\text{id}_G \times f), \quad \phi \circ \Lambda_M = \Lambda_N \circ (\text{id}_G \times \phi).$$

The following lemma is immediate.

Lemma 6.3.1. *Let $f : Y \rightarrow X$ be a G -equivariant morphism and let M be a G -equivariant $\mathbf{O}_X$ -module. Then the inverse image $f^*(M) = Y \times_f M$ is a G -equivariant $\mathbf{O}_Y$ -module.*

On the other hand, G acts on $\text{Hom}_{\widehat{\mathcal{C}}}(Y, X)$. Indeed, let $T \in \text{Ob}(\mathcal{C})$, $g \in G(T)$, and let ϕ be a T -morphism $Y_T \rightarrow X_T$. Then g defines automorphisms $\lambda_Y(g)$ and $\lambda_X(g)$ of Y_T and X_T , and we write $g \cdot \phi$, or $g\phi g^{-1}$, for the morphism

$$\lambda_X(g) \circ \phi \circ \lambda_Y(g^{-1}).$$

This defines an action of $G(T)$ on $\text{Hom}_T(Y_T, X_T)$, functorially in T .

Definition 6.3.2. *If $\phi : Y \rightarrow X$ is an arbitrary morphism in $\widehat{\mathcal{C}}$, we may therefore consider its stabilizer $\text{Stab}_G(\phi)$ (cf. 2.3.3.1). For every $T \in \text{Ob}(\mathcal{C})$, $\text{Stab}_G(\phi)(T)$ is the subgroup of $G(T)$ formed by those g for which $g \circ \phi_T = \phi_T \circ g$, that is, for which the following diagram commutes.*

$$\begin{array}{ccc} Y_T & \xrightarrow{\phi_T} & X_T \\ g \downarrow & & \downarrow g \\ Y_T & \xrightarrow{\phi_T} & X_T \end{array}$$

Figure 11: The defining square for $\text{Stab}_G(\phi)$.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 41;
combined-reader p. 51.*

Then $\phi : Y \rightarrow X$ is equivariant under $\text{Stab}_G(\phi)$.

6.4. Global sections

Let M be a G -equivariant $\mathbf{O}_X$ -module. Let S_0 denote the final object of $\mathcal{C}$, and denote by $\prod_{X/S_0} M$ the *functor of sections of M over X* (cf. Exposé II, 1.1): to every $T \in \text{Ob}(\mathcal{C})$ it assigns

$$\text{Hom}_X(X_T, M) = \text{Hom}_{X_T}(X_T, M_T) = \Gamma(M_T/X_T).$$

Recall also that every morphism $g : Z \rightarrow Y$ of $\widehat{\mathcal{C}}$ -objects over X induces a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Z),$$

compatible with the ring homomorphism $g^* : \mathbf{O}(Y) \rightarrow \mathbf{O}(Z)$. In particular, when $Z = Y$, so that g is an X -endomorphism of Y , one obtains a morphism of abelian groups

$$M(g) : M(Y) \longrightarrow M(Y)$$

which is not generally a morphism of $\mathbf{O}(Y)$ -modules, but which satisfies

$$M(g)(\alpha \cdot m) = g^*(\alpha) \cdot M(g)(m)$$

for every $m \in M(Y)$ and $\alpha \in \mathbf{O}(Y)$. In what follows, we simply write g^* instead of $M(g)$.

Let $T \in \text{Ob}(\mathcal{C})$, put $X_T = X \times T$, and let $\text{pr}_X : X_T \rightarrow X$ be the projection. For $\alpha \in \mathbf{O}(X_T)$ and $g \in G(T)$, set

$$g(\alpha) = (g^{-1})^*(\alpha).$$

Set-theoretically, this element of $\mathbf{O}(X_T)$ is defined, for every $S \in \text{Ob}(\mathcal{C})$, $x \in X(S)$, and $t \in T(S)$, by

$$g(\alpha)(x, t) = \alpha(g^{-1}x, t).$$

This gives a left action of $G(T)$ by ring automorphisms on $\mathbf{O}(X_T)$, functorially in T , and $g(\alpha) = \alpha$ when α is the image in $\mathbf{O}(X_T)$ of an element of $\mathbf{O}(T)$.

Let ϕ now be the identity morphism of X_T (cf. 6.3 and the generalization below to a morphism $\phi : Y \rightarrow X$). Denote $\text{Hom}_X(X_T, M)$ by $M(\phi g^{-1})$ or $M(\phi)$ according as X_T is regarded as an X -object through $\text{pr}_X \circ \lambda(g^{-1})_T$ or through pr_X , respectively.

Then $\lambda(g^{-1})_T$ is an X -morphism between these two X -objects. Hence there is a morphism of abelian groups

$$(g^{-1})^* : M(\phi) \longrightarrow M(\phi g^{-1}), \quad m \longmapsto m \circ \lambda(g^{-1})_T,$$

and

$$(g^{-1})^*(\alpha \cdot m) = (g\alpha) \cdot (g^{-1})^*(m)$$

for $m \in M(\phi)$ and $\alpha \in \mathbf{O}(X_T)$. If m is a section of M_T over X_T , then $(g^{-1})^*(m)$ is the section defined set-theoretically by $(x, t) \mapsto m(g^{-1}x, t)$. In particular, $(g^{-1})^*$ is a morphism of $\mathbf{O}(T)$ -modules.

The functoriality of the $\mathbf{O}(X_T)$ -module morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{array}{ccc}
\mathbf{M}(\phi) & \xrightarrow{(g^{-1})^*} & \mathbf{M}(\phi g^{-1}) \\
\Lambda_\phi(g) \downarrow & & \downarrow \Lambda_{\phi g^{-1}}(g) \\
\mathbf{M}(g\phi) & \xrightarrow{(g^{-1})^*} & \mathbf{M}(g\phi g^{-1})
\end{array}$$

Figure 12: Naturality of the maps $\Lambda_x(g)$ on sections.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 42;
combined-reader p. 52.*

Moreover, $g\phi g^{-1} = \phi$, because ϕ is the identity. Writing

$$A(g) = (g^{-1})^* \circ \Lambda_\phi(g),$$

we obtain a morphism of abelian groups

$$A(g) : M(\phi) \longrightarrow M(\phi)$$

which is compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$, namely

$$A(g)(\alpha \cdot m) = (g\alpha) \cdot A(g)(m).$$

Finally, if h is a second element of $G(T)$, functoriality of M and of the morphisms $\Lambda_x(g)$ gives the following commutative diagram.

$$\begin{array}{ccccc}
\mathbf{M}(\phi) & & & & \\
\Lambda_\phi(g) \downarrow & \searrow \Lambda_\phi(hg) & & & \\
\mathbf{M}(g\phi) & \xrightarrow{\Lambda_\phi(h)} & \mathbf{M}(hg\phi) & & \\
(g^{-1})^* \downarrow & & (g^{-1})^* \downarrow & \searrow ((hg)^{-1})^* & \\
\mathbf{M}(\phi) & \xrightarrow{\Lambda_\phi(h)} & \mathbf{M}(h\phi) & \xrightarrow{(h^{-1})^*} & \mathbf{M}(\phi)
\end{array}$$

Figure 13: Composition of the induced action on global sections.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 42;
combined-reader p. 52.*

It follows that $A(h) \circ A(g) = A(hg)$, and hence:

Proposition 6.4.1. *For every T , the $\mathbf{O}(X_T)$ -module*

$$\left(\prod_{X/S_0} M \right) (T) = \Gamma(M_T/X_T)$$

is endowed, functorially in T , with an action of $G(T)$ compatible with the action of $G(T)$ on $\mathbf{O}(X_T)$; that is,

$$A(g)(\alpha \cdot m) = g(\alpha) \cdot A(g)(m).$$

Since $g(\alpha) = \alpha$ for $\alpha \in \mathbf{O}(T)$, it follows in particular that $\prod_{X/S_0} M$ is a $G\text{-}\mathbf{O}_{S_0}$ -module.

More generally, let Y be a second G -equivariant $\widehat{\mathcal{C}}$ -object, let $\phi : Y \rightarrow X$ be a morphism in $\widehat{\mathcal{C}}$, and put $H = \text{Stab}_G(\phi)$, as in 6.3. Then

$$M_\phi = Y \times_\phi M$$

is an H -equivariant $\mathbf{O}_Y$ -module. We obtain:

Corollary 6.4.2. *The functor $\prod_{Y/S_0} M_\phi$ is a $\text{Stab}_G(\phi)\text{-}\mathbf{O}_{S_0}$ -module.*

6.5. G -equivariant $\mathcal{O}_X$ -modules

Apply the preceding discussion in the following situation. Let S be a scheme, let G be an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be an $\mathcal{O}_X$ -module, not necessarily quasi-coherent.

Definition 6.5.1. *We say that $\mathcal{F}$ is a G -equivariant $\mathcal{O}_X$ -module if the $\mathbf{O}_X$ -module $M = W(\mathcal{F})$ is G -equivariant. Thus, for every morphism $(g, x) : U \rightarrow G \times_S X$, one is given isomorphisms of $\mathcal{O}_U$ -modules*

$$\Lambda_x(g) : W(x^*(\mathcal{F})) \xrightarrow{\sim} W((gx)^*(\mathcal{F})),$$

functorial in U and satisfying $\Lambda_{hx}(g) \circ \Lambda_x(h) = \Lambda_x(gh)$. Because W is fully faithful (cf. 4.6.1.1 and 4.6.2), these yield isomorphisms of $\mathcal{O}_U$ -modules

$$\Lambda_x(g) : x^*(\mathcal{F}) \xrightarrow{\sim} (gx)^*(\mathcal{F}),$$

where gx denotes $\lambda \circ (g, x) : U \rightarrow X$. Applying this to the identity morphism of $G \times_S X$ gives an isomorphism of $\mathcal{O}_{G \times_S X}$ -modules

$$(\star) \quad \theta : \text{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

such that the following diagram $(\star\star)$ of morphisms of sheaves on $G \times_S G \times_S X$ commutes.

$$\begin{array}{ccccc}
 \text{pr}_{2,3}^* \circ \text{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{\text{pr}_{2,3}^*(\theta)} & \text{pr}_{2,3}^* \circ \lambda^*(\mathcal{F}) & \xlongequal{\quad} & (\text{id}_G \times \lambda)^* \circ \text{pr}_X^*(\mathcal{F}) \\
 \parallel & & (\star\star) & & \downarrow \wr (\text{id}_G \times \lambda)^*(\theta) \\
 (\mu \times \text{id}_X)^* \circ \text{pr}_X^*(\mathcal{F}) & \xrightarrow[\sim]{(\mu \times \text{id}_X)^*(\theta)} & (\mu \times \text{id}_X)^* \circ \lambda^*(\mathcal{F}) & \xlongequal{\quad} & (\text{id}_G \times \lambda)^* \circ \lambda^*(\mathcal{F}) .
 \end{array}$$

Figure 14: The cocycle condition $(\star\star)$ for a G -equivariant $\mathcal{O}_X$ -module.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 43;
combined-reader p. 53.*

If $\mathcal{E}$ is a second G -equivariant $\mathcal{O}_X$ -module, a morphism $\phi : \mathcal{E} \rightarrow \mathcal{F}$ of $\mathcal{O}_X$ -modules is called G -equivariant if $W(\phi) : W(\mathcal{E}) \rightarrow W(\mathcal{F})$ is G -equivariant. With the notation above, this is equivalent to

$$\theta_{\mathcal{F}} \circ \text{pr}_X^*(\phi) = \lambda^*(\phi) \circ \theta_{\mathcal{E}}.$$

Remark 6.5.2. If $f : Y \rightarrow X$ is G -equivariant, Lemma 6.3.1 shows that $f^*(\mathcal{F})$ is a G -equivariant $\mathcal{O}_Y$ -module.

Remark 6.5.3. Suppose in addition that $\mathcal{F}$ is quasi-coherent. Giving a G -equivariant $\mathcal{O}_X$ -module structure on $M = W(\mathcal{F})$ is then equivalent to giving such a structure on $V(\mathcal{F})$, and this in turn is equivalent to giving an action of G on the vector fibration $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

Indeed, let

$$\varphi : \lambda^*(\mathcal{F}) \xrightarrow{\sim} \mathrm{pr}_X^*(\mathcal{F})$$

be the inverse of θ . For every morphism $(g, x) : U \rightarrow G \times_S X$, the isomorphism of $\mathcal{O}_U$ -modules

$$\varphi_x(g) = \Lambda_x(g)^{-1} = \Lambda_{gx}(g^{-1})$$

from $(gx)^*(\mathcal{F})$ to $x^*(\mathcal{F})$ induces an isomorphism of $\mathcal{O}_U$ -modules

$$V((gx)^*(\mathcal{F})) \xrightarrow{\sim} V(x^*(\mathcal{F})).$$

Denote the latter by ${}^t\varphi_x(g)$. Then

$${}^t\varphi_{gx}(h) \circ {}^t\varphi_x(g) = {}^t(\Lambda_{gx}(h) \circ \varphi_x(g))^{-1} = \Lambda_x(hg)^{-1} = {}^t\varphi_x(hg).$$

For every X -scheme $f : Y \rightarrow X$, one has

$$V(\mathcal{F}) \times_f Y \simeq V(f^*(\mathcal{F})) \quad (\text{cf. 4.6.2}).$$

Therefore the isomorphism

$$\begin{aligned} {}^t\varphi : G \times_S V(\mathcal{F}) &= (G \times_S X) \times_{\mathrm{pr}_X} V(\mathcal{F}) = V(\mathrm{pr}_X^*(\mathcal{F})) \\ &\xrightarrow{\sim} V(\lambda^*(\mathcal{F})) = (G \times_S X) \times_{\lambda} V(\mathcal{F}) \end{aligned}$$

endows $V(\mathcal{F})$ with a G -equivariant $\mathcal{O}_X$ -module structure. Identifying every functor $V(-)$ with the vector fibration which represents it, and composing ${}^t\varphi$ with the projection onto $V(\mathcal{F})$, gives an action of G on $V(\mathcal{F})$, compatible with the action on X and linear on the fibers.

This recovers the definition in [GIT, Chapter 1, Section 3], except that Mumford there considers a locally free $\mathcal{O}_X$ -module $\mathcal{E}$ of finite rank and defines an action of G on $V(\mathcal{E}) = W(\mathcal{E}^\vee)$. Indeed, the preceding diagram $(\star\star)$, with θ replaced by φ and the directions of the arrows reversed, is exactly the diagram in *loc. cit.*, Definition 1.6, and the isomorphism ${}^t\varphi$ above coincides with the isomorphism Φ of *loc. cit.*, p. 31.

Remark 6.5.4. Consider in particular the case $X = S$, with the trivial action of G . A G -equivariant $\mathcal{O}_S$ -module $\mathcal{F}$ is then the same thing as a G - $\mathcal{O}_S$ -module (cf. 4.7.1). If $f : G \rightarrow S$ denotes the morphism which here equals both pr_X and λ , the isomorphism

$$(\star) \quad \theta : f^*(\mathcal{F}) \xrightarrow{\sim} f^*(\mathcal{F})$$

is an element of

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{F}), f^*(\mathcal{F})) &= \mathrm{Hom}_{\mathcal{O}_G}(W(f^*(\mathcal{F})), W(f^*(\mathcal{F}))) \\ &= \mathrm{End}_{\mathcal{O}_S}(W(\mathcal{F}))(G). \end{aligned}$$

It is precisely the morphism

$$\rho : G \longrightarrow \mathrm{End}_{\mathcal{O}_S}(W(\mathcal{F}))$$

which defines the action of G on $W(\mathcal{F})$. Moreover, θ corresponds by adjunction to the morphism of $\mathcal{O}_S$ -modules

$$f_*(\theta) \circ \tau : \mathcal{F} \longrightarrow f_*f^*(\mathcal{F}),$$

where $\tau : \mathcal{F} \rightarrow f_*f^*(\mathcal{F})$ is the unit morphism of the adjunction. This will be used in Exposé VI_B, 11.10 bis.

6.6. The functors $\prod_{X/S} W(\mathcal{F})$ and $W(p_*(\mathcal{F}))$

Let S be a scheme, let G be an S -group scheme acting on S -schemes X and Y through

$$\lambda : G \times_S X \rightarrow X, \quad \mu : G \times_S Y \rightarrow Y,$$

and let $p : X \rightarrow Y$ be a G -equivariant morphism. Suppose that $p : X \rightarrow Y$ and $\nu_Y : Y \rightarrow S$ are quasi-compact and quasi-separated and that $\pi : G \rightarrow S$ is flat.

The projection $\text{pr}_Y : G \times_S Y \rightarrow Y$ is then flat, as is μ , since μ is the composite of pr_Y and the automorphism $(g, y) \mapsto (g, gy)$. For a variable S -scheme $f : T \rightarrow S$, denote by $p_T : X_T \rightarrow T$ and $f_X : X_T \rightarrow X$ the morphisms obtained from p and f by base change.

Let $\mathcal{F}$ be a quasi-coherent G -equivariant $\mathcal{O}_X$ -module, and let

$$\theta : \text{pr}_X^*(\mathcal{F}) \xrightarrow{\sim} \lambda^*(\mathcal{F})$$

be the isomorphism of 6.5.1 ($\star$). Since p is quasi-compact and quasi-separated, $p_*(\mathcal{F})$ is quasi-coherent (cf. EGA I, 9.2.1).

Lemma 6.6.1. *The quasi-coherent $\mathcal{O}_Y$ -module $p_*(\mathcal{F})$ is G -equivariant.*

Indeed, one has the following two cartesian squares.

$$\begin{array}{ccccc} G \times_S X & \xrightarrow{\text{pr}_X} & X & \xleftarrow{\lambda} & G \times_S X \\ \downarrow q & & \downarrow p & & \downarrow q \\ G \times_S Y & \xrightarrow{\text{pr}_Y} & Y & \xleftarrow{\mu} & G \times_S Y. \end{array}$$

Figure 15: The two cartesian base-change squares used in Lemma 6.6.1.

*Temporary Loop-1 image; source locator: Polo-Gille Exp1-13oct24.pdf, component-local p. 45;
combined-reader p. 55.*

Since p is quasi-compact and quasi-separated and pr_Y and μ are flat, EGA III, 1.4.15, completed by EGA IV₁, 1.7.21, gives

$$q_* \text{pr}_X^*(\mathcal{F}) = \text{pr}_Y^* p_*(\mathcal{F}), \quad q_* \lambda^*(\mathcal{F}) = \mu^* p_*(\mathcal{F}).$$

Consequently, θ induces an isomorphism

$$\theta_Y : \text{pr}_Y^* p_*(\mathcal{F}) \xrightarrow{\sim} \mu^* p_*(\mathcal{F}).$$

The same argument shows that θ_Y satisfies the cocycle condition 6.5.1 ($\star\star$), because the morphisms $G \times_S G \times_S Y \rightarrow G \times_S Y$ involved are flat. This proves the lemma.

In particular, take $Y = S$ with the trivial action of G . By Remark 6.5.4, $p_*(\mathcal{F})$ is then a $G\text{-}\mathcal{O}_S$ -module. If, in addition, G is affine over S , and if

$$\mathcal{A}(G) = \pi_*(\mathcal{O}_G),$$

then $p_*(\mathcal{F})$ is an $\mathcal{A}(G)$ -comodule, by 4.7.2.

On the other hand, by Proposition 6.4.1, the functor $\prod_{X/S} W(\mathcal{F})$, which assigns to every $f : T \rightarrow S$

$$W(f_X^*(\mathcal{F}))(X_T) = \Gamma(X_T, f_X^*(\mathcal{F})) = \Gamma(T, p_{T*} f_X^*(\mathcal{F})),$$

is a $G\text{-}\mathcal{O}_S$ -module. There is moreover a canonical morphism

$$\tau : W(p_*(\mathcal{F})) \longrightarrow \prod_{X/S} W(\mathcal{F}),$$

which, for each $f : T \rightarrow S$, is given by the canonical morphism

$$\Gamma(T, f^*p_*(\mathcal{F})) \longrightarrow \Gamma(T, p_{T*}f_X^*(\mathcal{F})).$$

It is an isomorphism when restricted to the full subcategory of schemes T flat over S , and it is immediate that τ is a morphism of $G\text{-}\mathcal{O}_S$ -modules. We have therefore proved the following proposition; for part (ii), compare [GIT], p. 32.

Proposition 6.6.2. *Let S be a scheme, let G be an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Suppose that $\pi : G \rightarrow S$ is flat and $p : X \rightarrow S$ is quasi-compact and quasi-separated.*

(i) *Then $p_*(\mathcal{F})$ is a quasi-coherent $G\text{-}\mathcal{O}_S$ -module. The canonical morphism*

$$W(p_*(\mathcal{F})) \longrightarrow \prod_{X/S} W(\mathcal{F})$$

is a morphism of $G\text{-}\mathcal{O}_S$ -modules, and the two functors coincide on the category of flat S -schemes.

(ii) *If, in addition, G is affine over S , and if $\mathcal{A}(G) = \pi_*(\mathcal{O}_G)$, then $p_*(\mathcal{F})$ has the structure of an $\mathcal{A}(G)$ -comodule.⁵⁶*

6.7. Stabilizers

Let S be a scheme, let G be an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Let Y be a second S -scheme with an action of G , possibly the trivial action, let $\phi : Y \rightarrow X$ be an S -morphism, not necessarily G -equivariant, and let $\text{Stab}_G(\phi)$ be the stabilizer of ϕ in G (cf. 6.3.2).

Let us point out at once (see Exposé VI_B, Section 6) that $\text{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S (cf. *loc. cit.*, Definition 6.2.1). Indeed, consider

$$r : G \times_S Y \longrightarrow X \times_S X, \quad r(g, y) = (\phi(y), g\phi(g^{-1}y)),$$

put $P = G \times_S Y$, and let P' be the inverse image under r of the diagonal $\Delta_{X/S}$. Then (cf. *loc. cit.*, 6.2.4(a))

$$\text{Stab}_G(\phi) = \prod_{P/G} P'.$$

It follows that $\text{Stab}_G(\phi)$ is representable by a closed subgroup scheme H of G if X is separated over S and Y is essentially free over S . The latter condition is automatic if S is the spectrum of a field or if $Y = S$. Under these hypotheses, $\phi^*(\mathcal{F})$ is a quasi-coherent H -equivariant $\mathcal{O}_Y$ -module (cf. 6.5.2).

If, in addition, $\pi : G \rightarrow S$ and $p : Y \rightarrow S$ are quasi-compact and quasi-separated over S , and π is flat, then $p_*\phi^*(\mathcal{F})$ is an $H\text{-}\mathcal{O}_S$ -module by Proposition 6.6.2. In particular:

Corollary 6.7.1. *Let S be a scheme, let G be an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be a quasi-coherent G -equivariant $\mathcal{O}_X$ -module. Suppose that $\pi : G \rightarrow S$ is flat and $X \rightarrow S$ is quasi-compact and separated. Let $\tau : S \rightarrow X$ be a section of X over S .*

⁵⁶Editor's note: The same holds if G is flat, quasi-compact, and quasi-separated over S , and if $p_*(\mathcal{F})$ is a flat $\mathcal{O}_S$ -module; cf. Exposé VI_B, 11.6.1(ii).

(i) The stabilizer $H = \text{Stab}_G(\tau)$ is a closed subgroup scheme of G , and $\tau^*(\mathcal{F})$ is a quasi-coherent H - $\mathcal{O}_S$ -module.

(ii) If H is moreover affine over S , for example if G is, and if $\mathcal{A}(H) = \pi_*(\mathcal{O}_H)$, then $\tau^*(\mathcal{F})$ is an $\mathcal{A}(H)$ -comodule.

6.8. G -equivariant sheaves on G

To conclude, we give two results, 6.8.1 and 6.8.6, which will be used in Exposés II and III, in particular in III, 4.25.

Proposition 6.8.1. *Let S be a scheme, let $\pi : G \rightarrow S$ be an S -group scheme, and let $\varepsilon : S \rightarrow G$ be the unit section. Consider the action of G on itself by left translations. The functors*

$$\mathcal{E} \longmapsto \pi^*(\mathcal{E}), \quad \mathcal{F} \longmapsto \varepsilon^*(\mathcal{F})$$

induce mutually quasi-inverse equivalences between the category of quasi-coherent $\mathcal{O}_S$ -modules and the category of quasi-coherent G -equivariant $\mathcal{O}_G$ -modules.

Proof. Let μ denote multiplication in G , and let $\text{pr}_2 : G \times_S G \rightarrow G$ be the second projection. Since $\pi \circ \mu = \pi \circ \text{pr}_2$, every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$ has a canonical isomorphism

$$\mu^* \pi^*(\mathcal{E}) = \text{pr}_2^* \pi^*(\mathcal{E}).$$

It is immediate that this isomorphism satisfies the cocycle condition 6.5.1 ($\star\star$), so $\pi^*(\mathcal{E})$ is a G -equivariant $\mathcal{O}_G$ -module. Since $\varepsilon^* \pi^*(\mathcal{E}) = \mathcal{E}$, the functor $\mathcal{E} \mapsto \pi^*(\mathcal{E})$ is

fully faithful. It remains to show that, for every G -equivariant $\mathcal{O}_G$ -module $\mathcal{F}$,

$$\mathcal{F} \simeq \pi^* \varepsilon^*(\mathcal{F}).$$

By hypothesis, there is an isomorphism

$$\theta : \text{pr}_2^*(\mathcal{F}) \xrightarrow{\sim} \mu^*(\mathcal{F}).$$

Pulling θ back along the morphism

$$\tau : G \longrightarrow G \times_S G, \quad \tau = (\text{id}_G, \varepsilon \circ \pi),$$

gives an isomorphism $\mathcal{F} \xrightarrow{\sim} \pi^* \varepsilon^*(\mathcal{F})$.

Remarks 6.8.2. (a) Consider the action of $G \times_S G$ on G defined by

$$(g_1, g_2) \cdot g = g_1 g g_2^{-1}.$$

The stabilizer of the unit section $\varepsilon : S \rightarrow G$ is the diagonal subgroup H of $G \times_S G$. Consequently, if $\mathcal{F}$ is a quasi-coherent $(G \times_S G)$ -equivariant $\mathcal{O}_G$ -module, then $\mathcal{E} = \varepsilon^*(\mathcal{F})$ has the structure of an H - $\mathcal{O}_S$ -module, by 6.5.2.

(b) One can show that $\mathcal{F} \mapsto \varepsilon^*(\mathcal{F})$ is an equivalence between the category of quasi-coherent $(G \times_S G)$ -equivariant $\mathcal{O}_G$ -modules and the category of quasi-coherent H - $\mathcal{O}_S$ -modules. This is a special case of more general descent results (cf. Exposé IV, Section 2, and SGA 1, VIII); see, for example, [Th87], 1.2–1.3.

Remark 6.8.3. Retain the notation of 6.8.1. For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$, let $\pi_*^G \pi^*(\mathcal{E})$ be the $\mathcal{O}_S$ -submodule of $\pi_* \pi^*(\mathcal{E})$ whose sections over an open subset V of S are the elements

$$\gamma \in \Gamma(\pi^{-1}(V), \pi^*(\mathcal{E}))$$

such that $g \cdot \gamma_{S'} = \gamma_{S'}$ for every $S' \rightarrow V$ and $g \in G(S')$. Then the natural morphism

$$\mathcal{E} \longrightarrow \pi_*^G \pi^*(\mathcal{E})$$

is an isomorphism. This is immediate if

$$\pi_* \pi^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_S} \pi_*(\mathcal{O}_G),$$

for example if $G \rightarrow S$ is affine, or if $G \rightarrow S$ is quasi-compact and quasi-separated and $\mathcal{E}$ is flat; the general case is verified without difficulty.

Remark 6.8.4. Let S be a scheme, let H be an S -group scheme acting on an S -scheme X , and let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. Suppose H is flat over S , and let $W_P(\mathcal{F})$ denote the restriction of $W(\mathcal{F})$ to the full subcategory of flat S -schemes. By 6.5.1, endowing $\mathcal{F}$ with an H -equivariant module structure is equivalent to giving an isomorphism

$$\theta : \mathrm{pr}_X^*(\mathcal{F}) \longrightarrow \lambda^*(\mathcal{F})$$

of sheaves on $H \times_S X$ which satisfies the cocycle condition ($\star\star$). Hence, to endow $\mathcal{F}$ with an H -equivariant module structure, it suffices to endow $W_P(\mathcal{F})$ with such a structure.

Recollection 6.8.5. Let X be an S -scheme and Y a closed S -subscheme of X . Let $\mathcal{N}_{Y/X}$ denote the conormal sheaf of the immersion $i : Y \hookrightarrow X$ (cf. EGA IV₄, 16.1.2). If $S' \rightarrow S$ is flat and $i' : Y' \hookrightarrow X'$ is obtained from i by base change, then *loc. cit.*, 16.2.2(iii), gives

$$\mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'} = \mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}.$$

Proposition 6.8.6. *Let S be a scheme, let X be an S -group scheme, and let Y be an S -subgroup scheme of X . Suppose that Y is flat over S . Then the conormal sheaf $\mathcal{N}_{Y/X}$ is a $(Y \times_S Y)$ -equivariant $\mathcal{O}_Y$ -module.*

Indeed, $H = Y \times_S Y$ is flat over S . By Remark 6.8.4, it is therefore enough to endow $W_P(\mathcal{N}_{Y/X})$ with an H -equivariant module structure. Let S' be a flat S -scheme, let $Y' \hookrightarrow X'$ be the immersion obtained by base change, and put

$$\mathcal{N}' = \mathcal{N}_{Y'/X'} \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'}.$$

By 6.8.5, $\mathcal{N}' = \mathcal{N}_{Y'/X'}$. Every $h \in H(S')$ induces an automorphism of Y' , and hence, for every $y \in Y'(S')$, isomorphisms

$$\Gamma(S', y^*(\mathcal{N}')) \xrightarrow{\sim} \Gamma(S', y^* h^*(\mathcal{N}'))$$

which endow $W_P(\mathcal{N}_{Y/X})$ with an H -equivariant module structure (cf. 6.1).

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⁵⁷Editor's note: These are additional references cited in this Exposé.

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Exposé II

Tangent Bundles – Lie Algebras

by *M. Demazure*

We propose in this exposé to construct, in the theory of schemes, the analogues of the tangent bundles and Lie algebras of classical theory. It will, however, be useful not to restrict ourselves to schemes proper, but also to consider certain functors on the category of schemes that are not necessarily representable (for example, the functors Hom, Norm, and so forth). As announced in the preceding exposé (cf. I 1.1), we shall identify a scheme with its associated functor.⁰

On the other hand, the constructions presented below go beyond the setting of scheme theory. They are equally valid, for example, in the theory of analytic spaces with nilpotent elements, apart from a few changes of detail.

Before beginning this construction, we must give a few general definitions that supplement those of I 1.7.

1. The functors $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$

We resume the notation of I 1.1. We identify the category $\mathcal{C}$ with a full subcategory of

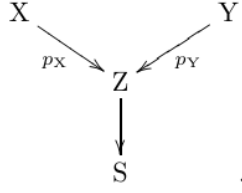
$$\widehat{\mathcal{C}} = \mathrm{Hom}(\mathcal{C}^\circ, \mathbf{Set})$$

(in particular, we suppress the underlinings that allowed us to distinguish graphically an object of $\widehat{\mathcal{C}}$ from an object of $\mathcal{C}$).¹

Consider the following situation: four objects of $\widehat{\mathcal{C}}$, denoted by S, X, Y, Z , the first of which is in fact an object of $\mathcal{C}$, with X and Y over Z , and Z over S .

⁰Editor's note: version of October 14, 2024.

¹Editor's note: in this edition these are rendered by bold letters **F, G, V, W**; cf. Exposé I. For functors such as Norm and Centr (cf. 5.2), however, the underlinings of the original have been retained, and one has been added to the functor Lie, in order to distinguish the functor Lie(G/S) from the $\Gamma(S, \mathcal{O}_S)$ -Lie algebra $\mathrm{Lie}(G/S) = \underline{\mathrm{Lie}}(G/S)(S)$, used, for example, in Exposé VIIA.

Figure 16: The objects X and Y over Z , and Z over S .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 2;
combined-reader p. 60.*

Definition 1.1.

We define an object of $\widehat{\mathcal{C}}_S$, denoted $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$, by

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}}(X_{S'}, Y_{S'}) = \mathrm{Hom}_Z(X \times_S S', Y), \quad (1)$$

for every object S' of $\mathcal{C}_S$. One sees at once that $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ is simply the subobject of $\underline{\mathrm{Hom}}_S(X, Y)$ formed by the morphisms compatible with p_X and p_Y ; that is, the kernel of the pair of morphisms

$$\underline{\mathrm{Hom}}_S(X, Y) \rightrightarrows \underline{\mathrm{Hom}}_S(X, Z),$$

the first defined by composition with p_Y , and the second being the constant morphism whose “image” is p_X .

As in I 1.7, for every object T of $\widehat{\mathcal{C}}$ over S , there is a natural bijection

$$\mathrm{Hom}_S(T, \underline{\mathrm{Hom}}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_Z(X \times_S T, Y). \quad (2)$$

Moreover, by I 1.7.1, if E, F are objects of $\widehat{\mathcal{C}}$ over Z , then

$$\mathrm{Hom}_Z(E, \underline{\mathrm{Hom}}_Z(F, Y)) \simeq \mathrm{Hom}_Z(E \times_Z F, Y) \simeq \mathrm{Hom}_Z(F, \underline{\mathrm{Hom}}_Z(E, Y)).$$

Applying this with $E = X$ and $F = Z \times_S T$, one obtains natural bijections, for every object T of $\widehat{\mathcal{C}}_S$,

$$\begin{aligned}
 \mathrm{Hom}_S(T, \underline{\mathrm{Hom}}_{Z/S}(X, Y)) &\simeq \mathrm{Hom}_Z(X \times_S T, Y) \\
 &\simeq \begin{cases} \mathrm{Hom}_Z(Z \times_S T, \underline{\mathrm{Hom}}_Z(X, Y)), \\ \mathrm{Hom}_Z(X, \underline{\mathrm{Hom}}_Z(Z \times_S T, Y)). \end{cases}
 \end{aligned} \quad (3)$$

These bijections are functorial in T , and hence give isomorphisms of S -functors.²

$$\begin{array}{ccc}
 \underline{\mathrm{Hom}}_S(T, \underline{\mathrm{Hom}}_{Z/S}(X, Y)) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(Z \times_S T, Y)) \\
 & \searrow \simeq & \nearrow \simeq \\
 & \underline{\mathrm{Hom}}_{Z/S}(X \times_S T, Y) &
 \end{array} \quad (4)$$

Figure 17: The functorial isomorphisms (4).

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 2;
combined-reader p. 60.*

²Editor’s note: points (2), (3), and (4) have been expanded; in particular, (4) will be used in 3.11.

Let us point out two special cases of the definition. If $Z = S$, then

$$\underline{\mathrm{Hom}}_{S/S}(X, Y) = \underline{\mathrm{Hom}}_S(X, Y).$$

On the other hand, when $X = Z$, we set

$$\prod_{Z/S} Y = \underline{\mathrm{Hom}}_{Z/S}(Z, Y), \quad (5)$$

so that, by definition,

$$\left(\prod_{Z/S} Y \right) (S') = \mathrm{Hom}_Z(Z \times_S S', Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

The functor $\prod_{Z/S} : \widehat{\mathcal{C}}_{/Z} \rightarrow \widehat{\mathcal{C}}_{/S}$ is right adjoint to the base-change functor from S to Z : for every S -functor U ,

$$\mathrm{Hom}_S \left(U, \prod_{Z/S} Y \right) = \mathrm{Hom}_Z(U \times_S Z, Y).$$

If $\mathcal{C} = \mathbf{Sch}$ and Z is an S -scheme, the functor $\prod_{Z/S}$ is called the “Weil restriction of scalars.”

³ One also has an isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y) \simeq \underline{\mathrm{Hom}}_{X/S}(X, Y \times_Z X) = \prod_{X/S} (Y \times_Z X), \quad (6)$$

which gives in particular, when $Z = S$, an isomorphism

$$\underline{\mathrm{Hom}}_S(X, Y) \simeq \prod_{X/S} Y_X. \quad (7)$$

Remark 1.2.

The functor $Y \mapsto \underline{\mathrm{Hom}}_{Z/S}(X, Y)$ commutes with products in the following sense: there is a functorial isomorphism

$$\underline{\mathrm{Hom}}_{Z/S}(X, Y \times_Z Y') \simeq \underline{\mathrm{Hom}}_{Z/S}(X, Y) \times_S \underline{\mathrm{Hom}}_{Z/S}(X, Y'). \quad (*)$$

It follows that if Y is a Z -group, respectively a Z -ring, and so on, then $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ is an S -group, respectively an S -ring, and so on.

Remark 1.3.

Let $\pi : M \rightarrow Y$ be a Y -functor in $\mathcal{O}_Y$ -modules (cf. I 4.3.3.1), and set $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$. Then $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ has a natural $\mathcal{O}_H$ -module structure; more precisely, for every H' over H , the set

$$\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$$

has a natural $\mathcal{O}(X \times_S H')$ -module structure. ⁴

Indeed, let $m : M \times_Y M \rightarrow M$ and $\lambda : \mathcal{O}_Y \times_Y M \rightarrow M$ denote the morphisms defining the structures of a Y -abelian group and an $\mathcal{O}_Y$ -module. Let H' be an S -scheme over $H =$

³Editor's note: the preceding two sentences have been added.

⁴Editor's note: this remark has been added; it is useful for Corollary 3.11.1.

$\underline{\mathrm{Hom}}_{Z/S}(X, Y)$; equivalently, one has been given a Z -morphism $f : X \times_S H' \rightarrow Y$, which makes $X \times_S H'$ a Y -object. Then

$$\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$$

is the set of Z -morphisms $\phi : X \times_S H' \rightarrow M$ such that $\pi \circ \phi = f$, that is, the set of Y -morphisms $X \times_S H' \rightarrow M$.

If ϕ, ψ are two such morphisms, define $\phi + \psi$ to be the composite Y -morphism

$$X \times_S H' \xrightarrow{\phi \times \psi} M \times_Y M \xrightarrow{m} M.$$

This makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ an abelian group over $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$.⁵

Similarly, if $a \in \mathcal{O}(X \times_S H')$, that is, if $a : X \times_S H' \rightarrow \mathcal{O}_S$ is an S -morphism, define $a\phi$ to be the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ is the Y -morphism from $X \times_S H'$ to $\mathcal{O}_Y \times_Y M \simeq \mathcal{O}_S \times_S M$ with components a and ϕ . This gives $\mathrm{Hom}_H(H', \underline{\mathrm{Hom}}_{Z/S}(X, M))$ an $\mathcal{O}(X \times_S H')$ -module structure, functorial in the H -object H' .

2. The schemes $I_S(\mathcal{M})$

Definition 2.1.

Let S be a scheme and $\mathcal{M}$ a quasi-coherent $\mathcal{O}_S$ -module. Denote by $\mathcal{D}_{\mathcal{O}_S}(\mathcal{M})$ the quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{O}_S \oplus \mathcal{M}$, where $\mathcal{M}$ is regarded as an ideal of square zero. Denote by

$$I_S(\mathcal{M}) = \mathrm{Spec} \mathcal{D}_{\mathcal{O}_S}(\mathcal{M})$$

the corresponding S -scheme.⁶ In particular, put

$$\mathcal{D}_{\mathcal{O}_S} = \mathcal{D}_{\mathcal{O}_S}(\mathcal{O}_S), \quad I_S = I_S(\mathcal{O}_S),$$

and call these, respectively, the algebra of dual numbers over S and the scheme of dual numbers over S .

The assignment $\mathcal{M} \mapsto I_S(\mathcal{M})$ is a contravariant functor from quasi-coherent $\mathcal{O}_S$ -modules to S -schemes. In particular, the morphisms $0 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow 0$ define, respectively, the structural morphism

$$\rho : I_S(\mathcal{M}) \longrightarrow I_S(0) = S$$

and a section $\varepsilon_{\mathcal{M}}$ of it, called the *zero section*.⁷

2.1.1.

Since $\mathcal{M} \mapsto I_S(\mathcal{M})$ is contravariant, every $a \in \mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ defines an S -endomorphism a^* of $I_S(\mathcal{M})$, and

$$1^* = \mathrm{id}, \quad (ab)^* = b^* \circ a^*, \quad 0^* = \varepsilon_{\mathcal{M}} \circ \rho, \quad a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}.$$

Consequently, $I_S(\mathcal{M})$ carries a right action of the multiplicative monoid of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$, commuting with the S -morphisms $I_S(\mathcal{M}) \rightarrow I_S(\mathcal{M}')$ arising from morphisms $\mathcal{M}' \rightarrow \mathcal{M}$; in particular, the operators a^* preserve the zero section.⁸

⁵Editor's note: if M is a Y -group, not necessarily abelian, and one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\underline{\mathrm{Hom}}_{Z/S}(X, M)$ a group over $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$.

⁶Editor's note: the underlying space of $I_S(\mathcal{M})$ is the same as that of S .

⁷Editor's note: compare 3.1.1.

⁸Editor's note: what follows expands the original; the numbering 2.1.1–2.1.3 has also been added.

For every $a \in \text{End}_{\mathcal{O}_S}(\mathcal{M})$, every $f : S' \rightarrow S$, and every $m \in I_S(\mathcal{M})(S')$, write $m \cdot a = a^*(m)$. Then

$$m \cdot 1 = m, \quad (m \cdot a) \cdot b = m \cdot (ab), \quad m \cdot 0 = \varepsilon_{\mathcal{M}}(\rho(m)),$$

and if $m = \varepsilon_{\mathcal{M}}(f)$, then $m \cdot a = m$.⁹

Remark 2.1.2.

Formation of $I_S(\mathcal{M})$ commutes with base extension: there are canonical isomorphisms

$$I_S(\mathcal{M})_{S'} \simeq I_{S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

For simplicity write $I_{S'}(\mathcal{M}) = I_S(\mathcal{M})_{S'}$. More generally, if X is an S -functor, not necessarily representable, write $I_X(\mathcal{M}) = I_S(\mathcal{M}) \times_S X$.

2.1.3.

By the preceding discussion, the multiplicative monoid of $\mathcal{O}(S')$ acts on the S' -scheme $I_{S'}(\mathcal{M})$, functorially in $\mathcal{M}$. In other words, the S -scheme $I_S(\mathcal{M})$ is an object with an operator-monoid structure $\mathcal{O}_S$, functorial in $\mathcal{M}$. Thus there is a morphism of S -schemes

$$\lambda : I_S(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_S(\mathcal{M}),$$

satisfying the evident conditions. For every S -functor X , base change gives a morphism of X -functors

$$\lambda_X : I_X(\mathcal{M}) \times_S \mathcal{O}_S \longrightarrow I_X(\mathcal{M}),$$

making $I_X(\mathcal{M})$ an object with operator monoid $\mathcal{O}(X)$. Every $a \in \mathcal{O}(X) = \text{Hom}_S(X, \mathcal{O}_S)$ defines the X -endomorphism

$$a^* = \lambda_X \circ (\text{id}_{I_X(\mathcal{M})} \times (a \circ \text{pr}_X)).$$

Explicitly, if $x \in X(S')$ and $m \in I_S(\mathcal{M})(S') = I_{S'}(\mathcal{M})(S')$, then $a(x) = a \circ x \in \mathcal{O}(S')$ and

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in $\mathcal{M}$ and preserves the zero section $\varepsilon_{\mathcal{M}} : X \rightarrow I_X(\mathcal{M})$; thus $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$.¹⁰

The operation is also “functorial in X ” in the following sense. If $\pi : Y \rightarrow X$ is a morphism of S -functors and $u : \mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ the corresponding ring morphism, so that $u(a) = a \circ \pi$, then the following diagram is commutative.

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{a^*} & I_X(\mathcal{M}) \\ \pi \uparrow & & \uparrow \pi \\ I_Y(\mathcal{M}) & \xrightarrow{u(a)^*} & I_Y(\mathcal{M}) \end{array} \quad .$$

Figure 18: Functoriality of the operator action under $Y \rightarrow X$.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 5;
combined-reader p. 63.*

⁹Editor's note: later we shall be chiefly interested in $a \in \mathcal{O}(S)$, acting by homotheties on $\mathcal{M}$. For example, if $\mathcal{M}$ is free of rank r , then for every $S' \rightarrow S$, $\text{Hom}_S(S', I_S(\mathcal{M}))$ identifies with the set of r -tuples $(\epsilon_1, \dots, \epsilon_r) \in \mathcal{O}(S')^r$ satisfying $\epsilon_i \epsilon_j = 0$ for all i, j , and $(\epsilon_1, \dots, \epsilon_r) \cdot a = (a\epsilon_1, \dots, a\epsilon_r)$ for every $a \in \mathcal{O}(S)$.

¹⁰Editor's note: this paragraph has been added; it will be useful in 3.4.2 and 4.6.2.

2.2.

Let $\mathcal{M}$ and $\mathcal{N}$ be two quasi-coherent $\mathcal{O}_S$ -modules. The commutative diagram below induces the corresponding diagram of S -schemes on the following page.

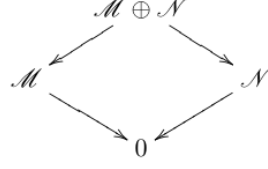


Figure 19: The direct-sum diagram of $\mathcal{O}_S$ -modules.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 5;
combined-reader p. 63.*

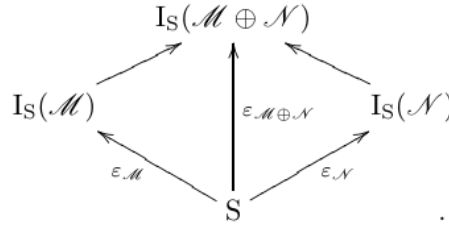


Figure 20: The induced commutative diagram of S -schemes.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 6;
combined-reader p. 64.*

Proposition 2.2.

For every S -scheme X , the diagram of functors over S obtained by applying $\underline{\mathrm{Hom}}_S(-, X)$ to the preceding diagram is cartesian.

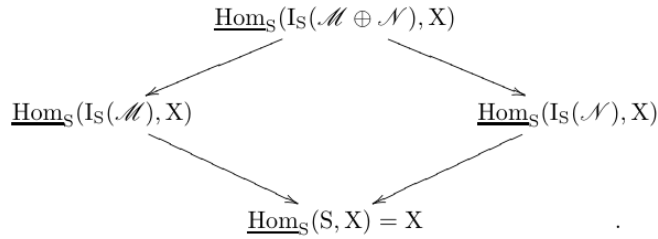
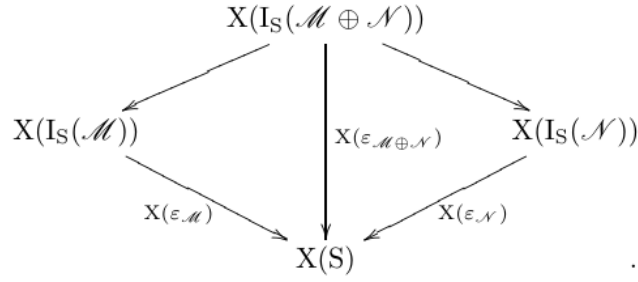


Figure 21: The cartesian diagram of $\underline{\mathrm{Hom}}_S$ -functors.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 6;
combined-reader p. 64.*

Indeed, for every $S' \rightarrow S$, one must verify that the diagram of sets obtained by evaluating the functors on S' is cartesian. Since formation of $I_S(\mathcal{P})$ commutes with base extension, it is enough to take $S' = S$, and hence to verify that the following diagram of sets is cartesian.

Figure 22: The cartesian diagram after evaluation on S .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 6;
combined-reader p. 64.*

If $x \in X(S)$, it follows from SGA 1, III 5.1 that $X(\varepsilon_{\mathcal{M}})^{-1}(x)$ is isomorphic, functorially in $\mathcal{M}$, to

$$\mathrm{Hom}_{\mathcal{O}_S}(x^*(\Omega_{X/S}^1), \mathcal{M}),$$

where $\Omega_{X/S}^1$ denotes the sheaf of relative differentials of X over S . This latter functor in $\mathcal{M}$ plainly carries a direct sum of $\mathcal{O}_S$ -modules to the product of the corresponding sets, which proves the proposition.¹¹

Corollary 2.2.1.

Let X be an S -scheme and $\mathcal{M}$ a finite free $\mathcal{O}_S$ -module. The functor $\underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)$, as a functor over X , is isomorphic to a finite product over X of copies of $\underline{\mathrm{Hom}}_S(I_S, X)$.

Nota 2.2.2.

It follows from the proof of the proposition that $\underline{\mathrm{Hom}}_S(I_S, X)$, as an X -functor, is isomorphic to $\mathbf{V}(\Omega_{X/S}^1)$ (I 4.6.1), and is therefore represented by the vector fibration¹²

$$V(\Omega_{X/S}^1).$$

3. The tangent bundle, condition (E)

Throughout this section, unless otherwise stated, the letters $\mathcal{M}, \mathcal{M}', \mathcal{N}, \dots$ will denote finite free $\mathcal{O}_S$ -modules (that is, modules isomorphic to finite direct sums of copies of $\mathcal{O}_S$).

We shall systematically use the identifications established in Exposé I. Thus, by a “functor over S ” we shall mean indifferently either a functor equipped with a morphism to $S = h_S$, or a functor on the category of objects over S . We shall use “group functor over S ,” and similar expressions, in the same way.

Definition 3.1.

Let S be a scheme, let $\mathcal{M}$ be a finite free $\mathcal{O}_S$ -module, and let X be a functor over S . The *tangent bundle of X over S , relative to the $\mathcal{O}_S$ -module $\mathcal{M}$* , is the S -functor

$$T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X).$$

¹¹Editor’s note: see also the addition 0.1.8 in Exposé III.

¹²Editor’s note: cf. editor’s note (33) of Exposé I.

In particular, the *tangent bundle of X over S* is ¹³

$$T_{X/S} = T_{X/S}(\mathcal{O}_S) = \underline{\mathrm{Hom}}_S(I_S, X).$$

The assignment $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$ -modules to S -functors. In particular, the morphisms $\mathcal{M} \rightarrow 0$ and $0 \rightarrow \mathcal{M}$ define, respectively, an S -morphism

$$\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \longrightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of it, called the *zero section*. ¹⁴

Remark 3.1.1.

¹⁵ The projection $\pi_{\mathcal{M}} : T_{X/S}(\mathcal{M}) \rightarrow X$ is induced by the zero section $\varepsilon_{\mathcal{M}} : S \rightarrow I_S(\mathcal{M})$, whereas the zero section $\tau_0 : X \rightarrow T_{X/S}(\mathcal{M})$ is induced by the structural morphism $\rho : I_S(\mathcal{M}) \rightarrow S$. More explicitly, let $t \in T_{X/S}(\mathcal{M})(S')$ correspond to the S -morphism $f : S' \times_S I_S(\mathcal{M}) \rightarrow X$, and let $x \in X(S')$ correspond to $g : S' \rightarrow X$. Then

$$\pi(t) = f \circ (\mathrm{id}_{S'} \times \varepsilon_{\mathcal{M}}), \quad \tau_0(x) = g \circ (\mathrm{id}_{S'} \times \rho).$$

It follows from 3.1 that $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from finite free $\mathcal{O}_S$ -modules to functors over X . In particular, $\mathcal{O}(S)$ is an operator monoid of the X -functor $T_{X/S}(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$.

Scholium 3.1.2.

¹⁵ The preceding discussion means, in particular, the following. For every S -morphism $\varphi : X' \rightarrow X$, put

$$\Sigma(X', \mathcal{M}) = \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})).$$

The multiplicative monoid $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ acts on $\Sigma(X', \mathcal{M})$; write the action as $(\lambda, x) \mapsto \lambda * x$. Then

$$\lambda * (\mu * x) = (\lambda\mu) * x, \quad 1 * x = x, \quad 0 * x = \tau_0 \circ \varphi.$$

There is likewise an action of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M} \oplus \mathcal{M})$ on $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$. ¹⁶

Let $m : \mathcal{M} \oplus \mathcal{M} \rightarrow \mathcal{M}$ be addition, and let $\delta : \mathcal{M} \rightarrow \mathcal{M} \oplus \mathcal{M}$ be the diagonal. Write

$$m_{X'} : \Sigma(X', \mathcal{M} \oplus \mathcal{M}) \rightarrow \Sigma(X', \mathcal{M}), \quad \delta_{X'} : \Sigma(X', \mathcal{M}) \rightarrow \Sigma(X', \mathcal{M} \oplus \mathcal{M})$$

for the induced maps. For $\lambda, \mu \in \mathcal{O}(S)$, let ℓ_{λ} denote multiplication by λ on $\mathcal{M}$, and let $\ell_{\lambda, \mu}$ denote multiplication by (λ, μ) on $\mathcal{M} \oplus \mathcal{M}$. Since

$$m \circ \ell_{\lambda, \lambda} = \ell_{\lambda} \circ m, \quad m \circ \ell_{\lambda, \mu} \circ \delta = \ell_{\lambda + \mu},$$

we have, for every $z \in \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ and $x \in \Sigma(X', \mathcal{M})$,

$$\lambda * m(z) = m((\lambda, \lambda) * z), \quad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x. \quad (\dagger)$$

¹³Editor's note: when $S = \mathrm{Spec}(k)$ and X is a k -scheme, one has $T_{X/S}(S') = \mathrm{Hom}_S(S' \otimes_k k[\epsilon], X) = X(S' \otimes_k k[\epsilon])$; this recovers one of the usual definitions of the tangent bundle.

¹⁴Editor's note: the original order " $0 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow 0$ " has been corrected to " $\mathcal{M} \rightarrow 0$ and $0 \rightarrow \mathcal{M}$."

¹⁵Editor's note: paragraphs 3.1.1 and 3.1.2 have been added.

¹⁶Editor's note: in fact, only the actions of $\mathcal{O}(S)$ (respectively $\mathcal{O}(S) \times \mathcal{O}(S)$) on $\Sigma(X', \mathcal{M})$ (respectively $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$) will be used; see below and the proof of 3.6.

Definition 3.2.

Let $u \in X(S) = \text{Hom}_S(S, X) = \Gamma(X/S)$. The *tangent space of X over S at u , relative to $\mathcal{M}$* , denoted $L_{X/S}^u(\mathcal{M})$, is the S -functor obtained from the X -functor $T_{X/S}(\mathcal{M})$ by pullback along $u : S \rightarrow X$.

$$\begin{array}{ccc} L_{X/S}^u(\mathcal{M}) & \longrightarrow & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow \pi \\ S & \xrightarrow{u} & X \end{array}$$

Figure 23: The cartesian square defining $L_{X/S}^u(\mathcal{M})$.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 8;
combined-reader p. 66.*

In particular, $L_{X/S}^u(\mathcal{O}_S)$ is denoted $L_{X/S}^u$ and is called the *tangent space of X over S at u* .

Remark 3.2.1.

¹⁷ It follows from 3.1.1 that, for every $t : S' \rightarrow S$, $L_{X/S}^u(\mathcal{M})(S')$ is the set of S -morphisms $f : I_{S'}(\mathcal{M}) \rightarrow X$ such that $f \circ \varepsilon_{\mathcal{M}} = u \circ t$, where $\varepsilon_{\mathcal{M}} : S' \rightarrow I_{S'}(\mathcal{M})$ is the zero section.

Proposition 3.3.

If X is represented by an S -scheme $\tilde{X}$, then $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ are representable. In particular, $T_{X/S}$ and $L_{X/S}^u$ are represented by the vector fibrations

$$V(\Omega_{\tilde{X}/S}^1) \quad \text{and} \quad V(u^* \Omega_{\tilde{X}/S}^1)$$

over $\tilde{X}$ and S , respectively.

It plainly suffices to prove the assertion for $T_{X/S}(\mathcal{M})$, since the corresponding assertion for $L_{X/S}^u(\mathcal{M})$ follows by pullback. By Corollary 2.2.1, it even suffices to treat $T_{X/S}$, in which case the result is exactly Nota 2.2.2.

Remark 3.3.1.

The proposition gives a particularly simple description of the vector fibration representing $L_{X/S}^u$. The image of the section u of X over S is locally closed,¹⁸ and hence is defined by a quasi-coherent ideal $\mathfrak{m}$ in the scheme induced on an open subscheme of X . The quotient $\mathfrak{m}/\mathfrak{m}^2$ may be regarded as a quasi-coherent module on S ; it is this module that defines the vector fibration in question.

For example, let X be an algebraic scheme over a field k , and let u be a k -rational point of X . Let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_{X,u}$, and let $\mathfrak{t}$ be the k -vector space dual to $\mathfrak{m}/\mathfrak{m}^2$; this is the Zariski tangent space of $\mathcal{O}_{X,u}$ at u . In the notation of I 4.6.5.1,

$$L_{X/k}^u = V(\mathfrak{m}/\mathfrak{m}^2) = W(\mathfrak{t}).$$

¹⁷Editor's note: this remark has been added.

¹⁸Editor's note: cf. EGA I, 5.3.11.

Returning to the general situation, observe first that $L_{X/S}^u(\mathcal{M})$ is covariant from finite free $\mathcal{O}_S$ -modules to functors over S . In particular, $\mathcal{O}(S)$ is a set of operators of the S -functor $L_{X/S}^u(\mathcal{M})$, compatible with functoriality in $\mathcal{M}$.¹⁹

Proposition 3.4.

Formation of $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ commutes with base extension. For every S -scheme S' , there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X_{S'}/S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} T_{X/S}(\mathcal{M})_{S'},$$

$$L_{X_{S'}/S'}^{u'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \xrightarrow{\sim} L_{X/S}^u(\mathcal{M})_{S'}, \quad u' = u_{S'}.$$

This follows immediately from the compatibility of Hom with base extension.

Corollary 3.4.1.

The X -functor $T_{X/S}(\mathcal{M})$, respectively the S -functor $L_{X/S}^u(\mathcal{M})$, is naturally an object with operators $\mathcal{O}_X$, respectively $\mathcal{O}_S$, and these structures are functorial in $\mathcal{M}$.

For $L_{X/S}^u(\mathcal{M})$, this is immediate. For each S' over S , the ring $\mathcal{O}(S')$ acts on $\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, hence on

$$L_{X_{S'}/S'}^{u'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = L_{X/S}^u(\mathcal{M})_{S'};$$

this action is functorial in S' .

The case of $T_{X/S}(\mathcal{M})$ requires a little more detail. For an X -object $f : X' \rightarrow X$, put $T_{X/S}(\mathcal{M})_{X'} = T_{X/S}(\mathcal{M}) \times_X X'$. We must equip

$$T_{X/S}(\mathcal{M})_{X'}(X') = \mathrm{Hom}_X(X', T_{X/S}(\mathcal{M}))$$

with an action of $\mathcal{O}(X')$, functorially in X' . Let $X_{X'} = X \times_S X'$, and let f' be the section of $X_{X'}$ over X' defined by f . The following diagram provides the required identification.

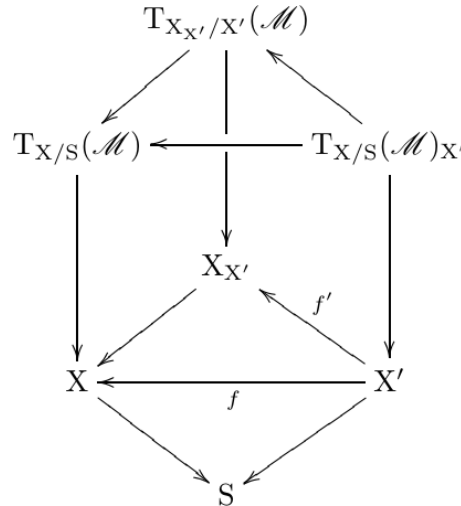


Figure 24: The base-change diagram used to define the $\mathcal{O}_X$ -operator structure.

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 10; combined-reader p. 68.

¹⁹Editor's note: cf. 3.1.2.

Together with 3.2.1, the diagram identifies $T_{X/S}(\mathcal{M})_{X'}(X')$ with

$$L_{X_{X'}/X'}^{f'}(\mathcal{M})(X') = \{ X' \text{-morphisms } \psi : I_{X'}(\mathcal{M}) \rightarrow X_{X'} \mid \psi \circ \varepsilon_{\mathcal{M}} = f' \}.$$

Every $a \in \mathcal{O}(X')$ acts through its action on $I_{X'}(\mathcal{M})$: in the notation of 2.1.1,

$$a\psi = \psi \circ a^*, \quad (a\psi)(x) = \psi(x \cdot a)$$

for every $X'' \rightarrow X'$ and every $x \in I_{X'}(\mathcal{M})(X'')$. This construction is plainly functorial in X' .²⁰ The isomorphisms of Proposition 3.4 are therefore, by construction, isomorphisms for the corresponding $\mathcal{O}_{X_{S'}}$ - and $\mathcal{O}_{S'}$ -operator structures.

Remark 3.4.2.

²¹ The action of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$ can be seen more directly as follows. For every $f : X' \rightarrow X$,

$$\mathrm{Hom}_X(X', T_{X/S}(\mathcal{M})) = \{ \varphi \in \mathrm{Hom}_S(I_{X'}(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = f \}.$$

By 2.1.3, the S -functor $I_{X'}(\mathcal{M})$ carries an action of the monoid $\mathcal{O}(X')$ preserving the zero section. Thus, if a^* denotes the X' -endomorphism defined by $a \in \mathcal{O}(X')$, then

$$a\varphi = \varphi \circ a^*, \quad (a\varphi)(m, x') = \varphi(m \cdot a(x'), x')$$

for every $S' \rightarrow S$ and every $(m, x') \in \mathrm{Hom}_S(S', I_S(\mathcal{M}) \times_S X')$. Notice that $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$, so that $(a\varphi) \circ \varepsilon_{\mathcal{M}} = f$.

Likewise, for $t : S' \rightarrow S$, the set $L_{X/S}^u(\mathcal{M})(S')$ consists of the S -morphisms $\varphi : I_{S'}(\mathcal{M}) \rightarrow X$ satisfying $\varphi \circ \varepsilon_{\mathcal{M}} = u \circ t$, and $a\varphi = \varphi \circ a^*$ for $a \in \mathcal{O}(S')$.

Remark 3.4.3.

²¹ The observations of 3.1.2 apply equally to the action of $\mathcal{O}_S$ on $L_{X/S}^u(\mathcal{M})$ and to that of $\mathcal{O}_X$ on $T_{X/S}(\mathcal{M})$.

Definition 3.5.

Let S be a scheme and X an S -functor. We say that X *satisfies condition (E) relative to S* if, for every S' over S and every pair of finite free $\mathcal{O}_{S'}$ -modules $\mathcal{M}, \mathcal{N}$, the diagram of sets obtained by applying X to diagram (*) of 2.1 is cartesian.²²

$$\begin{array}{ccc} & X(\mathrm{Is}'(\mathcal{M} \oplus \mathcal{N})) & \\ \swarrow & & \searrow \\ X(\mathrm{Is}'(\mathcal{M})) & & X(\mathrm{Is}'(\mathcal{N})) \\ \searrow & & \swarrow \\ & X(S') & \end{array},$$

Figure 25: The cartesian diagram in condition (E).

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 11;
combined-reader p. 69.*

²⁰Editor's note: the original has been expanded in what follows.

²¹Editor's note: Remarks 3.4.2 and 3.4.3 have been added.

²²Editor's note: for examples of S -functors and S -groups that do not satisfy (E), see 6.2 and 6.3.

3.5.1.

²³ Equivalently, the functor $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ carries direct sums of finite free $\mathcal{O}_S$ -modules to products of X -functors. In that case, the same is true of

$$\mathcal{M} \longmapsto L_{X/S}^u(\mathcal{M}) = S \times_X T_{X/S}(\mathcal{M})$$

for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that every representable functor satisfies condition (E). We shall sometimes abbreviate “ X satisfies condition (E) relative to S ” to “ X/S satisfies condition (E).”

If X/S satisfies (E), then $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ commutes with products and hence carries groups to groups. In particular, $T_{X/S}(\mathcal{M})$ is a commutative X -group; for the same reason, $L_{X/S}^u(\mathcal{M})$ is a commutative S -group.

Proposition 3.6.

If X/S satisfies (E), then the abelian-group structure on $T_{X/S}(\mathcal{M})$, respectively on $L_{X/S}^u(\mathcal{M})$, together with the action of $\mathcal{O}_X$, respectively of $\mathcal{O}_S$, makes it an $\mathcal{O}_X$ -module, respectively an $\mathcal{O}_S$ -module.

The operator action is functorial in $\mathcal{M}$, and therefore respects the abelian-group structure obtained by functoriality from that of $\mathcal{M}$. ²⁴ Indeed, retaining the notation of 3.1.2, the abelian X -group law on $T_{X/S}(\mathcal{M})$ is the composite

$$T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M}) \simeq T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \xrightarrow{m} T_{X/S}(\mathcal{M}),$$

whereas

$$T_{X/S}(\mathcal{M}) \xrightarrow{\delta} T_{X/S}(\mathcal{M} \oplus \mathcal{M}) \simeq T_{X/S}(\mathcal{M}) \times_X T_{X/S}(\mathcal{M})$$

is the diagonal morphism.

By Remark 3.4.3 and the identities (†) of 3.1.2,

$$\lambda(x + y) = \lambda x + \lambda y, \quad (\lambda + \mu)x = \lambda x + \mu x$$

for every $f : X' \rightarrow X$, every $x, y \in \text{Hom}_X(X', T_{X/S}(\mathcal{M}))$, and every $\lambda, \mu \in \mathcal{O}(X')$.

Remark 3.6.1.

If X is representable, then it satisfies (E), and $T_{X/S}$ and $L_{X/S}^u$ are represented by vector fibrations. The preceding laws are then exactly those induced by the vector-fibration structures (cf. I 4.6). ²⁵

Proposition 3.4 bis.

If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E), and the isomorphisms of 3.4 respect the $\mathcal{O}_{X_{S'}}$ -module and $\mathcal{O}_{S'}$ -module structures, respectively.

No further comment is needed.

²³Editor's note: what follows has been added.

²⁴Editor's note: what follows has been expanded.

²⁵Editor's note: explicitly, for every S -morphism $f : X' \rightarrow X$, the action of $\mathcal{O}(X')$ on $\text{Hom}_X(X', T_{X/S}(\mathcal{M}))$ corresponds, under the identification

$$\text{Hom}_X(X', T_{X/S}(\mathcal{M})) \simeq \text{Hom}_{\mathcal{O}_{X'}}(f^* \Omega_{X/S}^1, \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{X'}),$$

to the natural action on the right. This follows from the proof of SGA 1, III 5.1; see also the addition 0.1.8 in Exposé III.

Proposition 3.7.

The functors $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ are functorial in X . Thus, if $f : X \rightarrow X'$ is an S -morphism, the following diagrams commute.

$$\begin{array}{ccc} T_{X/S}(\mathcal{M}) & \xrightarrow{T(f)} & T_{X'/S}(\mathcal{M}) \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & X' \end{array} \quad , \quad \begin{array}{ccc} L_{X/S}^u(\mathcal{M}) & \xrightarrow{L(f)} & L_{X'/S}^{f \circ u}(\mathcal{M}) \\ & \searrow & \swarrow \\ & S & \end{array}$$

Figure 26: Functoriality of tangent bundles and tangent spaces in X .

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 12; combined-reader p. 70.

Moreover, if f is a monomorphism, then so are $T(f)$ and $L(f)$.²⁶ The existence of $T(f)$ and $L(f)$, and the last assertion, follow immediately from the definitions. Commutativity follows from functoriality in $\mathcal{M}$ and from $T_{X/S}(0) = X$.

Remark 3.7.1.

²⁷ Suppose that X and X' are representable, and let r be the rank of $\mathcal{M}$. By 2.2.2, $T_{X/S}(\mathcal{M})$ is isomorphic to the product over X of r copies of $V(\Omega_{X/S}^1)$, and similarly for $T_{X'/S}(\mathcal{M})$. Hence the square in 3.7 is cartesian when f is an open immersion, or more generally when $f^*\Omega_{X'/S}^1 = \Omega_{X/S}^1$, for example when f is étale. Under these conditions there is an isomorphism of S -functors

$$L_{X/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{X'/S}^{f \circ u}(\mathcal{M}).$$

In general, the square in 3.7 defines the following morphism of X -functors.

$$\begin{array}{ccc} T_{X/S}(\mathcal{M}) & \longrightarrow & T_{X'/S}(\mathcal{M}) \times_{X'} X \\ & \searrow & \swarrow \\ & X & \end{array} .$$

Figure 27: The induced morphism over X in the general case.

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 13; combined-reader p. 71.

Proposition 3.7 bis.

Let $f : X \rightarrow X'$ be an S -morphism. If X and X' satisfy (E) relative to S , then

$$T_{X/S}(\mathcal{M}) \xrightarrow{T(f)} T_{X'/S}(\mathcal{M})_X, \quad \text{respectively} \quad L_{X/S}^u(\mathcal{M}) \xrightarrow{L(f)} L_{X'/S}^{f \circ u}(\mathcal{M}),$$

is a morphism of $\mathcal{O}_X$ -modules, respectively of $\mathcal{O}_S$ -modules.

This follows from Proposition 3.7 by functoriality in $\mathcal{M}$.

²⁶Editor's note: the preceding sentence has been added.

²⁷Editor's note: the hypothesis that X and X' be representable has been added, and what follows has been expanded.

Proposition 3.8.

For two functors X, Y over S , there are isomorphisms, functorial in $\mathcal{M}$,

$$T_{X/S}(\mathcal{M}) \times_S T_{Y/S}(\mathcal{M}) \xrightarrow{\sim} T_{(X \times_S Y)/S}(\mathcal{M}), \quad (3.8.1)$$

$$L_{X/S}^u(\mathcal{M}) \times_S L_{Y/S}^v(\mathcal{M}) \xrightarrow{\sim} L_{(X \times_S Y)/S}^{(u,v)}(\mathcal{M}). \quad (3.8.2)$$

The first isomorphism follows from 1.2 (*); ²⁸ the second follows from it by base change along $(u, v) : S \rightarrow X \times_S Y$.

Remark 3.8.0.

The isomorphism (3.8.1) may also be read as an isomorphism of $X \times_S Y$ -functors

$$(T_{X/S}(\mathcal{M}) \times_X (X \times_S Y)) \times_{X \times_S Y} (T_{Y/S}(\mathcal{M}) \times_Y (X \times_S Y)) \xrightarrow{\sim} T_{(X \times_S Y)/S}(\mathcal{M}).$$

Corollary 3.8.1.

If X/S carries an algebraic structure defined by finite cartesian products, then $T_{X/S}(\mathcal{M})$ carries a structure of the same kind, and the projection $T_{X/S}(\mathcal{M}) \rightarrow X$ is a morphism for that structure.

Proposition 3.8 bis.

If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E), and (3.8.1), respectively (3.8.2), is an isomorphism of $\mathcal{O}_{X \times_S Y}$ -modules, respectively of $\mathcal{O}_S$ -modules.

Proof. ²⁹ By 3.5.1 and (3.8.1), the product $(X \times_S Y)/S$ satisfies (E). Let us prove the assertion about (3.8.1). Given an S -morphism $(x, y) : Z \rightarrow X \times_S Y$, Remark 3.4.2 reduces the claim to the $\mathcal{O}(Z)$ -linearity of the map

$$\begin{aligned} & \{\varphi \in \text{Hom}_S(I_Z(\mathcal{M}), X) \mid \varphi \circ \varepsilon_{\mathcal{M}} = x\} \times \{\psi \in \text{Hom}_S(I_Z(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = y\} \\ & \longrightarrow \{\theta \in \text{Hom}_S(I_Z(\mathcal{M}), X \times_S Y) \mid \theta \circ \varepsilon_{\mathcal{M}} = (x, y)\}, \end{aligned}$$

which sends (φ, ψ) to $\varphi \times \psi$. This is clear, since for $a \in \mathcal{O}(Z)$,

$$(\varphi \circ a^*) \times (\psi \circ a^*) = (\varphi \times \psi) \circ a^* = a \cdot (\varphi \times \psi).$$

The assertion about (3.8.2) follows similarly from 3.2.1.

3.9.0.

³⁰ If X is an S -group with unit section $e : S \rightarrow X$, write

$$\underline{\text{Lie}}(X/S, \mathcal{M}) = L_{X/S}^e(\mathcal{M}).$$

Thus $\underline{\text{Lie}}(X/S, \mathcal{M})$ is defined by the following cartesian square.

²⁸Editor's note: what follows has been expanded.

²⁹Editor's note: this proof has been added.

³⁰Editor's note: this paragraph has been added to explain the introduction of the notion of an S - H -functor.

$$\begin{array}{ccc}
\underline{\mathrm{Lie}}(X/S, \mathcal{M}) & \xrightarrow{i} & T_{X/S}(\mathcal{M}) \\
\downarrow & & \downarrow p \\
S & \xrightarrow{e} & X.
\end{array}$$

Figure 28: The cartesian square defining $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 14;
combined-reader p. 72.*

By Corollary 3.8.1, the projection $p : T_{X/S}(\mathcal{M}) \rightarrow X$ is a morphism of S -groups. Hence $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ is an S -group, identified by i with the kernel of p .

If X/S also satisfies (E), Proposition 3.9 will show that this group structure, induced by that of X , agrees with the abelian-group structure induced by functoriality in $\mathcal{M}$ (cf. 3.5.1). In fact, the result holds under the weaker assumption that X be an S -functor in monoids, or more generally an S -functor in H -sets, as defined next.

Definitions 3.9.0.1.

- (a) An H -set is a set X equipped with a composition law having a two-sided identity, denoted e_X or simply e .³¹ If $f : X \rightarrow Y$ is a morphism of H -sets, its kernel is $\ker f = f^{-1}(e_Y)$, a sub- H -set of X .
- (b) An H -object in a category $\mathcal{C}$ is an object X equipped with a morphism $X \times X \rightarrow X$ and a section of X over the final object that is a two-sided identity. Every $\mathcal{C}$ -monoid, and in particular every $\mathcal{C}$ -group, is a $\mathcal{C}$ - H -object. An H -object in the category of functors over S will be called an S - H -functor.
- (c) If X is an S - H -functor (for example, an S -group), and $e : S \rightarrow X$ is its identity section, write

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) = L_{X/S}^e(\mathcal{M}), \quad \underline{\mathrm{Lie}}(X/S) = \underline{\mathrm{Lie}}(X/S, \mathcal{O}_S).$$

We now spell out the following special case of 3.8.1.³²

Corollary 3.9.0.2.

If X is an S - H -functor, respectively an S -group, then $T_{X/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ are likewise S - H -functors, respectively S -groups, and there are morphisms of S - H -functors, respectively of S -groups, as follows.

$$\underline{\mathrm{Lie}}(X/S, \mathcal{M}) \xrightarrow{i} T_{X/S}(\mathcal{M}) \begin{array}{c} \xrightarrow{p} \\ \xleftarrow{s} \end{array} X,$$

Figure 29: The inclusion, projection, and zero section for the tangent functor.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 15;
combined-reader p. 73.*

Here i identifies $\underline{\mathrm{Lie}}(X/S, \mathcal{M})$ with $\ker p$, and s is a section of p .

³¹Editor's note: this terminology is inspired by the notion of an H -space in topology.

³²Editor's note: Corollary 3.9.0.2 has been added; it will be used in 4.1.

Proposition 3.9.

Let X be an S - H -functor satisfying (E) relative to S . The S - H -functor structure on $\underline{\text{Lie}}(X/S, \mathcal{M})$ induced by that of X agrees with the S -group structure defined in 3.5.1.

The preceding discussion shows that $\underline{\text{Lie}}(X/S, \mathcal{M})$ is an H -object in the category of $\mathcal{O}_S$ -modules. The proposition follows from the next lemma.³³

Lemma 3.10.

Let $\mathcal{C}$ be a category. Let G be an H -object in the category of $\mathcal{C}$ - H -objects. Thus G is a $\mathcal{C}$ - H -object, whose composition law we denote $f : G \times G \rightarrow G$, together with a morphism of $\mathcal{C}$ - H -objects $h : G \times G \rightarrow G$.³⁴ Then $f = h$, and f is commutative.

Evaluating the functors on a variable argument reduces the proof, as usual, to the case in which $\mathcal{C}$ is the category of sets. We then have a set G and two maps $f, h : G \times G \rightarrow G$ satisfying

$$h(f(x, y), f(z, t)) = f(h(x, z), h(y, t)).$$

Let $e, u \in G$ be the respective identities for f and h , so that

$$f(e, x) = f(x, e) = x, \quad h(u, x) = h(x, u) = x.$$

First,

$$h(f(u, y), f(x, u)) = f(x, y) = h(f(x, u), f(u, y)).$$

Putting $y = e$, respectively $x = e$, gives

$$\begin{aligned} x = f(x, e) &= h(f(u, e), f(x, u)) = h(u, f(x, u)) = f(x, u), \\ y = f(e, y) &= h(f(e, u), f(u, y)) = h(u, f(u, y)) = f(u, y). \end{aligned}$$

Substituting these identities into the original relation yields

$$h(y, x) = f(x, y) = h(x, y).$$

This proves the lemma and Proposition 3.9. The following corollaries now follow from 3.9.

Corollary 3.9.1.

If X is an S - H -functor satisfying (E) relative to S , every element of $X(I_S(\mathcal{M}))$ that maps to the identity element of $X(S)$ is invertible.

Corollary 3.9.2.

If X is an S -monoid satisfying (E) relative to S , an element of $X(I_S(\mathcal{M}))$ is invertible if and only if its image in $X(S)$ is invertible.

Corollary 3.9.3.

If X is an S -group satisfying (E) relative to S , the two S -group laws on $\underline{\text{Lie}}(X/S, \mathcal{M})$ coincide.

³³Editor's note: this is inspired by the standard proof that the fundamental group of an H -space is abelian.

³⁴Editor's note: $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \rightarrow G \times G, ((x, z), (y, t)) \mapsto (f(x, y), f(z, t))$.

Corollary 3.9.4.

³⁵ Let G be an S -group satisfying (E) relative to S . For $n \in \mathbb{Z}$, let $n_G : G \rightarrow G$ be the morphism of S -functors defined by $g \mapsto g^n$. Then the derived morphism

$$L(n_G) : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S)$$

is multiplication by n ; that is, it sends $x \in \underline{\mathrm{Lie}}(G/S)(S')$ to nx .

The map n_G is not generally a group homomorphism, but it preserves the identity section $e : S \rightarrow G$, so its derivative indeed carries $\underline{\mathrm{Lie}}(G/S) = L_{G/S}^e$ to itself. If i denotes the inclusion $\underline{\mathrm{Lie}}(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is characterized by

$$i(L(n_G)(x)) = i(x)^n.$$

By 3.9, the group laws on $\underline{\mathrm{Lie}}(G/S)$ coming from (E) and from the law on G agree. Hence $i(x)^n = i(nx)$, and therefore $L(n_G)(x) = nx$.

Before drawing further consequences from Proposition 3.9, we prove one more functoriality result.

Proposition 3.11.

In the situation of Section 1, there is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Indeed, by definition,

$$T_{\underline{\mathrm{Hom}}_{Z/S}(X,Y)/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X, Y)).$$

Applying the isomorphism (4) of 1.1 with $T = I_S(\mathcal{M})$ gives

$$\begin{aligned} \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X, Y)) \\ \simeq \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(Z \times_S I_S(\mathcal{M}), Y)). \end{aligned}$$

Since $Z \times_S I_S(\mathcal{M}) \simeq I_Z(\mathcal{M})$, this is

$$\underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(I_Z(\mathcal{M}), Y)) = \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Corollary 3.11.1.

If Y/Z satisfies (E), then $\underline{\mathrm{Hom}}_{Z/S}(X, Y)/S$ satisfies (E), and the isomorphism of 3.11 respects the module structures over $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$. ³⁶

Proof. Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$ -modules. If Y/Z satisfies (E), then

$$T_{Y/Z}(\mathcal{M} \oplus \mathcal{N}) \simeq T_{Y/Z}(\mathcal{M}) \times_Y T_{Y/Z}(\mathcal{N}).$$

The right-hand side is a subfunctor of $T_{Y/Z}(\mathcal{M}) \times_S T_{Y/Z}(\mathcal{N})$, and 1.2 (*) therefore gives

$$\begin{aligned} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M} \oplus \mathcal{N})) \\ \simeq \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \times_{\underline{\mathrm{Hom}}_{Z/S}(X, Y)} \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{N})). \end{aligned}$$

³⁵ Editor's note: this corollary has been added; it amplifies Remark 4.1.1.2 below.

³⁶ Editor's note: writing $H = \underline{\mathrm{Hom}}_{Z/S}(X, Y)$, this implies in particular that $T_{H/S}(\mathcal{M})$ is naturally a module over $\prod_{X \times_S H/H} \mathcal{O}_H$. This result appeared somewhat surprising to the editors, so the proof has been given in detail.

Together with 3.11, this says that $\underline{\text{Hom}}_{Z/S}(X, Y)$ satisfies (E), proving the first assertion.

For the second, put $H = \underline{\text{Hom}}_{Z/S}(X, Y)$, and let $\Delta : H' \rightarrow H$ be an S -morphism. Equivalently, let $\delta : H' \times_S X \rightarrow Y$ be the corresponding Z -morphism; it makes $H' \times_S X$ a Y -object. On the one hand, we have the following commutative diagram.

$$\begin{array}{ccc}
 \text{Hom}_H(H', \underline{\text{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))) & \hookrightarrow & \text{Hom}_S(H', \underline{\text{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))) \\
 \parallel & & \parallel \\
 \text{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M})) & \hookrightarrow & \text{Hom}_Z(H' \times_S X, T_{Y/Z}(\mathcal{M})) \\
 \parallel & & \parallel \\
 \{\psi \in \text{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\} & \hookrightarrow & \text{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y).
 \end{array}$$

Figure 30: The first comparison diagram in the proof of Corollary 3.11.1.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 17;
combined-reader p. 75.*

By 1.3, the action of $\alpha \in \mathcal{O}(H' \times_S X)$ on $\Psi \in \text{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M}))$ is given, for every $U \rightarrow S$ and every $(h, x) \in \text{Hom}_S(U, H' \times_S X)$, by

$$(\alpha\Psi)(h, x) = \alpha(h, x)\Psi(h, x),$$

where U is viewed over Y through $\delta \circ (h, x)$, and $\alpha(h, x) \in \mathcal{O}(U)$ acts through the $\mathcal{O}_Y$ -module structure on $T_{Y/Z}(\mathcal{M})$. By 3.4.2, under the identification

$$\text{Hom}_Y(H' \times_S X, T_{Y/Z}(\mathcal{M})) = \{\psi \in \text{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\},$$

this action is

$$(\alpha\psi)(m, h, x) = \psi(m \cdot \alpha(h, x), h, x) \tag{1}$$

for every $(m, h, x) \in \text{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$.

On the other hand, consider $T_{H/S}(\mathcal{M}) = \underline{\text{Hom}}_S(I_S(\mathcal{M}), H)$. We have a second commutative diagram.

$$\begin{array}{ccc}
 \text{Hom}_H(H', T_{H/S}(\mathcal{M})) & \hookrightarrow & \text{Hom}_S(H', T_{H/S}(\mathcal{M})) \\
 \parallel & & \parallel \\
 \{\Phi \in \text{Hom}_S(I_{H'}(\mathcal{M}), H) \mid \Phi \circ \varepsilon_{\mathcal{M}} = \Delta\} & \hookrightarrow & \text{Hom}_S(I_{H'}(\mathcal{M}), H) \\
 (*) \parallel & & \parallel \\
 \{\phi \in \text{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y) \mid \phi \circ \varepsilon_{\mathcal{M}} = \delta\} & \hookrightarrow & \text{Hom}_Z(I_{H' \times_S X}(\mathcal{M}), Y),
 \end{array}$$

Figure 31: The second comparison diagram in the proof of Corollary 3.11.1.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 18;
combined-reader p. 76.*

The bijection $(*)$ in the diagram is described as follows (cf. 1.1(2) and I 1.7.1). For every $U \rightarrow S$ and $(m, h, x) \in \text{Hom}_S(U, I_S(\mathcal{M}) \times_S H' \times_S X)$, where U lies over Z through $U \xrightarrow{x} X \rightarrow Z$, one has $\Phi(m, h) \in \text{Hom}_Z(X \times_S U, Y)$ and

$$\varphi(m, h, x) = \Phi(m, h) \circ (x \times \text{id}_U) \in \text{Hom}_Z(U, Y). \tag{\dagger}$$

By 3.4.2, with X replaced by H and X' by H' , the action of $a \in \mathcal{O}(H')$ on $\Phi \in \text{Hom}_S(I_{H'}(\mathcal{M}), H)$ is

$$(a\Phi)(m, h) = \Phi(m \cdot a(h), h).$$

Consequently, if φ , respectively $a\varphi$, corresponds to Φ , respectively $a\Phi$, then $(\dagger)$ gives

$$(a\varphi)(m, h, x) = \Phi(m \cdot a(h), h) \circ (x \times \text{id}_U) = \varphi(m \cdot a(h), h, x). \quad (2)$$

Together with (1), this proves that the isomorphism

$$T_{H/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M}))$$

of 3.11.1 is an isomorphism of $\mathcal{O}(H)$ -modules. Moreover, for every $H' \rightarrow H$, the $\mathcal{O}(H')$ -module structure on $\text{Hom}_H(H', T_{H/S}(\mathcal{M}))$ extends, functorially in H' , to an $\mathcal{O}(H' \times_S X)$ -module structure.

Taking $Z = S$ yields the next corollary.

Corollary 3.11.2.

There is an isomorphism, functorial in $\mathcal{M}$,

$$T_{\underline{\text{Hom}}_S(X, Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_S(X, T_{Y/S}(\mathcal{M})).$$

If Y/S satisfies (E), then $\underline{\text{Hom}}_S(X, Y)/S$ satisfies (E), and this isomorphism respects the module structures over $\underline{\text{Hom}}_S(X, Y)$.

Let $u : X \rightarrow Y$ be an S -morphism, identified with the constant morphism $u : S \rightarrow \underline{\text{Hom}}_S(X, Y)$ such that $u(f) = u$ for every $f : S' \rightarrow S$. The fiber product of u with

$$\underline{\text{Hom}}_S(X, T_{Y/S}(\mathcal{M})) \longrightarrow \underline{\text{Hom}}_S(X, Y)$$

identifies with $\underline{\text{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M}))$, where X lies over Y through u . Hence the definition of the tangent space and Corollary 3.11.2 give the following result. ³⁷

Corollary 3.11.3.

For an S -morphism $u : X \rightarrow Y$, there is an isomorphism, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Hom}}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})),$$

where on the right X lies over Y through u . If Y/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$ -modules. ³⁸

In particular, for $Y = X$, the functor $\underline{\text{End}}_S(X)$ is an S -functor in monoids, hence an S - H -functor. Recall that $\underline{\text{Lie}}(\underline{\text{End}}_S(X)/S, \mathcal{M})$ denotes $L_{\underline{\text{End}}_S(X)/S}^e(\mathcal{M})$, with e the identity section.

Corollary 3.11.4.

There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\text{Lie}}(\underline{\text{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}).$$

If X/S satisfies (E), this is an isomorphism of $\mathcal{O}_S$ -modules. ³⁸

³⁷Editor's note: the preceding sentences have been added.

³⁸Editor's note: the preceding sentence has been added.

Remark 3.11.5.

³⁹ Suppose that X/S satisfies (E). Then

$$\prod_{X/S} T_{X/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))$$

is a module over $\prod_{X/S} \mathcal{O}_X$. Explicitly, for every $S' \rightarrow S$,

$$\begin{aligned} \underline{\mathrm{Hom}}_{X/S}(X, T_{X/S}(\mathcal{M}))(S') \\ = \{\psi \in \mathrm{Hom}_X(I_{S'}(\mathcal{M}) \times_S X, X) \mid \psi \circ (\varepsilon_{\mathcal{M}} \times \mathrm{id}_X) = \mathrm{pr}_X\} \end{aligned}$$

is functorially an $\mathcal{O}(X \times_S S')$ -module. This follows either from 3.6 and the properties of $\prod_{X/S}$ (cf. 1.2), or from the proof of 3.11.1.

We now give a geometric interpretation of the tangent bundle. Let U be an S -functor.

⁴⁰ By I 1.7.2, there are isomorphisms, functorial in $\mathcal{M}$,

$$\begin{aligned} T_{X/S}(\mathcal{M})(U) &= \mathrm{Hom}_S(U, \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), X)) \\ &\simeq \mathrm{Hom}_S(I_S(\mathcal{M}), \underline{\mathrm{Hom}}_S(U, X)) \\ &\simeq \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}). \end{aligned}$$

In particular, the morphism $\mathcal{M} \rightarrow 0$ gives the following commutative diagram.

$$\begin{array}{ccc} \mathrm{Hom}_S(U, T_{X/S}(\mathcal{M})) & \xrightarrow{\sim} & \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \\ \pi_{\mathcal{M}} \downarrow & & \downarrow \\ \mathrm{Hom}_S(U, X) & \xrightarrow{\text{identité}} & \mathrm{Hom}_S(U, X) \end{array} \quad ,$$

Figure 32: The geometric interpretation of the projection $\pi_{\mathcal{M}}$.

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 19; combined-reader p. 77.

The second vertical arrow is obtained by base change along $\varepsilon_{\mathcal{M}} : S \rightarrow I_S(\mathcal{M})$. ⁴¹

Proposition 3.12.

Let $h_0 : U \rightarrow X$ be an S -morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(\mathcal{M}))$ identifies with the set of $I_S(\mathcal{M})$ -morphisms

$$U_{I_S(\mathcal{M})} \longrightarrow X_{I_S(\mathcal{M})}$$

whose restriction to U is h_0 , where U is regarded as a subobject of $U \times_S I_S(\mathcal{M})$ through $\mathrm{id}_U \times_S \varepsilon_{\mathcal{M}}$.

Taking $U = X$ and $h_0 = \mathrm{id}_X$ gives the following. ⁴²

³⁹ Editor's note: this remark has been added.

⁴⁰ Editor's note: the original has been simplified in what follows.

⁴¹ Editor's note: via

$$\mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \simeq \mathrm{Hom}_S(U \times_S I_S(\mathcal{M}), X),$$

this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

⁴² Editor's note: the original has been expanded in what follows.

Corollary 3.12.1.

The set $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$ -endomorphisms ϕ of $X_{I_S(\mathcal{M})}$ inducing the identity on X ; equivalently, the following diagram commutes.⁴³

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{\phi} & I_X(\mathcal{M}) \\ \varepsilon_{\mathcal{M}} \uparrow & & \uparrow \varepsilon_{\mathcal{M}} \\ X & \xrightarrow{\text{id}} & X. \end{array}$$

Figure 33: An infinitesimal endomorphism inducing the identity on X .

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 20; combined-reader p. 78.

On the other hand, by 3.11.2,

$$\Gamma(T_{X/S}(\mathcal{M})/X) \simeq \underline{\text{Lie}}(\underline{\text{End}}_S(X)/S, \mathcal{M})(S).$$

If X/S satisfies (E), then $\underline{\text{End}}_S(X)/S$ satisfies (E), so the latter Lie functor is an $\mathcal{O}_S$ -module by 3.6 (indeed, by 3.11.5, it is a $\prod_{X/S} \mathcal{O}_X$ -module). Applying 3.9 gives the next proposition.⁴⁴

Proposition 3.13.

If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(\mathcal{M})/X)$ identifies with the set of $I_S(\mathcal{M})$ -endomorphisms of $X_{I_S(\mathcal{M})}$ that induce the identity on X . Consequently, every such endomorphism is an automorphism.

Corollary 3.13.1.

Let $u : X \rightarrow Y$ be an S -isomorphism, with Y/S satisfying (E). Every $I_S(\mathcal{M})$ -morphism $X_{I_S(\mathcal{M})} \rightarrow Y_{I_S(\mathcal{M})}$ extending u is an isomorphism.

Corollary 3.13.2.

If Y/S satisfies (E), the monomorphism

$$\underline{\text{Isom}}_S(X, Y) \longrightarrow \underline{\text{Hom}}_S(X, Y)$$

induces, for every $u \in \underline{\text{Isom}}_S(X, Y)$, an isomorphism

$$L_{\underline{\text{Isom}}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\underline{\text{Hom}}_S(X, Y)/S}^u(\mathcal{M}).$$

*Proof.*⁴⁵ For every $S' \rightarrow S$, we must show that this map is a bijection on S' -points. By base change (cf. 3.4), it suffices to take $S' = S$. The right-hand side then parametrizes the $I_S(\mathcal{M})$ -morphisms $X_{I_S(\mathcal{M})} \rightarrow Y_{I_S(\mathcal{M})}$ extending u , whereas the left-hand side parametrizes those that are isomorphisms. The assertion is therefore Corollary 3.13.1.

⁴³Editor's note: when $\mathcal{M} = \mathcal{O}_X$, these are called *infinitesimal endomorphisms* of X ; see 6.2 at the end of this exposé.

⁴⁴Editor's note: the original has been expanded in what follows.

⁴⁵Editor's note: the following proof has been added.

Corollary 3.13.3.

If X/S satisfies (E), the monomorphism $\underline{\mathrm{Aut}}_S(X) \rightarrow \underline{\mathrm{End}}_S(X)$ induces, for every $u \in \underline{\mathrm{Aut}}_S(X)$, an isomorphism

$$L_{\underline{\mathrm{Aut}}_S(X)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\underline{\mathrm{End}}_S(X)/S}^u(\mathcal{M}).$$

In particular,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(\underline{\mathrm{End}}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}),$$

so that $\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S, \mathcal{M})$ is a $\prod_{X/S} \mathcal{O}_X$ -module.⁴⁶

3.14.

Finally, suppose that X is representable.⁴⁷ By 2.2.2, the X -functor $T_{X/S}$ is represented by $V(\Omega_{X/S}^1)$, whence bijections

$$\Gamma(T_{X/S}/X) \simeq \mathrm{Hom}_X(\Omega_{X/S}^1, \mathcal{O}_X) \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X).$$

This also follows from the preceding results. By 3.13, $\Gamma(T_{X/S}/X)$ identifies with the set of infinitesimal endomorphisms of X , that is, the I_S -endomorphisms of X_{I_S} that induce the identity on X . The schemes X and X_{I_S} have the same underlying topological space, and their structure sheaves are $\mathcal{O}_X$ and

$$\mathcal{D}_{\mathcal{O}_X} = \mathcal{O}_X \oplus \mathcal{M}, \quad \mathcal{M} = \mathcal{O}_X,$$

where $\mathcal{M}$ is a square-zero ideal. Let $\pi : \mathcal{D}_{\mathcal{O}_X} \rightarrow \mathcal{O}_X$ be the morphism of $\mathcal{O}_X$ -algebras vanishing on $\mathcal{M}$. Giving an infinitesimal endomorphism of X is equivalent to giving a morphism of $\mathcal{O}_S$ -algebras $\phi : \mathcal{O}_X \rightarrow \mathcal{D}_{\mathcal{O}_X}$ satisfying $\pi \circ \phi = \mathrm{id}_{\mathcal{O}_X}$, hence to giving an $\mathcal{O}_S$ -derivation of $\mathcal{O}_X$.

If $D, D' \in \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$, and ϕ_D denotes the infinitesimal endomorphism corresponding to D , then

$$\phi_{D+D'} = \phi_D \circ \phi_{D'}.$$

Thus the identification

$$\{\text{infinitesimal endomorphisms of } X\} \simeq \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X)$$

is an isomorphism of abelian groups. In view of 3.13 and 3.11.5, we have therefore constructed an isomorphism of abelian groups, indeed of $\mathcal{O}(X)$ -modules,

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} \mathrm{Der}_{\mathcal{O}_S}(\mathcal{O}_X),$$

recovering the classical interpretation of tangent vector fields as derivations of the structure sheaf.⁴⁸ Finally,

$$\Gamma(T_{X/S}/X) = H^0(X, \mathfrak{g}_{X/S}),$$

where $\mathfrak{g}_{X/S}$ is the dual of $\Omega_{X/S}^1$.

⁴⁶Editor's note: the preceding assertion has been added.

⁴⁷Editor's note: the reminder of 2.2.2 has been added, and the sequel has been expanded.

⁴⁸Editor's note: this isomorphism is also functorial in S . If $S' \rightarrow S$ and $X' = X_{S'}$, then

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq \mathrm{Der}_{\mathcal{O}_{S'}}(\mathcal{O}_{X'}).$$

4. Tangent Space to a Group — Lie Algebras

4.1.

Let G be a group functor over S . By 3.9.0.2, $T_{G/S}(\mathcal{M})$ and $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ are endowed with group structures over S , and there are group morphisms

$$\underline{\mathrm{Lie}}(G/S, \mathcal{M}) \xrightarrow{i} T_{G/S}(\mathcal{M}) \xrightleftharpoons[s]{p} G \quad ;$$

Figure 34: The split tangent sequence of the group functor G .

*Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 22;
combined-reader p. 80; printed p. 70.*

By definition, i identifies $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ with the kernel of p , and s is a section of p . It follows from I 2.3.7 that this sequence identifies $T_{G/S}(\mathcal{M})$ with the semidirect product of G by $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Definition 4.1.A.

The corresponding action of G on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is denoted

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

and is called the *adjoint representation* of G , relative to $\mathcal{M}$. Thus, by definition, for $x \in G(S')$ and $X \in \underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$,

$$\mathrm{Ad}(x)X = i^{-1}(s(x)i(X)s(x)^{-1}).$$

If G is commutative, then $T_{G/S}(\mathcal{M})$ is commutative as well and $\mathrm{Ad}(x)X = X$.⁴⁹

Definition 4.1.B.

Let G, H be group functors over S , and let $f : G \rightarrow H$ be a group morphism. Functoriality gives a morphism of exact sequences compatible with the canonical sections.

$$\begin{array}{ccccccc} 1 & \longrightarrow & \underline{\mathrm{Lie}}(G/S, \mathcal{M}) & \longrightarrow & T_{G/S}(\mathcal{M}) & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow L(f) & & \downarrow T(f) & & \downarrow f \\ 1 & \longrightarrow & \underline{\mathrm{Lie}}(H/S, \mathcal{M}) & \longrightarrow & T_{H/S}(\mathcal{M}) & \longrightarrow & H \longrightarrow 1 \end{array} \quad ;$$

Figure 35: Functoriality of the split tangent sequence.

*Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 22;
combined-reader p. 80; printed p. 70.*

The morphism $L(f)$, also denoted $\underline{\mathrm{Lie}}(f)$, is called the *derived morphism* of f .⁵⁰

⁴⁹Editor's note: the numbering 4.1.A and 4.1.B has been added in order to bring these definitions into relief.

⁵⁰Editor's note: if f is a monomorphism, then so are $L(f)$ and $T(f)$; cf. 3.7.

Remark 4.1.C.

If G/S and H/S satisfy condition E , then $L(f)$ respects the $\mathcal{O}_S$ -module structures obtained from “functoriality in $\mathcal{M}$ ” (cf. 3.6).⁵¹

Proposition 4.1.1.

Let $g \in G(S)$. Then

$$\mathrm{Ad}(g) : \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

is the derived morphism of $\mathrm{Int}(g) : G \rightarrow G$.

Indeed, $\mathrm{Ad}(g)X = i^{-1}(\mathrm{Int}(g)i(X))$, which is precisely $L(\mathrm{Int}(g))(X)$ by the definition of the derived morphism.

Suppose that G/S satisfies E . By Proposition 3.9, the group structure just defined on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$ is exactly the one induced by its $\mathcal{O}_S$ -module structure (which is defined by (E)). Proposition 4.1.1 and the functoriality of the $\mathcal{O}_S$ -action (cf. 3.6) therefore yield the following corollary.

Corollary 4.1.1.1.

Suppose that G/S satisfies E . Then Ad maps G into the subgroup

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

of $\underline{\mathrm{Aut}}_{S\text{-gr.}}(\underline{\mathrm{Lie}}(G/S, \mathcal{M}))$. In other words, for every $g \in G(S')$, the automorphism $\mathrm{Ad}(g)$ respects the $\mathcal{O}_{S'}$ -module structure on $\underline{\mathrm{Lie}}(G'/S', \mathcal{M})$, where $G' = G \times_S S'$. Thus Ad is a *linear representation* (cf. I 3.2) of G on the $\mathcal{O}_S$ -module $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$.

Remarks 4.1.1.2.

a) The functor G/S satisfies E if and only if, for every pair $(\mathcal{M}, \mathcal{N})$ of finite free $\mathcal{O}_S$ -modules, the following diagram, obtained by applying $\underline{\mathrm{Lie}}(G/S, -)$ to diagram (*) of 2.1, is cartesian.

$$\begin{array}{ccc} & \underline{\mathrm{Lie}}(G/S, \mathcal{M} \oplus \mathcal{N}) & \\ \swarrow & & \searrow \\ \underline{\mathrm{Lie}}(G/S, \mathcal{M}) & & \underline{\mathrm{Lie}}(G/S, \mathcal{N}) \\ \searrow & & \swarrow \\ & \underline{\mathrm{Lie}}(G/S, 0) = S & \end{array},$$

Figure 36: The cartesian criterion for condition E .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 23;
combined-reader p. 81; printed p. 71.*

b) Suppose that G/S satisfies E . The derived morphism of the group law $\pi : G \times_S G \rightarrow G$ is the addition law on $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$. Notice that π is not a group morphism, but $\pi(e, e) = e$; hence the derived morphism $L(\pi)$ maps

$$T_{(G \times_S G)/S}^{(e,e)}(\mathcal{M}) = \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(G/S, \mathcal{M})$$

to $\underline{\mathrm{Lie}}(G/S, \mathcal{M})$; cf. 3.7 and 3.8. Likewise, for $n \in \mathbb{Z}$, let $n_G : G \rightarrow G$ denote the S -functor morphism $g \mapsto g^n$. Then $L(n_G)$ is multiplication by n on $\underline{\mathrm{Lie}}(G/S)$; cf. 3.9.4.

⁵¹Editor's note: this remark has been added.

4.1.2.0.

Consider the S -functor $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$.⁵² For every $S' \rightarrow S$, one has $T_{G/S}(\mathcal{M})_{S'} \simeq T_{G_{S'}/S'}(\mathcal{M})$ (cf. 3.4), and therefore

$$\begin{aligned} & \underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))(S') \\ & \simeq \mathrm{Hom}_{G_{S'}}(G_{S'}, T_{G_{S'}/S'}(\mathcal{M})) = \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}). \end{aligned}$$

First note the isomorphism, functorial in S' ,

$$\mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})) \xrightarrow{\sim} \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}), \quad (*)$$

which sends $f : G_{S'} \rightarrow \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ to the section $s_f : G_{S'} \rightarrow T_{G_{S'}/S'}(\mathcal{M})$ defined, for every $S'' \rightarrow S'$ and $g \in G(S'')$, by

$$s_f(g) = i(f(g))s(g).$$

Let h be an automorphism of the functor $G_{S'}$ over S' , not necessarily preserving the group structure. To every section τ of $T_{G_{S'}/S'}(\mathcal{M})$ one associates, by transport of structure, a section $h(\tau)$: it is the unique section that makes the following diagram commute.

$$\begin{array}{ccc} G_{S'} & \xrightarrow{\tau} & T_{G_{S'}/S'}(\mathcal{M}) \\ h \downarrow & & \downarrow T(h) \\ G_{S'} & \xrightarrow{h(\tau)} & T_{G_{S'}/S'}(\mathcal{M}) \end{array} \quad .$$

Figure 37: Transport of a tangent section by an automorphism h .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 24;
combined-reader p. 82; printed p. 72.*

In particular, take $h = t_x$, right translation by an element $x \in G(S')$; thus $h(g) = t_x(g) = g \cdot x$ for all $g \in G(S'')$, $S'' \rightarrow S'$. One then has

$$t_x(s_f) = s_{t_x(f)},$$

where $t_x(f) : G_{S'} \rightarrow \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$ is given by

$$t_x(f)(g) = f(g \cdot x^{-1}).$$

Let G act by right translations on

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \quad \text{and} \quad \underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$$

as follows. For $S' \rightarrow S$ and $x \in G(S')$, let

$$\begin{aligned} \sigma & \in \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}), \\ f & \in \mathrm{Hom}_{S'}(G_{S'}, \underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})). \end{aligned}$$

Set

$$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x), \quad (f \cdot x)(g) = f(g \cdot x^{-1})$$

⁵²Editor's note: the original has been expanded in what follows to display the action of G on the functors $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))$ and $\underline{\mathrm{Hom}}_S(G, \underline{\mathrm{Lie}}(G/S, \mathcal{M}))$.

for every $g \in G(S'')$, $S'' \rightarrow S'$. The isomorphism $(*)$ respects these G -actions.

Consequently, under this isomorphism, the elements of

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$$

correspond to the constant morphisms from $G_{S'}$ to $\underline{\mathrm{Lie}}(G_{S'}/S', \mathcal{M})$, that is, to the morphisms factoring through $G_{S'} \rightarrow S'$; equivalently, they correspond to the elements of $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S')$.

Terminology. The elements of $\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ are called the sections of $T_{G_{S'}/S'}(\mathcal{M})$ invariant under right translation.

Proposition 4.1.2.

The map

$$\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S) \longrightarrow \Gamma(T_{G/S}(\mathcal{M})/G), \quad X \longmapsto (x \mapsto X \cdot x),$$

is a bijection onto the set of sections invariant under right translation.⁵³

Likewise, let G act on the right on $\underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$ by

$$(u \cdot x)(g) = u(g \cdot x^{-1}) \cdot x,$$

for $S' \rightarrow S$, $x \in G(S')$, $u \in \underline{\mathrm{End}}_{I_{S'}(\mathcal{M})}(G_{I_{S'}(\mathcal{M})})$, and $g \in G(S'')$, $S'' \rightarrow I_{S'}(\mathcal{M})$. The morphism of 3.12.1

$$\underline{\mathrm{Hom}}_{G/S}(G, T_{G/S}(\mathcal{M})) \longrightarrow \underline{\mathrm{End}}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$$

respects the G -actions. For each $S' \rightarrow S$, it therefore induces a bijection between $\Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and the set of $I_{S'}(\mathcal{M})$ -endomorphisms u of $G_{I_{S'}(\mathcal{M})}$ that induce the identity on G and “commute with right translations,” meaning that $u_{S''} \cdot x = u_{S''}$ for every $S'' \rightarrow S'$ and $x \in G(S'')$.

Proposition 4.1.3.

There is a bijection, functorial in G , between $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S)$ and the set of $I_S(\mathcal{M})$ -endomorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G , in the sense just specified.

Taking 3.13 into account gives the following theorem.

Theorem 4.1.4.

Let G be an S -functor in groups, and suppose that G/S satisfies E . Then $\underline{\mathrm{Lie}}(G/S, \mathcal{M})(S)$ is identified, functorially in G , with the group of $I_S(\mathcal{M})$ -automorphisms of $G_{I_S(\mathcal{M})}$ that induce the identity on G and commute with the right translations of G .

For $\mathcal{M} = \mathcal{O}_S$, this recovers one of the classical definitions of the Lie algebra of a group.

⁵³Statements 4.1.2, 4.1.3, and 4.1.4 can be obtained more simply by observing that the automorphisms of G invariant under right translations are the left translations. Editor’s note: see, for example, the proofs of Propositions 4.6 and 2.5 of [DG70], §II.4.

4.2.0.

Before proceeding, let us establish further consequences of 3.11.⁵⁴ Let X, Y be over Z , with Z over S , as in Section 1. As was seen in 3.11, the isomorphisms 1.1(4)

$$\begin{array}{ccc} \underline{\mathrm{Hom}}_S(\mathrm{I}_S(\mathcal{M}), \underline{\mathrm{Hom}}_{Z/S}(X, Y)) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}_{Z/S}(X, \underline{\mathrm{Hom}}_Z(\mathrm{I}_Z(\mathcal{M}), Y)) \\ & \searrow \simeq & \nearrow \simeq \\ & \underline{\mathrm{Hom}}_{Z/S}(X \times_S \mathrm{I}_S(\mathcal{M}), Y) & \end{array}$$

Figure 38: The adjunction isomorphisms of 1.1(4).

*Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 25;
combined-reader p. 83; printed p. 73.*

induce the following isomorphism θ .

$$\begin{array}{ccc} T_{\underline{\mathrm{Hom}}_{Z/S}(X, Y)}(\mathcal{M}) & \xrightarrow[\sim]{\theta} & \underline{\mathrm{Hom}}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \\ & \searrow \simeq & \nearrow \simeq \\ & \underline{\mathrm{Hom}}_{Z/S}(X \times_S \mathrm{I}_S(\mathcal{M}), Y) & \end{array} .$$

Figure 39: The induced tangent-space isomorphism θ .

*Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 25;
combined-reader p. 83; printed p. 73.*

By 1.3, if Y is a Z -group, then $\underline{\mathrm{Hom}}_Z(V, Y)$ is a Z -group for every $V \rightarrow Z$, in particular for $V = \mathrm{I}_Z(\mathcal{M})$. Explicitly, if $Z'' \rightarrow Z' \rightarrow Z$ and $\varphi, \psi \in \mathrm{Hom}_Z(V_{Z'}, Y)(Z')$, their product is defined by

$$(\varphi \cdot \psi)(v) = \varphi(v)\psi(v) \quad (v \in V_{Z'}(Z'')).$$

Suppose now that X and Y are Z -groups.

Definition 4.2.0.1.

We denote by $\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)$ the subfunctor of $\underline{\mathrm{Hom}}_{Z/S}(X, Y)$ defined, for every $S' \rightarrow S$, by

$$\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)(S') = \mathrm{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}). \quad (3)$$

The same definition applies with Y replaced by the Z -group $T_{Y/Z}(\mathcal{M})$.

Under the isomorphisms (2) above, $T_{\underline{\mathrm{Hom}}_{(Z/S)\text{-gr.}}(X, Y)/S}(\mathcal{M})(S')$ corresponds to the $Z_{S'}$ -morphisms

$$\varphi : X_{S'} \times_{S'} \mathrm{I}_{S'}(\mathcal{M}) \longrightarrow Y_{S'}$$

that are multiplicative in X , that is,

$$\varphi(x_1 x_2, m) = \varphi(x_1, m) \varphi(x_2, m).$$

These in turn correspond to the morphisms of $Z_{S'}$ -groups $X_{S'} \rightarrow T_{Y/Z}(\mathcal{M})_{S'}$. Hence:

⁵⁴Editor's note: paragraph 4.2.0 has been added; its results are used implicitly in 4.2 and 4.7 of the original.

Proposition 4.2.0.2.

Let X, Y be Z -groups, where Z is over S . There are isomorphisms of S -functors, functorial in $\mathcal{M}$,

$$T_{\underline{\text{Hom}}_{(Z/S)\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{(Z/S)\text{-gr.}}(X, T_{Y/Z}(\mathcal{M})).$$

Taking $Z = S$ gives the following consequence. If Y is a commutative S -group, then so is $T_{Y/S}(\mathcal{M})$, and hence so are $H = \underline{\text{Hom}}_{S\text{-gr.}}(X, Y)$, $\underline{\text{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M}))$, and $T_{H/S}(\mathcal{M})$.

Corollary 4.2.0.3.

Let X, Y be S -groups. There are isomorphisms of S -functors, functorial in $\mathcal{M}$,

$$T_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M})).$$

If Y is commutative, these are moreover isomorphisms of abelian S -groups.

Definition 4.2.0.4.

If Y is an $\mathcal{O}_S$ -module, the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\text{Lie}}(Y/S, \mathcal{M})$, carries an $\mathcal{O}_S$ -module structure induced by that of Y . Equipped with this structure, it will be denoted $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\text{Lie}}'(Y/S, \mathcal{M})$.⁵⁵

Consequently, if X, Y are $\mathcal{O}_S$ -modules, then $T'_{Y/S}(\mathcal{M}) = \underline{\text{Hom}}_S(I_S(\mathcal{M}), Y)$. If $H = \underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)$, then $\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M}))$ and $T'_{H/S}(\mathcal{M})$ also carry $\mathcal{O}_S$ -module structures.

Corollary 4.2.0.5.

If X, Y are $\mathcal{O}_S$ -modules, there are isomorphisms of $\mathcal{O}_S$ -modules, functorial in $\mathcal{M}$,

$$T'_{\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})).$$

Definition 4.2.A.

Let X, L be S -groups, with X acting on L by group automorphisms (cf. I 2.3.5). We define the subfunctor $\underline{Z}_S^1(X, L)$ of $\underline{\text{Hom}}_S(X, L)$ by

$$\underline{Z}_S^1(X, L)(S') = \{ \varphi \in \text{Hom}_{S'}(X_{S'}, L_{S'}) \mid \varphi(x_1 x_2) = \varphi(x_1)(x_1 \cdot \varphi(x_2)), \\ x_1, x_2 \in X(S''), S'' \rightarrow S' \}.$$

It is called the *functor of crossed homomorphisms* from X to L .⁵⁶

Remark 4.2.B.

a) If L' is another S -group on which X acts by group automorphisms, then

$$\underline{Z}_S^1(X, L \times_S L') \simeq \underline{Z}_S^1(X, L) \times_S \underline{Z}_S^1(X, L').$$

b) If L is an X - $\mathcal{O}_S$ -module, then $\underline{Z}_S^1(X, L)$ is the kernel of the differential

$$\partial : \underline{\text{Hom}}_S(X, L) \longrightarrow \underline{\text{Hom}}_S(X^2, L)$$

defined in I 5.1. In particular, $\underline{Z}_S^1(X, L)$ is then an $\mathcal{O}_S$ -module.

⁵⁵Editor's note: the prime is used to distinguish the $\mathcal{O}_S$ -module structure induced by that of Y from the one that will be obtained from condition E.

⁵⁶Editor's note: the original has been expanded below; in particular, Definition 4.2.A and Remark 4.2.B have been added.

4.2.

Let $u : X \rightarrow Y$ be a morphism of S -groups. We describe $L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M})$. First, 3.11.3 gives an isomorphism of S -functors, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Hom}}_{S(X,Y)/S}}^u(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{Y/S}(X, T_{Y/S}(\mathcal{M})). \quad (\dagger)$$

Since Y is an S -group,

$$T_{Y/S}(\mathcal{M}) = \underline{\text{Lie}}(Y/S, \mathcal{M}) \cdot Y = \underline{\text{Lie}}(Y/S, \mathcal{M})_Y,$$

and one obtains an isomorphism, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Hom}}_{S(X,Y)/S}}^u(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_S(X, \underline{\text{Lie}}(Y/S, \mathcal{M})).$$

For $S' \rightarrow S$, let $u' : X' \rightarrow Y'$ be the base change of u . The S -functor⁵⁷

$$\underline{\text{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\text{Lie}}(Y/S, \mathcal{M}) \cdot Y)$$

is defined by

$$\begin{aligned} & \underline{\text{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\text{Lie}}(Y/S, \mathcal{M}) \cdot Y)(S') \\ &= \text{Hom}_{Y'\text{-gr.}}(X', (\underline{\text{Lie}}(Y/S, \mathcal{M}) \cdot Y)_{S'}) \\ &= \text{Hom}_{Y'\text{-gr.}}(X', \underline{\text{Lie}}(Y'/S', \mathcal{M}) \cdot Y'). \end{aligned}$$

Thus $(\dagger)$ induces an isomorphism

$$L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{(Y/S)\text{-gr.}}(X, \underline{\text{Lie}}(Y/S, \mathcal{M}) \cdot Y). \quad (\dagger')$$

On the other hand, the composite

$$X \xrightarrow{u} Y \xrightarrow{\text{Ad}} \underline{\text{Aut}}_{S\text{-gr.}}(\underline{\text{Lie}}(Y/S, \mathcal{M}))$$

Figure 40: The action of X on $\underline{\text{Lie}}(Y/S, \mathcal{M})$.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 27;
combined-reader p. 85; printed p. 75.*

defines an action of X on $L = \underline{\text{Lie}}(Y/S, \mathcal{M})$ by group automorphisms.

Let $\Phi \in \underline{\text{Hom}}_{Y/S}(X, \underline{\text{Lie}}(Y/S, \mathcal{M}) \cdot Y)$. For $S'' \rightarrow S' \rightarrow S$ and $x \in X(S'')$, there is a unique expression

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \quad \varphi(S')(x) \in \underline{\text{Lie}}(Y'/S', \mathcal{M})(S'').$$

This determines an element $\varphi \in \underline{\text{Hom}}_S(X, \underline{\text{Lie}}(Y/S, \mathcal{M}))$. The map $\Phi(S')$ is a group morphism if and only if, for all $x_1, x_2 \in X(S'')$,

$$\begin{aligned} \varphi(S')(x_1 x_2) &= \varphi(S')(x_1) (u(x_1) \varphi(S')(x_2) u(x_1)^{-1}) \\ &= \varphi(S')(x_1) (x_1 \cdot \varphi(S')(x_2)), \end{aligned}$$

that is, precisely when $\varphi \in \underline{Z}_S^1(X, \underline{\text{Lie}}(Y/S, \mathcal{M}))$.

⁵⁷ Editor's note: this is Definition 4.2.0.1 applied with $Z = Y$, to the Y -groups $u : X \rightarrow Y$ and $T_{Y/S}(\mathcal{M}) \rightarrow Y$.

Proposition 4.2.

Let $u : X \rightarrow Y$ be a morphism of S -groups. There is an isomorphism of S -functors, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \underline{Z}_S^1(X, \underline{\text{Lie}}(Y/S, \mathcal{M})).$$

Suppose in addition that Y/S satisfies E . It follows from 4.2.0.3, just as in the proof of 3.11.1, that $\underline{\text{Hom}}_{S\text{-gr.}}(X, Y)/S$ satisfies E . Thus (cf. 3.5.1)

$$\begin{aligned} L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M} \oplus \mathcal{N}) &\simeq L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \\ &\quad \times_S L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{N}). \end{aligned}$$

This also follows directly from Proposition 4.2 and Remark 4.2.Ba. Consequently, both sides of the isomorphism in Proposition 4.2 carry $\mathcal{O}_S$ -module structures obtained from functoriality in $\mathcal{M}$, and that isomorphism respects them.

Proposition 4.2 bis.

Let $u : X \rightarrow Y$ be a morphism of S -groups, and suppose that Y/S satisfies E . There is an isomorphism of $\mathcal{O}_S$ -modules, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \underline{Z}_S^1(X, \underline{\text{Lie}}(Y/S, \mathcal{M})).$$

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When Y/S satisfies E , 3.13.1 also gives, just as in the proof of 3.13.2, an isomorphism functorial in $\mathcal{M}$, for every $u \in \underline{\text{Isom}}_{S\text{-gr.}}(X, Y)$,

$$L_{\underline{\text{Isom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}). \quad (*)$$

This yields the following two corollaries.

Corollary 4.2.1.

Let $u : X \rightarrow Y$ be a morphism of S -groups. If Y/S satisfies E , there is an isomorphism of $\mathcal{O}_S$ -modules, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Isom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \underline{Z}_S^1(X, \underline{\text{Lie}}(Y/S, \mathcal{M})).$$

Corollary 4.2.2.

Let X be an S -group. If X/S satisfies E , there is an isomorphism of $\mathcal{O}_S$ -modules, functorial in $\mathcal{M}$,

$$\underline{\text{Lie}}(\underline{\text{Aut}}_{S\text{-gr.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{Z}_S^1(X, \underline{\text{Lie}}(X/S, \mathcal{M})).$$

If Y is commutative, its adjoint action on $L = \underline{\text{Lie}}(Y/S, \mathcal{M})$ is trivial, so $\underline{Z}_S^1(X, L) = \underline{\text{Hom}}_{S\text{-gr.}}(X, L)$.⁵⁹

Corollary 4.2.3.

Let Y be a commutative S -group. There is an isomorphism of S -functors, functorial in $\mathcal{M}$,

$$L_{\underline{\text{Hom}}_{S\text{-gr.}}(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{S\text{-gr.}}(X, \underline{\text{Lie}}(Y/S, \mathcal{M})).$$

⁵⁸Editor's note: the original has been expanded from this point; in particular, Proposition 4.2 bis has been added.

⁵⁹Editor's note: the following sentence has been added.

4.3.

We now consider the case in which X and Y are $\mathcal{O}_S$ -modules. Recall (cf. 4.2.0.4) that $T'_{Y/S}(\mathcal{M})$, respectively $\underline{\text{Lie}}'(Y/S, \mathcal{M})$, denotes the functor $T_{Y/S}(\mathcal{M})$, respectively $\underline{\text{Lie}}(Y/S, \mathcal{M})$, equipped with the $\mathcal{O}_S$ -module structure induced by that of Y .

When Y/S satisfies E , we shall always write $\underline{\text{Lie}}(Y/S, \mathcal{M})$ for this functor equipped instead with the $\mathcal{O}_S$ -module structure defined for every functor satisfying E . By 3.9, the underlying abelian-group structures of $\underline{\text{Lie}}(Y/S, \mathcal{M})$ and $\underline{\text{Lie}}'(Y/S, \mathcal{M})$ coincide, but their module structures need not coincide (see the counterexample in 6.3). For $S' \rightarrow S$, $a \in \mathcal{O}(S')$, and $m \in \underline{\text{Lie}}(Y/S, \mathcal{M})(S')$, we write $a \cdot' m$, respectively $a \cdot m$, for the action coming from Y , respectively the action obtained from condition E . We use the same convention for the two actions on $T_{Y/S}(\mathcal{M})$.

There is an isomorphism of $\mathcal{O}_S$ -modules

$$T'_{Y/S}(\mathcal{M}) \simeq \underline{\text{Lie}}'(Y/S, \mathcal{M}) \oplus Y.$$

Exactly as for Propositions 4.2 and 4.2 bis, one therefore obtains:

Proposition 4.3.

Let $u : X \rightarrow Y$ be a morphism of $\mathcal{O}_S$ -modules. There is an isomorphism of S -functors, functorial in $\mathcal{M}$,

$$L^u_{\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S}(\mathcal{M}) \xrightarrow{\sim} \underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\text{Lie}}'(Y/S, \mathcal{M})). \quad (*)$$

If Y/S satisfies E , then $\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S$ satisfies E , and $(*)$ is an isomorphism of $\mathcal{O}_S$ -modules when both sides carry the structures obtained from functoriality in $\mathcal{M}$.^{60 61}

Remark 4.3 bis.

Let $u : X \rightarrow Y$ be a morphism of $\mathcal{O}_S$ -modules.⁶² Denote by τ_u the map that sends each $\mathcal{O}_S$ -linear morphism $\varphi : X \rightarrow \underline{\text{Lie}}'(Y/S, \mathcal{M})$ to

$$u \oplus \varphi : X \rightarrow T'_{Y/S}(\mathcal{M}) = Y \oplus \underline{\text{Lie}}'(Y/S, \mathcal{M}).$$

The isomorphism of 4.3 belongs to the following commutative diagram, functorial in $\mathcal{M}$.

$$\begin{array}{ccc} L^u_{\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S}(\mathcal{M}) & \xrightarrow{\sim} & \underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\text{Lie}}'(Y/S, \mathcal{M})) \\ \downarrow & & \downarrow \tau_u \\ T_{\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S}(\mathcal{M}) & \xrightarrow{\sim} & \underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})). \end{array}$$

Figure 41: Compatibility of the isomorphism of 4.3 with τ_u .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 30;
combined-reader p. 88; printed p. 78.*

Moreover, when Y/S satisfies E , 3.13.1 gives, just as in the proof of 3.13.2, for every $u \in \underline{\text{Isom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)$,

$$L^u_{\underline{\text{Isom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S}(\mathcal{M}) = L^u_{\underline{\text{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, Y)/S}(\mathcal{M}). \quad (*)$$

⁶⁰Editor's note: the following sentence has been added.

⁶¹Editor's note: on the right-hand side, the scalar action is $(a\varphi)(x) = a\varphi(x)$, where $a\varphi(x)$ denotes the action of a on $\varphi(x) \in \underline{\text{Lie}}(Y/S, \mathcal{M})$. This differs from the action $(a \cdot' \varphi)(x) = a \cdot' \varphi(x) = \varphi(ax)$, but the two agree when $\underline{\text{Lie}}(Y/S, \mathcal{M}) = \underline{\text{Lie}}'(Y/S, \mathcal{M})$.

⁶²Editor's note: this remark has been added; it will be useful in 4.5.1.

Corollary 4.3.1.

Let X be an $\mathcal{O}_S$ -module satisfying E over S . There is an isomorphism, functorial in $\mathcal{M}$,

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(X, \underline{\mathrm{Lie}}'(X/S, \mathcal{M})),$$

and this isomorphism respects the $\mathcal{O}_S$ -module structures obtained from functoriality in $\mathcal{M}$. In particular, $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S$ satisfies E .⁶³

Proof. The first assertion follows from $(*)$ and 4.3. For the second, since X/S satisfies E , there are isomorphisms of $\mathcal{O}_S$ -modules

$$\underline{\mathrm{Lie}}'(X/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(X/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(X/S, \mathcal{N}).$$

Consequently,

$$\begin{aligned} & \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M} \oplus \mathcal{N}) \\ & \simeq \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(X)/S, \mathcal{N}). \end{aligned}$$

Remark 4.1.1.2a now proves the assertion.

4.3.2.

Before continuing in this direction, let us examine more closely the relations among Y , $\underline{\mathrm{Lie}}(Y/S)$, and $\underline{\mathrm{Lie}}'(Y/S)$. First,

$$\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(\mathcal{O}_S/S, \mathcal{M}) = \mathbf{W}(\mathcal{M}), \quad (1)$$

where $\mathbf{W}(\mathcal{M})$ is defined in I 4.6. There is therefore a canonical isomorphism

$$d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S). \quad (2)$$

Let F be an $\mathcal{O}_S$ -module. For every $S_2 \rightarrow S_1 \rightarrow S$, there is a dihomomorphism

$$\begin{cases} F(S_1) \longrightarrow F(S_2), \\ \mathcal{O}(S_1) \longrightarrow \mathcal{O}(S_2), \end{cases} \quad (3)$$

and hence a morphism of $\mathcal{O}(S_2)$ -modules

$$F(S_1) \otimes_{\mathcal{O}(S_1)} \mathcal{O}(S_2) \longrightarrow F(S_2).$$

⁶⁴

In particular, take $S_1 = S'$ and $S_2 = I_{S'}(\mathcal{M})$. One obtains morphisms of $\mathcal{O}(S')$ -modules, functorial in $\mathcal{M}$,

$$F(S') \otimes_{\mathcal{O}(S')} T_{\mathcal{O}_S/S}(\mathcal{M})(S') \longrightarrow T'_{F/S}(\mathcal{M})(S').$$

As S' varies, these give morphisms of $\mathcal{O}_S$ -modules

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}), \quad (4)$$

functorial in $\mathcal{M}$. Compatibility with the tangent-bundle projections onto their bases then yields morphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}), \quad (5)$$

⁶³Editor's note: compare editor's note (61). The final sentence and its proof have also been added.

⁶⁴Editor's note: $S' \rightarrow S$ has been replaced by $S_2 \rightarrow S_1 \rightarrow S$, since both S_2 and S_1 must vary below. The original has also been expanded in what follows.

and the following diagram commutes.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) & \longrightarrow & F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \parallel \\
 0 & \longrightarrow & \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) & \longrightarrow & T'_{F/S}(\mathcal{M}) & \longrightarrow & F \longrightarrow 0.
 \end{array}$$

Figure 42: The morphisms (4) and (5) in the split tangent sequences.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 31;
combined-reader p. 89; printed p. 79.*

The morphisms (5) may be regarded as morphisms of abelian S -groups

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}). \quad (6)$$

Tensoring F with the isomorphism $d : \mathcal{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathcal{O}_S/S)$ gives, when $\mathcal{M} = \mathcal{O}_S$, a morphism of abelian S -groups

$$F \xrightarrow{\sim} F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S) \longrightarrow \underline{\mathrm{Lie}}(F/S), \quad (7)$$

also denoted $d : F \rightarrow \underline{\mathrm{Lie}}(F/S)$.

Remark 4.3.3.

When F/S satisfies E , the morphisms (6) and (7) need not be $\mathcal{O}_S$ -linear if both sides are equipped with the module structures obtained from $\mathcal{M}$ by condition E .

For example, let k be a field of characteristic $p > 0$, let $S = \mathrm{Spec}(k)$, and let F be the $\mathcal{O}_S$ -module that assigns to every S -scheme T the set $F(T) = \Gamma(T, \mathcal{O}_T)$, with scalars acting through the p -th power:

$$r \cdot f = r^p f \quad (r \in \mathcal{O}(T), f \in F(T)).$$

As an S -functor in groups, $F \simeq \mathbf{G}_{a,S}$. Hence F satisfies E and $\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbf{G}_{a,S}/S) \simeq \mathcal{O}_S$. For every $T \rightarrow S$, the canonical morphism $d : F(T) \rightarrow \mathcal{O}(T)$ is the identity: it respects the abelian-group structure but not the $\mathcal{O}_S$ -module structure. ⁶⁵

Remark 4.3.4.

The morphisms (4) and (5) can be made explicit as follows. The morphism

$$\Theta : F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}) = \underline{\mathrm{Hom}}_S(I_S(\mathcal{M}), F)$$

is defined, for $S' \rightarrow S$, $\alpha \in \mathcal{O}(I_{S'}(\mathcal{M}))$, and $f : S' \rightarrow F$, by

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where $\tau_0 : F \rightarrow T'_{F/S}(\mathcal{M})$ is the zero section and $\rho : I_{S'}(\mathcal{M}) \rightarrow S'$ is the structural morphism.

The morphism Θ induces

$$\theta : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}).$$

⁶⁵Editor's note: the original asserted that, when F/S satisfies E , (6), and hence (7), are morphisms of $\mathcal{O}_S$ -modules. The counterexample in 6.3 contradicts this. That assertion has been removed, the preceding example inserted, and Definition 4.4 modified accordingly. This does not affect what follows.

This follows from the functoriality in $\mathcal{M}$ noted after (4), and can also be seen directly. Indeed,

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M})(S') = \{ \varphi \in \mathrm{Hom}_S(I_{S'}(\mathcal{M}), F) \mid \varphi \circ \varepsilon_{\mathcal{M}} = e \},$$

where $e : S' \rightarrow F$ is the unit section. On the other hand, $\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})(S')$ equals $\Gamma(S', \mathcal{M})$, the kernel of the augmentation

$$\eta : \mathcal{O}(I_{S'}(\mathcal{M})) \longrightarrow \mathcal{O}(S').$$

It remains to see that if $f \in F(S')$ and $\alpha \in \Gamma(S', \mathcal{M})$, then $\alpha \cdot (f \circ \rho) \circ \varepsilon_{\mathcal{M}} = e$.

Apply the dihomomorphism (3) to the S -morphism $\varepsilon_{\mathcal{M}} : S' \rightarrow I_{S'}(\mathcal{M})$. The following diagram commutes.

$$\begin{array}{ccc} F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S') \\ \alpha \downarrow & & \downarrow \eta(\alpha) \\ F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S') \end{array} \quad .$$

Figure 43: Compatibility of scalar action with $F(\varepsilon_{\mathcal{M}})$.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 32;
combined-reader p. 90; printed p. 80.*

Thus, for every $\varphi : I_{S'}(\mathcal{M}) \rightarrow F$,

$$(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = \eta(\alpha) \cdot (\varphi \circ \varepsilon_{\mathcal{M}}),$$

and therefore $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = e$ when $\eta(\alpha) = 0$.

In particular, taking $\mathcal{M} = t\mathcal{O}_S$, one has $\underline{\mathrm{Lie}}(\mathcal{O}_S/S) = t\mathcal{O}_S$, and the morphism $F \rightarrow \underline{\mathrm{Lie}}'(F/S)$ is $f \mapsto t \cdot (f \circ \rho)$.

⁶⁶

Remark 4.3.5.

Let F be an $\mathcal{O}_S$ -module, set

$$E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

and denote by d_F and d_E the $\mathcal{O}_S$ -linear morphisms given by 4.3.2(5):

$$d_F : F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(F/S, \mathcal{M}),$$

$$d_E : E \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}'(E/S, \mathcal{M}).$$

Remark 4.3.4 gives the following commutative diagram of $\mathcal{O}_S$ -linear morphisms.

$$\begin{array}{ccc} E \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \underline{\mathrm{Lie}}'(E/S, \mathcal{M}) \\ \parallel & & \uparrow \simeq (*) \\ \underline{\mathrm{Hom}}_{\mathcal{O}_S}(F, F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \underline{\mathrm{Hom}}_{\mathcal{O}_S}(F, \underline{\mathrm{Lie}}'(F/S, \mathcal{M})) \end{array}$$

Figure 44: Compatibility of d_E and d_F .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 32;
combined-reader p. 90; printed p. 80.*

⁶⁶Editor's note: this remark has been added; it will be useful in 4.6.2.

The right vertical arrow is the isomorphism $(*)$ of 4.3. Thus, by that result, if F/S satisfies E , then E/S satisfies E , and $(*)$ is also an isomorphism of $\mathcal{O}_S$ -modules when the right-hand terms are equipped with the module structures obtained from E .⁶⁷

Remark 4.4.0.

In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. If these equivalent conditions hold, then F/S satisfies E . Indeed, it is enough to verify that

$$\underline{\mathrm{Lie}}'(F/S, \mathcal{M} \oplus \mathcal{N}) \simeq \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(F/S, \mathcal{N}).$$

Under the hypothesis, the horizontal arrows in the following commutative diagram are isomorphisms.

$$\begin{array}{ccc} F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M} \oplus \mathcal{N}) & \xrightarrow{\sim} & \underline{\mathrm{Lie}}'(F/S, \mathcal{M} \oplus \mathcal{N}) \\ \downarrow \wr & & \downarrow \\ F \otimes_{\mathcal{O}_S} (\underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{N})) & \xrightarrow{\sim} & \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \times_S \underline{\mathrm{Lie}}'(F/S, \mathcal{N}); \end{array}$$

Figure 45: The direct-sum comparison used to verify condition E .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 33;
combined-reader p. 91; printed p. 81.*

The second vertical arrow is therefore an isomorphism as well, which is exactly condition E for F/S .⁶⁸

Definition 4.4.

An $\mathcal{O}_S$ -module F is called *good* if the morphisms

$$F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow T_{F/S}(\mathcal{M}),$$

or, equivalently,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(F/S, \mathcal{M}),$$

are isomorphisms of abelian S -groups (so that F/S satisfies (E)), and if moreover they respect the $\mathcal{O}_S$ -module structures obtained from condition E .

Corollary 4.4.1.

Let F be an $\mathcal{O}_S$ -module, and consider the following conditions:⁶⁹

- (i) F is a good $\mathcal{O}_S$ -module;
- (ii) F/S satisfies E , and $d : F \rightarrow \underline{\mathrm{Lie}}(F/S)$ is an isomorphism of $\mathcal{O}_S$ -modules;
- (iii) $\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$.

Then

$$(i) \iff (ii) \implies (iii).$$

⁶⁷Editor's note: this remark has been added; it will be useful in 4.5 and 4.7.

⁶⁸Editor's note: in view of the changes made in Definition 4.4 (cf. editor's note (65)), this remark has been inserted here; in the original it occurred in the proof of 4.4.1.

⁶⁹Editor's note: the original proved that (i) implies (ii) and (iii); the implication (ii) implies (i) has been added.

Proof. The implication (i) $\Rightarrow$ (ii) follows from the definition. To prove (ii) $\Rightarrow$ (i), it remains to show that the morphisms of abelian S -groups, functorial in $\mathcal{M}$,

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}(F/S, \mathcal{M})$$

are isomorphisms of $\mathcal{O}_S$ -modules. Since F/S satisfies E , both sides carry finite direct sums of copies of $\mathcal{O}_S$ to finite products of abelian S -groups. We are therefore reduced to the case $\mathcal{M} = \mathcal{O}_S$, which is precisely the hypothesis.

Finally, (i) $\Rightarrow$ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Lie}}'(F/S, \mathcal{M})$$

of 4.3.2(5) are $\mathcal{O}_S$ -linear.

Examples 4.4.2.

For every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$, the $\mathcal{O}_S$ -modules $\mathbf{V}(\mathcal{E})$ and $\mathbf{W}(\mathcal{E})$ defined in I 4.6 are good.

Indeed, for every $f : S' \rightarrow S$, the morphisms

$$\begin{aligned} \mathbf{V}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) &\longrightarrow T_{\mathbf{V}(\mathcal{E})/S}(\mathcal{M})(S'), \\ \mathbf{W}(\mathcal{E})(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) &\longrightarrow T_{\mathbf{W}(\mathcal{E})/S}(\mathcal{M})(S') \end{aligned}$$

correspond respectively to

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathcal{O}_{S'}) \otimes_{\mathcal{O}(S')} \Gamma(S', D_{\mathcal{O}_{S'}}(\mathcal{M})) \\ \longrightarrow \mathrm{Hom}_{\mathcal{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), D_{\mathcal{O}_{S'}}(\mathcal{M})) \quad (\mathrm{V}) \\ \Gamma(S', f^*(\mathcal{E})) \otimes_{\mathcal{O}(S')} \Gamma(S', D_{\mathcal{O}_{S'}}(\mathcal{M})) \\ \longrightarrow \Gamma(S', f^*(\mathcal{E}) \otimes_{\mathcal{O}_{S'}} D_{\mathcal{O}_{S'}}(\mathcal{M})) \quad (\mathrm{W}). \end{aligned}$$

These are isomorphisms because, as an $\mathcal{O}_{S'}$ -module, $D_{\mathcal{O}_{S'}}(\mathcal{M})$ is isomorphic to a finite direct sum of copies of $\mathcal{O}_{S'}$.⁷⁰

Proposition 4.5. *Let F be a good $\mathcal{O}_S$ -module. Then:*

(i) *The S -group $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ satisfies condition (E), and there is a functorial isomorphism*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\mathcal{O}_S\text{-mod.}}(F, \underline{\mathrm{Lie}}(F/S, \mathcal{M})),$$

which respects the $\mathcal{O}_S$ -module structures obtained from condition (E). In particular, there is an isomorphism of $\mathcal{O}_S$ -modules

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \xrightarrow{\sim} \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

(ii)⁷¹ *Moreover, $\underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$ is a good $\mathcal{O}_S$ -module.*

Indeed, by 4.4.1, F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S, \mathcal{M}) = \underline{\mathrm{Lie}}'(F/S, \mathcal{M}) \simeq F \otimes_{\mathcal{O}_S} \underline{\mathrm{Lie}}(\mathcal{O}_S/S, \mathcal{M}). \quad (1)$$

Assertion (i) now follows from 4.3.1. Put $E = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$. By (1) and Remark 4.3.5, one has the following commutative diagram of morphisms of abelian S -groups.

⁷⁰Editor's note: the preceding proof has been added.

⁷¹Editor's note: part (ii), an immediate consequence of what precedes, has been added.

$$\begin{array}{ccc}
\underline{\text{End}}_{\mathcal{O}_S\text{-mod.}}(F) \otimes_{\mathcal{O}_S} \underline{\text{Lie}}(\mathcal{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \underline{\text{Lie}}(\underline{\text{End}}_{\mathcal{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \\
\parallel & & \simeq \uparrow (*) \\
\underline{\text{Hom}}_{\mathcal{O}_S}(F, F \otimes_{\mathcal{O}_S} \underline{\text{Lie}}(\mathcal{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \underline{\text{Hom}}_{\mathcal{O}_S}(F, \underline{\text{Lie}}(F/S, \mathcal{M}))
\end{array}$$

Figure 46: The comparison diagram for the differential d_E .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 34;
combined-reader p. 92; printed p. 82.*

Here d_F and the vertical isomorphism $(*)$ are isomorphisms of $\mathcal{O}_S$ -modules; consequently, so is d_E . This proves (ii).

Scholium 4.5.1. ⁷² Put (cf. 2.1) $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$, and let F be a good $\mathcal{O}_S$ -module. For every $S' \rightarrow S$, the morphism

$$F(S') \oplus tF(S') = F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}) \longrightarrow F(I_{S'}) = F(S') \oplus \underline{\text{Lie}}(F/S)(S'),$$

which is the identity on $F(S')$, induces an isomorphism of $\mathcal{O}(S')$ -modules

$$tF(S') \xrightarrow{\sim} \underline{\text{Lie}}(F/S)(S').$$

As S' varies, this gives an isomorphism that we may write

$$\underline{\text{Lie}}(F/S) \simeq tF.$$

Thus, for every $S' \rightarrow S$, Proposition 4.5 gives a commutative diagram.

$$\begin{array}{ccccc}
\underline{\text{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}) & \xrightarrow{\sim} & \underline{\text{Hom}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}, tF_{S'}) & \xrightarrow{\sim} & \underline{\text{Lie}}(\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S)(S') \\
& & \downarrow & & \downarrow \\
& & \underline{\text{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}}) & \xlongequal{\quad} & T_{\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S}(S')
\end{array}$$

Figure 47: Endomorphisms and automorphisms over the dual numbers.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 35;
combined-reader p. 93; printed p. 83.*

It follows from 4.3 bis that every

$$X \in \underline{\text{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$$

corresponds to $\text{id} + tX \in \underline{\text{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$.

Definition 4.6. An S -functor in groups G is called *good* if G/S satisfies condition (E) and $\underline{\text{Lie}}(G/S)$ is a good $\mathcal{O}_S$ -module.

⁷³ Notice that if F is a good $\mathcal{O}_S$ -module, then it is a good S -group: indeed, F/S satisfies (E), and $\underline{\text{Lie}}(F/S) \simeq F$ is a good $\mathcal{O}_S$ -module.

⁷²Editor's note: this scholium, implicit in the original, has been added for use in 4.7.1. Here and below, t, t' , and so forth denote square-zero variables; the letter ε is reserved for the unit section of a group.

⁷³Editor's note: the following sentence has been added.

Example 4.6.1. *If G is representable, it is good. Indeed, G/S satisfies (E), and $\underline{\text{Lie}}(G/S)$ has the form $V(E)$, hence is good by 4.4.2.*

Lemma 4.6.2. ⁷⁴ *Let G be an S -functor in groups such that G/S satisfies (E), and put $F = \underline{\text{Lie}}(G/S)$. Then F satisfies (E), and the morphism of abelian groups*

$$d : F \longrightarrow \underline{\text{Lie}}(F/S)$$

respects the $\mathcal{O}_S$ -module structures. Consequently, G is good if and only if $F \rightarrow \underline{\text{Lie}}(F/S)$ is bijective.

Proof. Suppose that G/S satisfies (E). Let $\mathcal{M}, \mathcal{N}$ be finite free $\mathcal{O}_S$ -modules, write

$$F(\mathcal{N}) = \underline{\text{Lie}}(G/S, \mathcal{N}),$$

and let e denote the unit section of G . For every $S' \rightarrow S$,

$$F(\mathcal{N})(S') = \{g \in \text{Hom}_S(I_{S'}(\mathcal{N}), G) \mid g \circ \varepsilon_{\mathcal{N}} = e\}.$$

Moreover, $\underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M})(S')$ identifies with the set of morphisms

$$\left\{ \Phi \in \text{Hom}_S(I_{S'}(\mathcal{N}) \times_{S'} I_{S'}(\mathcal{M}), G) \left| \begin{array}{l} \Phi \circ (\varepsilon_{\mathcal{N}} \times \text{id}_{I_{S'}(\mathcal{M})}) = e \in G(I_{S'}(\mathcal{M})), \\ \Phi \circ (\text{id}_{I_{S'}(\mathcal{N})} \times \varepsilon_{\mathcal{M}}) = e \in G(I_{S'}(\mathcal{N})). \end{array} \right. \right\}.$$

This proves

$$\underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}) \simeq \underline{\text{Lie}}(F(\mathcal{M})/S, \mathcal{N}). \quad (1)$$

Since G/S satisfies (E), it follows that

$$\begin{aligned} \underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) &\simeq \underline{\text{Lie}}(F(\mathcal{M}_1 \oplus \mathcal{M}_2)/S, \mathcal{N}) \\ &\simeq \underline{\text{Lie}}((F(\mathcal{M}_1) \times_S F(\mathcal{M}_2))/S, \mathcal{N}). \end{aligned}$$

By 3.8, the last term is isomorphic to

$$\begin{aligned} \underline{\text{Lie}}(F(\mathcal{M}_1)/S, \mathcal{N}) \times_S \underline{\text{Lie}}(F(\mathcal{M}_2)/S, \mathcal{N}) \\ \simeq \underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2). \end{aligned}$$

Hence

$$\underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) \simeq \underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}_1) \times_S \underline{\text{Lie}}(F(\mathcal{N})/S, \mathcal{M}_2), \quad (2)$$

so $F(\mathcal{N})/S$ satisfies (E), by Remark 4.1.1.2(a).

It remains to show that the morphism of abelian groups

$$d : F(\mathcal{N}) \longrightarrow \underline{\text{Lie}}(F(\mathcal{N})/S)$$

respects the $\mathcal{O}_S$ -module structures. Consider the free $\mathcal{O}_S$ -module $\mathcal{M} = t\mathcal{O}_S$. Then

$$\mathcal{O}(I_{S'}(\mathcal{M})) = \mathcal{O}(S')[t]/(t^2) = \mathcal{O}(S') \oplus t\mathcal{O}(S'), \quad \underline{\text{Lie}}(\mathcal{O}_S/S) \simeq t\mathcal{O}_S.$$

Let $\rho_t : I_S(\mathcal{M}) \rightarrow S$ be the structural morphism, corresponding to the injection

$$u_t : \mathcal{O}_S \hookrightarrow D_{\mathcal{O}_S}(\mathcal{M}) = \mathcal{O}_S \oplus t\mathcal{O}_S.$$

⁷⁴Editor's note: this lemma has been added.

Recall (cf. 3.4.2) that for every S -functor X , $F(\mathcal{N})(X)$ is the set of S -morphisms $\phi : I_X(\mathcal{N}) \rightarrow G$ such that $\phi \circ \varepsilon_{\mathcal{N}} = e$, and the action of $a \in \mathcal{O}(X)$ is

$$a \cdot \phi = \phi \circ a^*,$$

where a^* is the endomorphism of $I_X(\mathcal{N})$ associated with a ; see 2.1.3.

Consequently, by 4.3.4, the composite

$$d : F(\mathcal{N}) \xrightarrow{\sim} F(\mathcal{N}) \otimes_{\mathcal{O}_S} t\mathcal{O}_S \longrightarrow \underline{\text{Lie}}(F(\mathcal{N})/S)$$

is given, for $S' \rightarrow S$ and $f \in \text{Hom}_S(S', F(\mathcal{N}))$, by

$$f \longmapsto f \otimes t \longmapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*.$$

Here $f \circ \rho_t \in F(\mathcal{N})(I_{S'}(\mathcal{M}))$ is viewed as an S -morphism

$$\phi : I_{I_{S'}(\mathcal{M})}(\mathcal{N}) \longrightarrow G$$

such that $\phi \circ \varepsilon_{\mathcal{N}} = e$. We must prove that d is $\mathcal{O}_S$ -linear, or equivalently that

$$d(a \cdot f) = u_t(a) \cdot d(f) \quad (a \in \mathcal{O}(S')).$$

Viewing $f \in F(\mathcal{N})(S')$ as a morphism $I_{S'}(\mathcal{N}) \rightarrow G$, one likewise has $a \cdot f = f \circ a^*$. Functoriality in X of the action of $\mathcal{O}(X)$ on $I_X(\mathcal{N})$ gives the following commutative diagram.

$$\begin{array}{ccc} I_{S'}(\mathcal{N}) & \xrightarrow{a^*} & I_{S'}(\mathcal{N}) \\ \rho_t \uparrow & & \uparrow \rho_t \\ I_{I_{S'}(\mathcal{M})}(\mathcal{N}) & \xrightarrow{u_t(a)^*} & I_{I_{S'}(\mathcal{M})}(\mathcal{N}) \end{array} .$$

Figure 48: Compatibility of scalar action with the structural morphism ρ_t .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 36;
combined-reader p. 94; printed p. 84.*

Therefore

$$\begin{aligned} d(a \cdot f) &= f \circ a^* \circ \rho_t \circ t^* \\ &= f \circ \rho_t \circ u_t(a)^* \circ t^* \\ &= f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f). \end{aligned}$$

The penultimate equality follows from the commutativity of $\mathcal{O}$. This completes the proof of Lemma 4.6.2.

Theorem 4.7. *If F is a good $\mathcal{O}_S$ -module, then the S -group $\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ is good.*

⁷⁵ Indeed, by 4.5, $\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S$ satisfies (E), while

$$\underline{\text{Lie}}(\underline{\text{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \simeq \underline{\text{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a good $\mathcal{O}_S$ -module.

⁷⁵Editor's note: the original has been simplified here by using the additions made in 4.3.1 and 4.5.

4.7.1.

⁷⁶ Let G be an S -group and F a good $\mathcal{O}_S$ -module. Suppose given a linear representation of G in F , that is (I 4.7.1), a morphism of S -groups

$$\rho : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F).$$

If G/S satisfies (E), then 4.1.C and 4.5 give a morphism of $\mathcal{O}_S$ -modules, denoted ρ' or $d\rho$,

$$\underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) \simeq \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

⁷⁷ Furthermore, put $\mathcal{O}_{I_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$, with $t^2 = 0$. It follows from 4.5.1 that, if $S' \rightarrow S$ and

$$X \in \underline{\mathrm{Lie}}(G/S)(S') \subset G(I_{S'}),$$

then in $\underline{\mathrm{Aut}}_{\mathcal{O}_{I_{S'}}\text{-mod.}}(F_{I_{S'}})$ one has

$$\rho(X) = \mathrm{id} + t\rho'(X). \quad (*)$$

In other words, for every $S'' \rightarrow I_{S'}$ and $f \in F(S'')$,

$$\rho(X)(f) = f + t\rho'(X)(f) \quad \text{in } F(S'').$$

Definition 4.7.2. Let G be a good S -group. Then $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$ -module, and there is a morphism of S -groups

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S)).$$

By 4.7.1, it induces a morphism of $\mathcal{O}_S$ -modules

$$\mathrm{ad} : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S)),$$

or, equivalently, an $\mathcal{O}_S$ -bilinear morphism

$$\underline{\mathrm{Lie}}(G/S) \times_S \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{Lie}}(G/S), \quad (x, y) \longmapsto [x, y] = \mathrm{ad}(x) \cdot y.$$

Here x, y are arbitrary elements of

$$\underline{\mathrm{Lie}}(G/S)(S') = \underline{\mathrm{Lie}}(G_{S'}/S')(S').$$

If G is commutative, then $[x, y] = 0$.

4.7.3.

The bracket admits the following equivalent description. It is enough to consider sections over S , that is,

$$x, y \in \underline{\mathrm{Lie}}(G/S)(S).$$

There is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$. To avoid confusion, let I and I' denote two copies of I_S , and put

$$\mathcal{O}_I = \mathcal{O}_S[t], \quad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \quad t^2 = 0 = (t')^2.$$

⁷⁶ Editor's note: the numbering 4.7.1, 4.7.2, and 4.7.3 has been added.

⁷⁷ Editor's note: the following sentence has been added.

There is then a commutative square

$$\begin{array}{ccc} I \times I' & \longrightarrow & I' \\ \downarrow & & \downarrow \\ I & \longrightarrow & S \end{array},$$

Figure 49: The two dual-number structures on $I \times_S I'$.

*Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 37;
combined-reader p. 95; printed p. 85.*

in which the two arrows from $I \times I'$ identify it with the scheme of dual numbers over I or over I' , respectively. Writing $L = \underline{\mathrm{Lie}}(G/S)$, one obtains the following commutative diagram of groups.

$$(1) \quad \begin{array}{ccccccc} & & 1 & & 1 & & \\ & & \downarrow & & \downarrow & & \\ & & L(I) & \longrightarrow & L(S) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & L(I') & \longrightarrow & G(I \times I') & \longrightarrow & G(I') \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & L(S) & \longrightarrow & G(I) & \longrightarrow & G(S) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 1 & & 1 & & 1 \end{array}.$$

Figure 50: The first exact-sequence diagram used to construct the bracket.

*Temporary Loop-1 image; source locator: Polo–Gille Exp2-14oct24.pdf, component-local p. 38;
combined-reader p. 96; printed p. 86.*

The ninth object in this diagram is $\underline{\mathrm{Lie}}(L/S)(S)$. If G is good, this is $L(S)$, and hence one obtains the next commutative diagram. Its rows and columns are exact sequences of groups; the five groups denoted $L(-)$ are commutative.⁷⁸ Under the identifications

$$L(I) = L(S) \oplus tL(S), \quad L(I') = L(S) \oplus t'L(S),$$

the injections $L(S) \hookrightarrow L(I)$ and $L(S) \hookrightarrow L(I')$ are $u \mapsto tu$ and $u \mapsto t'u$, respectively.

⁷⁸Editor's note: the original has been expanded in what follows.

$$(2) \quad \begin{array}{ccccc} z \in L(S) & \xrightarrow{t} & L(I) & \longrightarrow & L(S) \ni x \\ \downarrow t' & & \downarrow & & \downarrow \\ L(I') & \longrightarrow & G(I \times I') & \longrightarrow & G(I') \\ \downarrow & & \downarrow & & \downarrow \\ y \in L(S) & \longrightarrow & G(I) & \longrightarrow & G(S) \end{array} .$$

Figure 51: The exact-sequence diagram defining $[x, y]$.

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 38; combined-reader p. 96; printed p. 86.

In such a diagram, take x, y as indicated and choose arbitrary lifts $\tilde{x} \in L(I)$ of x and $y' \in L(I')$ of y . The commutator

$$\tilde{x} y' \tilde{x}^{-1} (y')^{-1} \quad \text{in } G(I \times I')$$

is independent of the chosen lifts and is the image of a uniquely determined element z as shown. One has $z = [x, y]$.⁷⁹

Indeed, use the same letters x and y for their images under the canonical sections $L(S) \rightarrow L(I)$ and $L(S) \rightarrow L(I')$. Then

$$\tilde{x} = xu, \quad y' = yv, \quad u, v \in L(S) = L(I) \cap L(I').$$

Since $L(I)$ and $L(I')$ are commutative,

$$\begin{aligned} \tilde{x} y' \tilde{x}^{-1} (y')^{-1} &= xuyvu^{-1}x^{-1}v^{-1}y^{-1} \\ &= xuyu^{-1}vx^{-1}v^{-1}y^{-1} \\ &= xyx^{-1}y^{-1}. \end{aligned}$$

This element maps to the unit in both $G(I)$ and $G(I')$, and therefore comes from the unique z indicated above. Finally, viewing y as an element of $L_{I'}(I')$, and x as an element of $L(S) \subset G(I')$, formula 4.7.1 (*) gives

$$xyx^{-1} = \text{Ad}(x)(y) = (\text{id} + t' \text{ad}(x))(y) = y + t'[x, y].$$

Thus $xyx^{-1}y^{-1} \in L(I')$ is the image of $z = [x, y] \in L(S)$.

Two properties are now apparent from this construction:

(i) the bracket is functorial in G . More precisely,

$$G \longmapsto \underline{\text{Lie}}(G/S)$$

is a functor from good S -groups to good $\mathcal{O}_S$ -modules equipped with an $\mathcal{O}_S$ -bilinear composition law;

(ii) $[x, y] + [y, x] = 0$, since the diagram is symmetric about its main diagonal.⁸⁰

⁷⁹Editor's note: the original said in a footnote that the author, at Gabriel's request, acknowledged that the exercise was not immediate, which was precisely why it had not been placed in the text. The details have been supplied here, taking account of the addition in 4.7.1; see also [DG70], §II.4, 4.2.

⁸⁰Editor's note: that is, x and y play symmetric roles. The image of $[y, x]$ in $G(I \times I')$ is the commutator $y'\tilde{x}(y')^{-1}\tilde{x}^{-1}$, the inverse of $\tilde{x}y'\tilde{x}^{-1}(y')^{-1} = [x, y]$. On the other hand (cf. 4.10 below), it is not known at this point whether $[x, x] = 0$ necessarily holds: if x is represented by $\tilde{x} \in L(I)$ and by $x' \in L(I')$, it is not a priori clear that $\tilde{x}$ and x' commute.

Proposition 4.8. *Let F be a good $\mathcal{O}_S$ -module. Under the identification*

$$\underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S) = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F),$$

one has

$$\mathrm{Ad}(g) \cdot Y = g \circ Y \circ g^{-1}, \quad [X, Y] = X \circ Y - Y \circ X,$$

for every $S' \rightarrow S$, $g \in \underline{\mathrm{Aut}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'})$, and

$$X, Y \in \underline{\mathrm{Lie}}(\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)/S)(S') = \underline{\mathrm{End}}_{\mathcal{O}_{S'}\text{-mod.}}(F_{S'}).$$

*Proof.*⁸¹ After base change, we may assume $S' = S$. Put $I = I_S$ and $\mathcal{O}_I = \mathcal{O}_S[t]$, with $t^2 = 0$. Recall from 4.5.1 that the inclusion

$$i : \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F) \hookrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_I\text{-mod.}}(F_I)$$

sends Y to $\mathrm{id} + tY$. By the definition of $\mathrm{Ad}(g)$ in 4.1.A,

$$\mathrm{id} + t \mathrm{Ad}(g)(Y) = g \circ (\mathrm{id} + tY) \circ g^{-1} = \mathrm{id} + t(g \circ Y \circ g^{-1}),$$

and therefore $\mathrm{Ad}(g)(Y) = g \circ Y \circ g^{-1}$.

Let I' be a second copy of I_S , and put $\mathcal{O}_{I'} = \mathcal{O}_S[t']$, with $(t')^2 = 0$. Apply 4.7.3 to

$$G = \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F), \quad L = \underline{\mathrm{Lie}}(G/S) = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F).$$

Here $L = \underline{\mathrm{End}}(F)$ is the differential identification of Proposition 4.5. The re-edition prints $\underline{\mathrm{Aut}}(F)$ at this point, but $\underline{\mathrm{End}}(F)$ is the type-correct object used by the ensuing endomorphism calculation. Identify X with its image under the canonical section $L(S) \hookrightarrow L(I)$. Its image in $G(I \times I')$ is $\mathrm{id} + t'X$, whose inverse is $\mathrm{id} - t'X$. Similarly, Y lifts to $\mathrm{id} + tY$, with inverse $\mathrm{id} - tY$. Their commutator is

$$\begin{aligned} & (\mathrm{id} + t'X) \circ (\mathrm{id} + tY) \circ (\mathrm{id} - t'X) \circ (\mathrm{id} - tY) \\ &= \mathrm{id} + tt'(X \circ Y - Y \circ X). \end{aligned}$$

This is the image in $G(I \times I')$ of $Z = X \circ Y - Y \circ X \in L(S)$: first Z maps to $tZ \in L(I)$, and then to $\mathrm{id} + t'tZ \in G(I \times I')$. By 4.7.3, $[X, Y] = X \circ Y - Y \circ X$.

Corollary 4.8.1. *Let G be a good S -group and let $x, y, z \in \underline{\mathrm{Lie}}(G/S)(S')$. Then*

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

⁸² Indeed, since G is good, $\underline{\mathrm{Lie}}(G/S)$ is a good $\mathcal{O}_S$ -module; hence Theorem 4.7 shows that

$$\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S))$$

is a good S -group. The morphism

$$\mathrm{Ad} : G \longrightarrow \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\underline{\mathrm{Lie}}(G/S))$$

and the functoriality in 4.7.3(i) yield

$$\mathrm{ad}[x, y] = [\mathrm{ad} x, \mathrm{ad} y].$$

⁸¹Editor's note: the original has been expanded and simplified in what follows.

⁸²Editor's note: the original has been expanded in what follows.

Together with Proposition 4.8, this gives

$$\mathrm{ad}[x, y] = \mathrm{ad} x \circ \mathrm{ad} y - \mathrm{ad} y \circ \mathrm{ad} x.$$

Applying both sides to z gives the Jacobi identity.

Corollary 4.8.2. *Let G be a good S -group acting linearly on a good $\mathcal{O}_S$ -module F ; equivalently, let F be a G - $\mathcal{O}_S$ -module, with G and F good. Then the linear map*

$$\rho' : \underline{\mathrm{Lie}}(G/S) \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod.}}(F)$$

is a representation; that is,

$$\rho'([x, y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholium 4.9. To every good S -group G (for example, every representable one) we have functorially associated a good $\mathcal{O}_S$ -module $\underline{\mathrm{Lie}}(G/S)$ equipped with a bilinear map satisfying

$$[x, y] + [y, x] = 0, \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

We shall call $\underline{\mathrm{Lie}}(G/S)$, with this structure, the *Lie algebra* of G over S . The qualification is that it is not yet known whether it is an $\mathcal{O}_S$ -Lie algebra in the strict sense.⁸³ Every linear representation of G in a good $\mathcal{O}_S$ -module F induces a representation of its Lie algebra. In particular, the adjoint representation of G induces the adjoint representation of its Lie algebra.

Definition 4.10. An S -functor in groups G is called *very good* if it is good and $\underline{\mathrm{Lie}}(G/S)$ is an $\mathcal{O}_S$ -Lie algebra, that is, if $[x, x] = 0$ identically.

Examples 4.10.1. The following S -groups are very good: $\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(F)$ for every good $\mathcal{O}_S$ -module F (by 4.7 and 4.8); every representable group (as shown below); every good S -group admitting a monomorphism into a very good S -group, in particular every good subgroup functor of a representable group; and every good S -group admitting a faithful linear representation in a good $\mathcal{O}_S$ -module, for example every good S -group for which Ad is a monomorphism.

4.11.

Suppose now that G is a group scheme over S . By 4.1.4, $\underline{\mathrm{Lie}}(G/S)(S)$ identifies with the group of infinitesimal automorphisms of G/S invariant under right translation; by 3.14, this is the group of derivations of $\mathcal{O}_G$ over $\mathcal{O}_S$ invariant under right translation. This identification respects the module structure and is an anti-isomorphism of Lie algebras.⁸⁴

To see this, argue as in 4.7.3. Put

$$\mathcal{O}_I = \mathcal{O}_S[t], \quad \mathcal{O}_{I'} = \mathcal{O}_S[t'], \quad t^2 = 0 = (t')^2,$$

and let $x \in L(I)$, $y \in L(I')$, where $L = \underline{\mathrm{Lie}}(G/S)$. Left translation λ_x , respectively λ_y , is an S -automorphism of $G_{I \times I'}$ inducing the identity on $G_{I'}$, respectively on G_I . It corresponds to the $\mathcal{O}_S$ -automorphism

$$u = \mathrm{id} + t d_x, \quad \text{respectively} \quad v = \mathrm{id} + t' d_y$$

⁸³Editor's note: the unresolved point is whether $[x, x] = 0$; see 4.10.

⁸⁴Editor's note: a sign error in the original has been corrected here; cf. [DG70], §II.4, 4.4 and 4.6.

of

$$\mathcal{O}_{G_{I \times I'}} = \mathcal{O}_G[t, t']/(t^2, (t')^2),$$

where d_x, d_y are $\mathcal{O}_S$ -derivations of $\mathcal{O}_G$ invariant under right translation. The correspondence between S -automorphisms of $G_{I \times I'}$ and $\mathcal{O}_S$ -automorphisms of $\mathcal{O}_{G_{I \times I'}}$ is contravariant. Therefore

$$\lambda_x \lambda_y \lambda_x^{-1} \lambda_y^{-1}$$

corresponds to

$$v^{-1}u^{-1}vu = \text{id} + tt'(d_y d_x - d_x d_y).$$

It follows from 4.7.3 that

$$x \longmapsto -d_x$$

is an isomorphism of Lie algebras; for further details, see [DG70], §II.4, 4.4 and 4.6. The same argument applies to

$$\underline{\text{Lie}}(G/S)(S') = \underline{\text{Lie}}(G_{S'}/S')(S')$$

for every $S' \rightarrow S$. Thus one recovers the classical definition.⁸⁵

Scholium 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $\underline{\text{Lie}}(G/S)$ identifies with the functor that assigns to each $S' \rightarrow S$ the $\mathcal{O}(S')$ -Lie algebra of derivations of $G_{S'}$ relative to S' invariant under right translation.*

Since every representable group is already known to be good by 4.6.1, the preceding discussion gives:

Corollary 4.11.2. *Every representable group is very good.*

Let $\varepsilon : S \rightarrow G$ be the unit section of G , and put

$$\omega_{G/S}^1 = \varepsilon^*(\Omega_{G/S}^1).$$

Recall from 3.3 that $\underline{\text{Lie}}(G/S)$ is represented by the vector fibration $V(\omega_{G/S}^1)$.

Scholium 4.11.3. *Thus to every S -group scheme G we have functorially associated a vector fibration*

$$\text{Lie}(G/S) = V(\omega_{G/S}^1)$$

over S , representing the functor $\underline{\text{Lie}}(G/S)$, and hence endowed with the structure of an S -scheme in $\mathcal{O}_S$ -Lie algebras. Moreover, this construction commutes with base extension and finite products (cf. 3.4 and 3.8).

Remarks 4.11.4. ⁸⁶ Let $\pi : G \rightarrow S$ be the structural morphism.

(a) The $\mathcal{O}_G$ -module $\Omega_{G/S}^1$ is naturally $(G \times_S G)$ -equivariant (cf. I, §6). Hence, by I 6.8.1,

$$\Omega_{G/S}^1 \simeq \pi^*(\omega_{G/S}^1).$$

For example, $\Omega_{G/S}^1$ is locally free, respectively locally free of finite type, whenever $\omega_{G/S}^1$ is. This holds in particular when S is the spectrum of a field, respectively when S is the spectrum of a field and G is locally of finite type over S .

(b) By I 6.8.2, $\omega_{G/S}^1$ has a canonical G - $\mathcal{O}_S$ -module structure, which induces the adjoint action on

$$V(\omega_{G/S}^1) = \text{Lie}(G/S).$$

⁸⁵Editor's note: the original has been expanded in what follows. In particular, the numbering 4.11.1–4.11.8 has been added for later reference.

⁸⁶Editor's note: what follows has been expanded, taking account of the additions made in Exposé I, §6.8.

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(c) By EGA I, 5.3.11 and 5.4.6, $\varepsilon : S \rightarrow G$ is an immersion; it is a closed immersion if G is separated over S . Thus $\omega_{G/S}^1$ identifies with $\mathcal{I}/\mathcal{I}^2$, where $\mathcal{I}$ is the quasi-coherent ideal defining $\varepsilon(S)$ in an open subscheme $U \subset G$ in which $\varepsilon(S)$ is closed. If $G \rightarrow S$ is separated, one may take $U = G$. If $G = \operatorname{Spec} A(G)$ is affine over S , then $\mathcal{I}$ is the augmentation ideal of $A(G)$, namely the kernel of

$$\varepsilon^\sharp : A(G) \longrightarrow \mathcal{O}_S$$

(cf. I 4.2).⁸⁸

Remark 4.11.5. The isomorphism $\Omega_{G/S}^1 \simeq \pi^*(\omega_{G/S}^1)$ identifies the $\mathcal{O}_S$ -module $\omega_{G/S}^1$ with the sheaf

$$\pi_*(\Omega_{G/S}^1)^G$$

of differentials of G relative to S invariant under right translation: its sections over an open $U \subset S$ are the sections of $\Omega_{G/S}^1$ over $\pi^{-1}(U)$ invariant under right translation (cf. I 6.8.3; compare also VII_A, 2.4).⁸⁹

Notation 4.11.6. Write $\mathcal{L}ie(G/S)$ for the sheaf of sections of the vector fibration $\operatorname{Lie}(G/S) \rightarrow S$. It is the $\mathcal{O}_S$ -module

$$\mathcal{L}ie(G/S) = (\omega_{G/S}^1)^\vee = \mathcal{H}om_{\mathcal{O}_S}(\omega_{G/S}^1, \mathcal{O}_S),$$

dual to $\omega_{G/S}^1$ (cf. EGA II, 1.7.9), and it carries a structure of $\mathcal{O}_S$ -Lie algebra.

In general this construction does not commute with base extension. Consequently, the Lie-algebra structure on this module does not determine the structure of an S -scheme in $\mathcal{O}_S$ -Lie algebras on $\operatorname{Lie}(G/S)$.⁹⁰

Nevertheless:

Lemma 4.11.7. *Suppose that $\omega_{G/S}^1$ is locally free of finite type. This is the case, in particular, if G is smooth over S (cf. EGA IV₄, 17.2.4), or if S is the spectrum of a field and G is locally of finite type over S . Then*

$$\mathcal{L}ie(G/S)^\vee \simeq (\omega_{G/S}^1)^{\vee\vee} \simeq \omega_{G/S}^1,$$

and therefore

$$\operatorname{Lie}(G/S) = V(\omega_{G/S}^1) = V(\mathcal{L}ie(G/S)^\vee) = W(\mathcal{L}ie(G/S)),$$

where the final equality follows from I 4.6.5.1.

Finally, let $G \rightarrow H$ be a monomorphism of functors in groups. Then

$$\operatorname{Lie}(G/S) \longrightarrow \operatorname{Lie}(H/S)$$

⁸⁷Editor's note: for this reason, the linear action of G on $\omega_{G/S}^1$ could be called the pre-adjoint action; by an abuse of terminology it will still be called the adjoint action. A slightly more general construction, used in Expose III (especially III 0.8), is as follows. Suppose that to every $Y \rightarrow S$ one assigns an $\mathcal{O}_Y$ -module $\mathcal{N}(Y)$, and to every S -morphism $\phi : Z \rightarrow Y$ a functorial morphism of $\mathcal{O}_Z$ -modules $\mathcal{N}(\phi) : \phi^*\mathcal{N}(Y) \rightarrow \mathcal{N}(Z)$. For example, if $\mathcal{J}$ is a quasi-coherent ideal of $\mathcal{O}_S$, one may take $\mathcal{N}(Y) = \mathcal{J}\mathcal{O}_Y$. The action of G on $\omega_{G/S}^1$ then induces a generalized adjoint action of G on the S -functor in abelian groups that assigns to $f : Y \rightarrow S$ the group $\operatorname{Hom}_{\mathcal{O}_Y}(f^*\omega_{G/S}^1, \mathcal{N}(Y))$.

⁸⁸Editor's note: the preceding description of $\omega_{G/S}^1$, which is useful later (for example in VII_A, 5.5), has been added.

⁸⁹Editor's note: this description of $\omega_{G/S}^1$ in terms of invariant differentials is not used below.

⁹⁰Editor's note: the reason is that $\mathcal{L}ie(G/S) \simeq (\omega_{G/S}^1)^\vee$ need not determine $\omega_{G/S}^1$. For example, let $S = \mathbb{A}_k^1$, where k is a field, and let G be the S -group whose fiber at $s = 0$ is $\mathbb{G}_{a,k}$, while all other fibers are the unit group. Then $\mathcal{L}ie(G/S) = 0$, but $\operatorname{Lie}(G_s/k)(k) = k$.

is also a monomorphism (cf. 3.7). Every vector monomorphism of vector fibrations is a closed immersion.⁹¹ It follows that:

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S -group schemes.*

(i) *The morphism*

$$\mathrm{Lie}(G/S) \longrightarrow \mathrm{Lie}(H/S)$$

is a closed immersion; consequently

$$\omega_{H/S}^1 \longrightarrow \omega_{G/S}^1$$

is an epimorphism.

(ii) *If $\omega_{G/S}^1$ is locally free of finite type, then the corresponding morphism*

$$\mathcal{L}ie(G/S) \longrightarrow \mathcal{L}ie(H/S)$$

identifies $\mathcal{L}ie(G/S)$ with an $\mathcal{O}_S$ -submodule of $\mathcal{L}ie(H/S)$ that is locally a direct summand.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: diagonalizable groups

Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). Since the formation of $\underline{\mathrm{Lie}}(G/S)$ commutes with base extension, it is enough to carry out the construction for $G = D(M)$. We then have

$$G(I_S) = \mathrm{Hom}_{\mathrm{gr}}(M, \Gamma(I_S, \mathcal{O}_{I_S})^\times) = \mathrm{Hom}_{\mathrm{gr}}(M, \Gamma(S, \mathcal{D}_{\mathcal{O}_S})^\times).$$

There is a split exact sequence

$$1 \longrightarrow \Gamma(S, \mathcal{O}_S) \longrightarrow \Gamma(S, \mathcal{D}_{\mathcal{O}_S})^\times \longrightarrow \Gamma(S, \mathcal{O}_S)^\times \longrightarrow 1,$$

which shows that $\underline{\mathrm{Lie}}(G)(S)$ identifies with $\mathrm{Hom}_{\mathrm{gr}}(M, \mathcal{O}(S))$, endowed with its evident $\mathcal{O}(S)$ -module structure. After base change we therefore obtain:

Proposition 5.1. *There are isomorphisms*

$$\underline{\mathrm{Hom}}_{S\text{-gr}}(M_S, \mathbf{O}_S) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S)$$

and

$$\mathcal{H}om_{\mathrm{gr}}(\widetilde{M}_S, \mathcal{O}_S) \xrightarrow{\sim} \mathcal{L}ie(D_S(M)/S),$$

where, in the second isomorphism, $\widetilde{M}_S$ denotes the constant sheaf of groups on S defined by M , and $\mathcal{H}om_{\mathrm{gr}}$ denotes the sheaf of homomorphisms of sheaves of groups.

Corollary 5.1.1. *If M is free of finite type—or, as we shall say later, if $D_S(M)$ is a split torus—then, with W as in I 4.6.5,*

$$W(\mathcal{L}ie(D_S(M)/S)) \xrightarrow{\sim} \underline{\mathrm{Lie}}(D_S(M)/S), \quad M^\vee \otimes_{\mathbb{Z}} \mathcal{O}_S \xrightarrow{\sim} \mathcal{L}ie(D_S(M)/S).$$

Here $M^\vee$ denotes the dual of the abelian group M . In particular,

$$\mathbf{O}_S \xrightarrow{\sim} \underline{\mathrm{Lie}}(\mathbb{G}_{m,S}/S), \quad \mathcal{O}_S \xrightarrow{\sim} \mathcal{L}ie(\mathbb{G}_{m,S}/S).$$

⁹¹Editor's note: let $f : M \rightarrow N$ be a morphism of $\mathcal{O}_S$ -modules and let $P = \mathrm{Coker}(f)$. If $V(N) \rightarrow V(M)$ is a monomorphism, the surjective morphism $\mathrm{Sym}(N) \rightarrow \mathrm{Sym}(P)$ factors through $\mathcal{O}_S$; hence $P = 0$.

5.2. Normalizers and centralizers

We first prove a few lemmas. Recall (I 3.1.1) that a sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

of $\mathbf{O}_S$ -modules is called *exact* if, for every $S' \rightarrow S$, the sequence

$$0 \longrightarrow F'(S') \longrightarrow F(S') \longrightarrow F''(S') \longrightarrow 0$$

of $\mathbf{O}(S')$ -modules is exact. Likewise, a sequence $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ of S -groups is called exact if the corresponding sequence of groups on S' -points is exact for every $S' \rightarrow S$.

Lemma 5.2.1. *Let $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ be an exact sequence of S -groups.* ⁹²

(i) *The sequences*

$$1 \longrightarrow T_{G'/S}(\mathcal{M}) \longrightarrow T_{G/S}(\mathcal{M}) \longrightarrow T_{G''/S}(\mathcal{M}) \longrightarrow 1$$

and

$$1 \longrightarrow \underline{\mathrm{Lie}}(G'/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(G/S, \mathcal{M}) \longrightarrow \underline{\mathrm{Lie}}(G''/S, \mathcal{M}) \longrightarrow 1$$

are exact.

(ii) *Let $1 \rightarrow H' \rightarrow H \rightarrow H'' \rightarrow 1$ be a second sequence of groups. It is exact if and only if*

$$1 \longrightarrow G' \times_S H' \longrightarrow G \times_S H \longrightarrow G'' \times_S H'' \longrightarrow 1$$

is exact.

(iii) *If two of G', G, G'' satisfy condition (E), then so does the third.*

(iv) *If $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is exact and two of the $\mathbf{O}_S$ -modules F', F, F'' are good, then so is the third.*

(v) *If two of the S -groups G', G, G'' are good, then so is the third.*

The first part of (i) is immediate, and the second follows from it. Likewise, (ii) is immediate. We prove (iii). For brevity, write $L(\mathcal{M}) = \underline{\mathrm{Lie}}(G, \mathcal{M})$, $L'(\mathcal{M}) = \underline{\mathrm{Lie}}(G', \mathcal{M})$, and similarly for L'' . We have the following commutative diagram. ⁹³

$$\begin{array}{ccccccc} 1 & \longrightarrow & L'(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & L(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & L''(\mathcal{M} \oplus \mathcal{N}) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & L'(\mathcal{M}) \times L'(\mathcal{N}) & \longrightarrow & L(\mathcal{M}) \times L(\mathcal{N}) & \longrightarrow & L''(\mathcal{M}) \times L''(\mathcal{N}) \longrightarrow 1 \end{array}$$

Figure 52: The exact-sequence diagram used to prove Lemma 5.2.1(iii).

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 44;
combined-reader p. 102.*

The upper row is exact by (i), and the lower row is exact by (i) and (ii). Assertion (iii) follows.

⁹²Editor's note: The statement and proof of this lemma have been expanded; this will be useful in 5.3.1.

⁹³Editor's note: The proofs of (iii) and (v) have been added.

We prove (iv). There is a commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & F' \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) & \longrightarrow & F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) & \longrightarrow & F'' \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & T_{F'/S}(\mathcal{M}) & \longrightarrow & T_{F/S}(\mathcal{M}) & \longrightarrow & T_{F''/S}(\mathcal{M}) \longrightarrow 0
 \end{array}$$

Figure 53: The exact-sequence diagram used to prove Lemma 5.2.1(iv).

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 44;
combined-reader p. 102.*

with exact rows: the first because $T_{\mathbf{O}_S/S}(\mathcal{M})$ is a free, hence flat, $\mathbf{O}_S$ -module, and the second by (i). Thus, if two of F', F, F'' are good, so is the third. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let G be an S -group, let E, H be two G -objects, and let F be a G - $\mathbf{O}_S$ -module.*

- (i) *The canonical homomorphism $E^G \times_S H^G \rightarrow (E \times_S H)^G$ is an isomorphism.*
- (ii) *If F is a good $\mathbf{O}_S$ -module, then so is F^G .*

The first assertion is immediate; we prove the second. For every $S' \rightarrow S$, there is a commutative diagram

$$\begin{array}{ccc}
 F^G(I_{S'}(\mathcal{M})) & \xhookrightarrow{\quad} & F(I_{S'}(\mathcal{M})) \\
 \uparrow \phi & & \uparrow \simeq \phi_1 \\
 F^G(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M})) & \xhookrightarrow{\quad} & F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M}))
 \end{array}$$

Figure 54: The invariants-and-base-change square in the proof of Lemma 5.2.2.

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 45;
combined-reader p. 103.*

and we must prove that ϕ is bijective. It is plainly injective. To prove surjectivity, let $(t_0, \dots, t_n)$ be a basis of the free $\mathbf{O}(S')$ -module $\mathbf{O}(I_{S'}(\mathcal{M}))$, and let

$$u = \sum_{i=0}^n f_i \otimes t_i$$

be an element of $F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M}))$ such that $\phi_1(u)$ belongs to $F^G(I_{S'}(\mathcal{M}))$. We show that each f_i belongs to $F^G(S')$.

Let $S'' \rightarrow S'$. Via the zero section $\varepsilon_{\mathcal{M}}$, we may regard S'' as lying over $I_{S'}(\mathcal{M})$. Then, for every $g \in G(S'')$,

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^n g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

Since the $(t_i)_{S''}$ form a basis of

$$\mathbf{O}(I_{S''}(\mathcal{M})) \simeq \mathbf{O}(I_{S'}(\mathcal{M})) \otimes_{\mathbf{O}(S')} \mathbf{O}(S'')$$

over $\mathbf{O}(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, hence $f_i \in F^G(S')$ for all i .

Notation. If E is an S -group and F a sub- S -group, write E/F for the S -functor sending $S' \rightarrow S$ to the set $E(S')/F(S')$ of classes $\bar{x} = xF(S')$. If E is commutative, then E/F is a commutative S -group.⁹⁴

Now let G be an S -group and K a sub- S -group, and set $E = \underline{\text{Lie}}(G/S, \mathcal{M})$ and $F = \underline{\text{Lie}}(K/S, \mathcal{M})$. The adjoint action of K on E stabilizes F , and therefore induces an action on E/F . For every $S' \rightarrow S$,

$$(E/F)^K(S') = \{\bar{x} \in E(S')/F(S') \mid \text{for every } f : S'' \rightarrow S' \text{ and } k \in K(S''), \\ f^*(\bar{x}^{-1}) \text{Ad}(k)(f^*(\bar{x})) \in F(S'')\},$$

where $f^*(\bar{x})$ denotes the image of $\bar{x}$ in $E(S'')$.

Theorem 5.2.3. *Let G be an S -group and K a sub- S -group. Put (I 2.3.3)*

$$N = \underline{\text{Norm}}_G(K), \quad Z = \underline{\text{Centr}}_G(K),$$

and let K act on $\underline{\text{Lie}}(G/S, \mathcal{M})$ through the adjoint representation of G .

(i) *If the group law of $\underline{\text{Lie}}(G/S, \mathcal{M})$ is commutative,^{95,96} then*

$$\frac{\underline{\text{Lie}}(N/S, \mathcal{M})}{\underline{\text{Lie}}(K/S, \mathcal{M})} = \left(\frac{\underline{\text{Lie}}(G/S, \mathcal{M})}{\underline{\text{Lie}}(K/S, \mathcal{M})} \right)^K.$$

(ii) *If the same group law is commutative, then*

$$\underline{\text{Lie}}(Z/S, \mathcal{M}) = \underline{\text{Lie}}(G/S, \mathcal{M})^K.$$

(iii) *If G satisfies (E), then Z satisfies (E); if G and K satisfy (E), then N satisfies (E).*

(iv) *If G is good, then Z is good; if, moreover, G is very good, then Z is very good.*

(v) *If G and K are good, then N is good; if, moreover, G is very good, then N is very good.*

To prove (i) and (ii),⁹⁷ we use the following lemma, which follows from the diagram of exact sequences considered in 4.1.B, with G and H interchanged.

Lemma 5.2.3.0. *Let G be an S -group, H a sub- S -group, and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. Then $T_{H/S}(\mathcal{M})$ and $\underline{\text{Lie}}(G/S, \mathcal{M})$ are sub- S -groups of $T_{G/S}(\mathcal{M})$, and*

$$T_{H/S}(\mathcal{M}) \cap \underline{\text{Lie}}(G/S, \mathcal{M}) = \underline{\text{Lie}}(H/S, \mathcal{M}),$$

where

$$T_{H/S}(\mathcal{M}) \cap \underline{\text{Lie}}(G/S, \mathcal{M}) := T_{H/S}(\mathcal{M}) \times_{T_{G/S}(\mathcal{M})} \underline{\text{Lie}}(G/S, \mathcal{M}).$$

⁹⁴Editor's note: These notations have been expanded.

⁹⁵This condition is automatic when G satisfies (E), for example when G is representable.

⁹⁶Editor's note: For an example in which the S -group $\underline{\text{Lie}}(G/S)$ is noncommutative, see 6.3.

⁹⁷Editor's note: The proof has been expanded, and in particular Lemma 5.2.3.0, implicit in the original, has been added.

Since the functors in (i) and (ii) commute with base extension, it is enough to prove the equalities on S -points. Put $H = N$ (respectively, $H = Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definitions of N and Z (I 2.3.3),

$$\begin{aligned} \underline{\text{Lie}}(H/S, \mathcal{M})(S) = \{ X \in \underline{\text{Lie}}(G/S, \mathcal{M})(S) \subset G(I_S(\mathcal{M})) \mid \\ \text{for every } f : S' \rightarrow I_S(\mathcal{M}) \text{ and } u \in K(S'), \\ f^*(X^\alpha) u f^*(X^{-\alpha}) u^{-1} \in K(S'), \\ \text{respectively } f^*(X) u f^*(X^{-1}) u^{-1} = 1 \}. \end{aligned} \quad (*)$$

Here $f^*(X)$ denotes the image of X in $G(S')$.

For brevity, write

$$\mathfrak{g} = \underline{\text{Lie}}(G/S, \mathcal{M}), \quad \mathfrak{k} = \underline{\text{Lie}}(K/S, \mathcal{M}), \quad \mathfrak{n} = \underline{\text{Lie}}(N/S, \mathcal{M}), \quad \mathfrak{z} = \underline{\text{Lie}}(Z/S, \mathcal{M}).$$

If $X \in \underline{\text{Lie}}(H/S, \mathcal{M})(S)$, the equalities (*) hold for every $f : S' \rightarrow S$, because f factors through $I_S(\mathcal{M})$ as

$$f = \rho \circ \varepsilon_{\mathcal{M}} \circ f,$$

where $\rho : I_S(\mathcal{M}) \rightarrow S$ is the structural morphism and $\varepsilon_{\mathcal{M}} : S \rightarrow I_S(\mathcal{M})$ is the zero section. Consequently,

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S), \quad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

For the reverse inclusions, assume henceforth that $\mathfrak{g}$ is commutative. Then $\mathfrak{g}^K$ and $(\mathfrak{g}/\mathfrak{k})^K$ are commutative S -groups. Let $X \in \mathfrak{g}(S)$, and write $\bar{X}$ for its image in $(\mathfrak{g}/\mathfrak{k})(S)$. Suppose that $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$ (respectively, $X \in \mathfrak{g}^K(S)$); we show that $X \in \mathfrak{n}(S)$ (respectively, $X \in \mathfrak{z}(S)$).

Let $f : S' \rightarrow I_S(\mathcal{M})$. Consider the cartesian square

$$\begin{array}{ccc} I_{S'}(\mathcal{M}) & \xrightarrow{p} & I_S(\mathcal{M}) \\ \rho' \downarrow & & \downarrow \rho \\ S' & \xrightarrow{\rho \circ f} & S \end{array}$$

Figure 55: The cartesian square used in the reverse-inclusion argument.

Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 47; combined-reader p. 105.

and let h be the section of ρ' defined by f . It is enough to show that, for every $v \in K(I_{S'}(\mathcal{M}))$,

$$\left. \begin{aligned} & p^*(X^\alpha) v p^*(X^{-\alpha}) v^{-1} \in K(I_{S'}(\mathcal{M})), \\ \text{respectively } & p^*(X) v p^*(X^{-1}) v^{-1} = 1. \end{aligned} \right\} \quad (**)$$

Indeed, take $v = \rho'^*(u)$ and apply h^* to (**). This gives (*), because $\rho' \circ h = \text{id}_{S'}$ and $p \circ h = f$.

We now prove (**), writing X for $p^*(X)$. Every $v \in K(I_{S'}(\mathcal{M}))$ is uniquely of the form Yk , with $Y \in \underline{\text{Lie}}(K/S, \mathcal{M})(S')$ and $k \in K(S')$. The expression $X^\alpha v X^{-\alpha} v^{-1}$ becomes

$$X^\alpha Y (k X^{-\alpha} k^{-1}) Y^{-1} = X^\alpha \text{Ad}(k)(X^{-\alpha}),$$

because $\underline{\text{Lie}}(G/S, \mathcal{M})$ is commutative. This element lies in $\underline{\text{Lie}}(G/S, \mathcal{M})(S')$. By Lemma 5.2.3.0, condition (**) is therefore equivalent, for every $k \in K(S')$, to

$$\begin{cases} \text{Ad}(k)(X) = X, & H = Z, \\ X^\alpha \text{Ad}(k)(X^{-\alpha}) \in \underline{\text{Lie}}(K/S, \mathcal{M})(S'), & H = N. \end{cases}$$

For $H = Z$, this follows from $X \in \mathfrak{g}^K(S)$. For $H = N$, the condition may equally be written

$$\text{Ad}(k)(\bar{X}) = \bar{X}, \quad \text{Ad}(k)(\bar{X}^{-1}) = \bar{X}^{-1}. \quad (*')$$

The second equality follows from the first because the action of K on $\mathfrak{g}/\mathfrak{k}$ respects its group structure. Thus (**) also follows from $\bar{X} \in (\mathfrak{g}/\mathfrak{k})^K(S)$, proving (i) and (ii).

For (iii)–(v), put

$$\mathfrak{g}(\mathcal{M}) = \underline{\text{Lie}}(G/S, \mathcal{M})$$

and define $\mathfrak{k}(\mathcal{M})$, $\mathfrak{z}(\mathcal{M})$, and $\mathfrak{n}(\mathcal{M})$ similarly. If G/S satisfies (E), then

$$\mathfrak{g}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{g}(\mathcal{M}) \times_S \mathfrak{g}(\mathcal{N}),$$

and hence, by Lemma 5.2.2(i),

$$\mathfrak{g}(\mathcal{M} \oplus \mathcal{N})^K \simeq \mathfrak{g}(\mathcal{M})^K \times_S \mathfrak{g}(\mathcal{N})^K.$$

It follows that $\mathfrak{z}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{z}(\mathcal{M}) \times_S \mathfrak{z}(\mathcal{N})$, so Z satisfies (E). If K/S also satisfies (E), then

$$\begin{aligned} (\mathfrak{g}/\mathfrak{k})(\mathcal{M} \oplus \mathcal{N}) &\simeq (\mathfrak{g}/\mathfrak{k})(\mathcal{M}) \times_S (\mathfrak{g}/\mathfrak{k})(\mathcal{N}), \\ (\mathfrak{g}/\mathfrak{k})^K(\mathcal{M} \oplus \mathcal{N}) &\simeq (\mathfrak{g}/\mathfrak{k})^K(\mathcal{M}) \times_S (\mathfrak{g}/\mathfrak{k})^K(\mathcal{N}), \end{aligned}$$

and we have the following commutative diagram.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{k}(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & \mathfrak{n}(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & (\mathfrak{n}/\mathfrak{k})(\mathcal{M} \oplus \mathcal{N}) \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow & & \downarrow \wr \\ 0 & \longrightarrow & \mathfrak{k}(\mathcal{M}) \times_S \mathfrak{k}(\mathcal{N}) & \longrightarrow & \mathfrak{n}(\mathcal{M}) \times_S \mathfrak{n}(\mathcal{N}) & \longrightarrow & (\mathfrak{n}/\mathfrak{k})(\mathcal{M}) \times_S (\mathfrak{n}/\mathfrak{k})(\mathcal{N}) \longrightarrow 0 \end{array}$$

Figure 56: The exact-sequence comparison proving condition (E) for N .

*Temporary Loop-1 image; source locator: Polo-Gille Exp2-14oct24.pdf, component-local p. 48;
combined-reader p. 106.*

It follows that

$$\mathfrak{n}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{n}(\mathcal{M}) \times_S \mathfrak{n}(\mathcal{N}),$$

so N satisfies (E).

Now write $\mathfrak{g} = \mathfrak{g}(\mathcal{O}_S) = \underline{\text{Lie}}(G/S)$, and similarly $\mathfrak{k}, \mathfrak{z}, \mathfrak{n}$. If G is good, then $\mathfrak{g}$ is a good $\mathbf{O}_S$ -module, and Lemma 5.2.2(ii) shows that $\mathfrak{z} \simeq \mathfrak{g}^K$ is good; since Z satisfies (E), Z is good.

If K is also good, then $\mathfrak{k}$ is good, and Lemmas 5.2.1(iv) and 5.2.2(ii) show that $\mathfrak{g}/\mathfrak{k}$ and $(\mathfrak{g}/\mathfrak{k})^K$ are good. From the exact sequence

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{n} \longrightarrow (\mathfrak{g}/\mathfrak{k})^K \longrightarrow 0$$

and Lemma 5.2.1(iv), $\mathfrak{n}$ is good. Finally, if G is very good, so that $[x, x] = 0$ identically for $x \in \mathfrak{g}$, then Z and N are plainly very good. This proves (iii)–(v).

Corollary 5.2.3.1. *If the group law of $\underline{\text{Lie}}(G/S)$ is commutative, then*

$$\underline{\text{Lie}}(\underline{\text{Centr}}(G)/S) = \underline{\text{Lie}}(G/S)^G.$$

Corollary 5.2.3.2. *If the group law of $\underline{\text{Lie}}(G/S)$ is commutative and K is a normal subgroup of G , then*

$$(\underline{\text{Lie}}(G/S)/\underline{\text{Lie}}(K/S))^K = \underline{\text{Lie}}(G/S)/\underline{\text{Lie}}(K/S).$$

5.3. Linear representations

Let G be a good S -group acting linearly on a good $\mathbf{O}_S$ -module F through

$$\rho : G \longrightarrow \underline{\text{Aut}}_{\mathbf{O}_S\text{-mod}}(F).$$

In 4.7.1 and 4.8.2 we defined the corresponding linear representation

$$\rho' : \underline{\text{Lie}}(G/S) \longrightarrow \underline{\text{End}}_{\mathbf{O}_S\text{-mod}}(F).$$

The sub- S -groups $\underline{\text{Norm}}_G(E)$ and $\underline{\text{Centr}}_G(E)$ are defined for every subset E of F , and in particular for every $\mathbf{O}_S$ -submodule E .

Definition 5.3.0. For every $S' \rightarrow S$, put

$$\underline{\text{Norm}}_{\underline{\text{Lie}}(G/S)}(E)(S') = \{X \in \underline{\text{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} \subset E_{S'}\},$$

$$\underline{\text{Centr}}_{\underline{\text{Lie}}(G/S)}(E)(S') = \{X \in \underline{\text{Lie}}(G/S)(S') \mid \rho'(X)E_{S'} = 0\}.$$

This construction works for every linear representation of an $\mathbf{O}_S$ -Lie algebra in the sense of 4.9, and the two subobjects obtained are $\mathbf{O}_S$ -submodules stable under the bracket.

Theorem 5.3.1. *Let G be a good S -group acting linearly on a good $\mathbf{O}_S$ -module F , and let E be an $\mathbf{O}_S$ -submodule of F .*

(i) *One has*

$$\underline{\text{Lie}}(\underline{\text{Centr}}_G(E)/S) = \underline{\text{Centr}}_{\underline{\text{Lie}}(G/S)}(E),$$

and $\underline{\text{Centr}}_G(E)$ is a good S -group; it is very good if G is very good.

(ii) *Suppose that E is a good $\mathbf{O}_S$ -module. Then⁹⁸*

$$\underline{\text{Lie}}(\underline{\text{Norm}}_G(E)/S) = \underline{\text{Norm}}_{\underline{\text{Lie}}(G/S)}(E),$$

and $\underline{\text{Norm}}_G(E)$ is a good S -group; it is very good if G is very good.

⁹⁸Editor's note: The original stated the following equality without a hypothesis on E ; this has been corrected. See 6.5.1 for counterexamples.

The proof is left to the reader.

5.3.2. Let G be a good S -group. The preceding results apply in particular to the adjoint representation. Let E be a good $\mathbf{O}_S$ -submodule of $\underline{\mathrm{Lie}}(G/S)$.⁹⁹ It has a centralizer and a normalizer in G , and by 5.3.1 their Lie algebras are the usual centralizer and normalizer of E , computed using the bracket:

$$\begin{aligned}\underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') &= \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] = 0\}, \\ \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(E)(S') &= \{X \in \underline{\mathrm{Lie}}(G/S)(S') \mid [X, E_{S'}] \subset E_{S'}\}.\end{aligned}$$

5.3.3. Let K be a sub- S -group of G . Then $\underline{\mathrm{Lie}}(K/S)$ is an $\mathbf{O}_S$ -submodule of $\underline{\mathrm{Lie}}(G/S)$. Suppose that it is a good $\mathbf{O}_S$ -module,¹⁰⁰ as happens when K is a good S -group. Plainly,

$$\begin{aligned}\underline{\mathrm{Centr}}_G(K) &\subset \underline{\mathrm{Centr}}_G(\underline{\mathrm{Lie}}(K/S)), \\ \underline{\mathrm{Norm}}_G(K) &\subset \underline{\mathrm{Norm}}_G(\underline{\mathrm{Lie}}(K/S)).\end{aligned}$$

Hence, by 5.3.1,

$$\begin{aligned}\underline{\mathrm{Lie}}(\underline{\mathrm{Centr}}_G(K)/S) &\subset \underline{\mathrm{Centr}}_{\underline{\mathrm{Lie}}(G/S)}(\underline{\mathrm{Lie}}(K/S)), \\ \underline{\mathrm{Lie}}(\underline{\mathrm{Norm}}_G(K)/S) &\subset \underline{\mathrm{Norm}}_{\underline{\mathrm{Lie}}(G/S)}(\underline{\mathrm{Lie}}(K/S)).\end{aligned}$$

None of these four inclusions is an equality in general; many examples will appear later. In particular, if K is a normal subgroup of G , then $\underline{\mathrm{Lie}}(K/S)$ is an ideal of $\underline{\mathrm{Lie}}(G/S)$.

6. Miscellaneous remarks

6.1.

The bracket of two infinitesimal automorphisms can be defined for an S -functor X which is not necessarily a group: apply the results of this exposé to $\underline{\mathrm{Aut}}_S(X)$. To obtain a convenient formalism, one is led to assume that X is good, meaning that the $\mathbf{O}_X$ -module $T_{X/S}$ is good. If X is an S -group, this agrees with Definition 4.6.

6.2.

There exist functors with infinitesimal endomorphisms that are not automorphisms and hence, a fortiori, do not satisfy condition (E).¹⁰¹ For a pointed set (E, x_0) , let $M_A(E)$ be the free abelian monoid generated by E , and let $M_{A,P}(E, x_0)$ be the quotient of $M_A(E)$ by the equivalence relation generated by $m \sim x_0 + m$. The assignment

$$(E, x_0) \longmapsto M_{A,P}(E, x_0)$$

is left adjoint to the forgetful functor from abelian monoids to pointed sets; $M_{A,P}(E, x_0)$ is the free abelian monoid on the pointed set (E, x_0) .

Now let X be the functor assigning to every scheme S the free abelian monoid on $\mathbf{O}(S)$, pointed by zero. Every morphism

$$f : S \longrightarrow I_{\mathbb{Z}} = \mathrm{Spec}(\mathbb{Z}[\varepsilon])$$

⁹⁹Editor's note: This hypothesis has been added; cf. 6.5.1.

¹⁰⁰Editor's note: This hypothesis has been added; cf. 6.5.1.

¹⁰¹Editor's note: The original has been expanded here. In particular, O. Gabber pointed out the correct definition of the free abelian monoid on a pointed set.

corresponds to a square-zero element $u_f \in \mathbf{O}(S)$, and hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$, where the sum is taken in $M_{A,P}(\mathbf{O}(S), 0)$. Thus we obtain an endomorphism ϕ of

$$X_{I_{\mathbb{Z}}} = X \times_{\mathbb{Z}} I_{\mathbb{Z}}$$

defined, for $f \in I_{\mathbb{Z}}(S)$ and $x \in X(S)$, by

$$\phi((x, f)) = (x + u_f, f).$$

If $f_0 : S \rightarrow I_{\mathbb{Z}}$ is the composite of the structural morphism $S \rightarrow \mathrm{Spec}(\mathbb{Z})$ and the zero section, then $u_{f_0} = 0$, so $\phi((x, f_0)) = (x, f_0)$. Thus ϕ induces the identity on X : it is an infinitesimal endomorphism of X which is plainly not an automorphism.

6.3.

There exist modules which are not good. First, the preceding counterexample can be modified slightly.¹⁰² Let $F(S)$ be the free $\mathbf{O}(S)$ -module with basis the elements of $\mathbf{O}(S)$. Then F does not satisfy (E) relative to $\mathrm{Spec}(\mathbb{Z})$. Moreover, let G be the $\mathbb{Z}$ -group defined by

$$G(S) = \mathrm{GL}_n(R(S)), \quad n \geq 2,$$

where $R(S)$ is the $\mathbf{O}(S)$ -algebra of the abelian group $\mathbf{O}(S)$. Then the $\mathbb{Z}$ -group $\underline{\mathrm{Lie}}(G/\mathbb{Z})$ is not commutative.

There are also the following counterexamples.¹⁰³ Let $S = \mathrm{Spec}(k)$, where k is a field of characteristic $p > 0$.

(a) Let F be the $\mathbf{O}_S$ -module assigning to every S -scheme T the group

$$F(T) = \Gamma(T, \mathcal{O}_T),$$

with its $\mathbf{O}(T)$ -module structure defined by $r \cdot f = r^p f$. As an S -functor of groups, F is isomorphic to $\mathbb{G}_{a,S}$. Thus F/S satisfies (E), and

$$\underline{\mathrm{Lie}}(F/S) \simeq \underline{\mathrm{Lie}}(\mathbb{G}_{a,S}/S) \simeq \mathbf{O}_S.$$

For each $T \rightarrow S$, the canonical morphism $F \rightarrow \underline{\mathrm{Lie}}(F/S)$ is the identity map $F(T) \rightarrow \mathbf{O}(T)$. It respects the abelian-group structure but not the module structure, so F is not good (4.4.1).

(b) Let $\alpha_{p,k}$ be the k -functor of groups defined by

$$\alpha_{p,k}(T) = \{x \in \mathbf{O}(T) \mid x^p = 0\}.$$

It is represented by $\mathrm{Spec}(k[X]/(X^p))$, hence is a very good S -group. It also carries an $\mathbf{O}_S$ -module structure, but is not a good $\mathbf{O}_S$ -module, because the canonical morphism

$$\alpha_{p,k} \longrightarrow \underline{\mathrm{Lie}}(\alpha_{p,k}/k) = \mathbb{G}_{a,k}$$

is not bijective.

6.4.

For a group-valued functor G over S , the definitions give

$$(G/S \text{ satisfies (E)}) \Longleftarrow (G \text{ is good}) \Longleftarrow (G \text{ is very good}).$$

It would be interesting to prove or disprove the reverse implications.¹⁰⁴

¹⁰²Editor's note: The following observation has been added.

¹⁰³Editor's note: Example (a) has been expanded and example (b) has been added.

¹⁰⁴Editor's note: A representable G is very good (4.11). For representability criteria see, for example, [MO67]. For automorphisms of an affine algebraic group over an algebraically closed field of characteristic zero, see [HM69].

6.5.

Let Nil be the $\mathbb{Z}$ -functor of groups defined by

$$\text{Nil}(S) = \{x \in \mathbf{O}(S) \mid \text{there exists } n \in \mathbb{N} \text{ with } x^n = 0\}.$$

¹⁰⁵ The functor Nil is very good but not representable. Let Nil^2 , $\mathbf{O}_{\text{red}}$, and F be the $\mathbb{Z}$ -functors of groups assigning to S , respectively, the ideal $\text{Nil}(S)^2$ and the groups

$$\mathbf{O}_{\text{red}}(S) = \mathbf{O}(S)/\text{Nil}(S), \quad F(S) = \mathbf{O}(S)/\text{Nil}(S)^2.$$

It is easy to see that

$$\underline{\text{Lie}}(\mathbf{O}_{\text{red}}/\mathbb{Z}) = 0,$$

so the $\mathbf{O}_{\mathbb{Z}}$ -module $\mathbf{O}_{\text{red}}$ is not good, although it is a good $\mathbb{Z}$ -group. If $\mathcal{M}, \mathcal{N}$ are free $\mathbb{Z}$ -modules of finite rank, then

$$\begin{aligned} \text{Nil}^2(I_S(\mathcal{M} \oplus \mathcal{N})) &\simeq \text{Nil}^2(S) \oplus \text{Nil}(S) \otimes_{\mathbb{Z}} \mathcal{M} \\ &\quad \oplus \text{Nil}(S) \otimes_{\mathbb{Z}} \mathcal{N}, \end{aligned}$$

and hence

$$\begin{aligned} F(I_S(\mathcal{M} \oplus \mathcal{N})) &\simeq F(S) \oplus \mathbf{O}_{\text{red}}(S) \otimes_{\mathbb{Z}} \mathcal{M} \\ &\quad \oplus \mathbf{O}_{\text{red}}(S) \otimes_{\mathbb{Z}} \mathcal{N}. \end{aligned}$$

Thus F satisfies condition (E), while

$$\underline{\text{Lie}}(F/\mathbb{Z}) = \mathbf{O}_{\text{red}}.$$

Since $\mathbf{O}_{\text{red}}$ is not a good $\mathbf{O}_{\mathbb{Z}}$ -module, F is a $\mathbb{Z}$ -group which satisfies (E) but is not good.

6.5.1. We now give the counterexamples announced in 5.3.1–5.3.3. Let S be a scheme, let $F = \mathbf{O}_S^{\oplus 2}$ with the natural action of the good S -group $G = \text{GL}_{2,S}$, and let E be the $\mathbf{O}_S$ -submodule of F formed by the pairs (x_1, x_2) for which x_2 is nilpotent. Put

$$N = \underline{\text{Norm}}_G(E).$$

Then

$$\underline{\text{Lie}}(N/S) = \underline{\text{Lie}}(G/S),$$

whereas, for every $S' \rightarrow S$,

$$\underline{\text{Norm}}_{\underline{\text{Lie}}(G/S)}(E)(S') = \left\{ \begin{pmatrix} a & b \\ x & c \end{pmatrix} \mid a, b, c, x \in \mathbf{O}(S'), \ x \text{ nilpotent} \right\}.$$

Consequently,

$$\underline{\text{Lie}}(\underline{\text{Norm}}_G(E)/S) \neq \underline{\text{Norm}}_{\underline{\text{Lie}}(G/S)}(E).$$

Taking the semidirect product $G_1 = F \rtimes G$ gives an analogous counterexample in which E is an $\mathbf{O}_S$ -submodule of $\underline{\text{Lie}}(G_1/S)$. Moreover, with the notation above,

$$E = \underline{\text{Lie}}(K/S),$$

where K is the subgroup $\mathbf{O}_S \oplus \text{Nil}^2$ of F ; that is, $K(S')$ consists of the pairs (x_1, x_2) with $x_2 \in \text{Nil}(S')^2$.

¹⁰⁵Editor's note: This paragraph has been added.

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¹⁰⁶Editor's note: Additional references cited in this exposé.